Variable-Delay Low-Pass, All-Pass Filters and their Applications

A thesis submitted in partial fulfillment of
the requirements for the degree of

Doctor of Philosophy

by

Mayur Shivamurthy
(Roll No. EE18DP003)

Under the guidance of
Prof. Naveen Kadayinti

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ಭಾ.ತಂ.ಸಂ. ಧಾರವಾಡ
भा. प्रौ. सं. धारवाड
I.I.T. DHARWAD

DEPARTMENT OF ELECTRICAL, ELECTRONICS AND
COMMUNICATION ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY DHARWAD
November, 2025

Thesis Approval

The thesis entitled

Variable Delay Low-Pass, All-Pass Filters and their Applications

by

[illegible]
[illegible]

is approved for the degree of

Doctor of Philosophy

Internal Examiner	**External Examiner**
[illegible]	[illegible]
	[illegible]
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Mayur Shivamurthy
Roll No. EE18DP003

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Abstract

Continuous-time filters introduce a certain amount of delay to an input signal. The delay of these filters can be varied by changing the filter's time constant. This work mainly focuses on delay tuning techniques for low-pass filters (LPFs) and all-pass filters (APFs). In the case of variable-delay LPFs, the delay of the filter is altered by changing the effective impedance of the circuit using the principle of the Miller effect. In the case of APFs realized with active G_m-C architecture, programmable delays are obtained using constant-C frequency-scaling techniques. These techniques of delay tuning are used in various real-time applications such as in realizing wideband tunable delay lines for beamforming, expansion, or compression of continuous-time pulses and voltage-controlled oscillators (VCOs).

The first application of a tunable-delay LPF is in realization of a true time-delay (TTD) circuit for wideband beamforming systems. A 15-stage TTD circuit with delay correction is implemented in the TSMC 65-nm CMOS technology with supply voltage of 1.2 V. Post layout simulation results show that 15 cascaded stages of the TTD with the proposed delay correction technique has a maximum delay variation of ± 5 ps across 1 GHz bandwidth with a power/delay-range factor of 0.1 mW/ps. A 3-stage VCO with a proposed variable-delay cell in current-modelogic (CML) is discussed. The proposed VCO architecture is implemented in TSMC 180-nm and 65-nm technologies. Results show that VCO has a tuning range of 56.4% with phase noise of -76 dBc/Hz at offset frequency of 1 MHz. The VCO dissipates power of 7.2 mW from 1.2 V supply with figure of merit of 181.69 dBc/Hz. The next application of variable delay LPF would be in the expansion and compression of continuous-time pulses. A 15-stage expansion circuit is implemented in TSMC 65-nm CMOS technology respectively. A stretch factor of 2.2 is obtained for a Gaussian pulse with a 3-dB bandwidth of 1 GHz. The circuit has a power consumption of 36 mW with RMS error of 1.5 mV between actual and ideal samples. A 15-stage pulse compression circuit using similar variable delay cells is implemented in TSMC 65-nm CMOS technology. A compression factor of 0.47 is obtained with an input pulse width

of 3.32 ns.

A G_m-C-based variable-delay APF as a TTD element is presented in this work. The APF group delay is varied by changing its bias current to modulate its Gm for different delay settings, which avoids the need for high-quality varactors. The proposed design is implemented in TSMC 65-nm technology. Simulation results show that the delay element with 3 stages of third-order APF maintains uniform magnitude response with the delay-range-bandwidth product of 0.8 and power/delay-range factor of 0.04 mW/ps upto bandwidth of 1 GHz. Expansion and compression of continuous-time pulses is achieved using a similar architecture consisting of 3 stages of low-order variable delay APFs. A Gilbert multiplier with high output impedance is used as a variable Gm block for enabling expansion and compression. The expansion and compression circuits are implemented in TSMC 65-nm CMOS technology. Simulation results show that the expansion circuit provides a stretch factor of 1.48 and compression circuit provides a compression factor of 0.84 for an input Gaussian pulse of 0.94 ns. A novel method for calibrating a negative conductance load for impedance cancellation in high-gain operational transconductance amplifiers (OTAs) is also presented in this work. The basic principle of tuning this negative conductance load is by minimizing the steady state error of the OTA that is configured in a closed loop in calibration mode. The proposed method is implemented in TSMC 65-nm CMOS technology. Simulation results show that an OTA with a gain of 18 dB achieves a gain of 54 dB with the tuned negative conductance load.

Finally, this work also discusses the practical implications of a TTD circuit on beamforming. TTD elements have been explored over phase shifters in wideband beamforming applications to overcome the effect of beam-squint. This work reviews different TTD architectures reported in the literature and discusses the non-ideal effects of TTD circuits on beam- pattern of timed antenna array receivers. This helps in understanding the inevitable beam-squints associated with practical TTDs.

Contents

List of Tables

List of Figures

List of Abbreviations

ADC analog-to-digital converter.

APF all-pass filter.

CML current-mode logic.

DAC digital-to-analog converter.

DPDT double pole double throw.

FIR finite impulse response.

FOM figure of merit.

LPF low-pass filter.

OTA operational transconductance amplifier.

PLL phase locked loop.

RMSE root mean square error.

SPDT single pole double throw.

TTD true time-delay.

VCO voltage-controlled oscillator.

Chapter 1

Introduction

On-chip continuous-time filters are used in several analog and mixed signal integrated circuits. For example, low-pass filters (LPFs) are used for anti-aliasing of signals before sampling in an analog-to-digital converter (ADC) and for smoothening analog signal at the output of digital-to-analog converter (DAC) [1]. They are used in intermediate frequency (IF) bandpass filters in wireless transceivers [2], as loop filters in the design of phase locked loop (PLL)'s [3] and continuous time $\Delta\Sigma$ converters [4] etc. Continuous-time filters are also used in disk drivers for noise cancellation [5], high-speed serial data links [6], and pulse-shaping applications [7].

One of the important characteristics of a continuous time filter is its group delay. The filter's group delay is defined as the time delay experienced by different frequency components of an input signal. Let $H(j\omega) = |H(j\omega)|\measuredangle H(j\omega)$ represents the transfer function of a continuous time filter, where $|H(j\omega)|$ and $\measuredangle H(j\omega)$ represent gain and phase of the filter respectively. Mathematically group delay $\tau(\omega)$ of the filter is expressed as

$$\tau(\omega) = -\frac{\mathrm{d}\measuredangle H(j\omega)}{\mathrm{d}\omega}. \tag{1.1}$$

A filter with frequency-independent delay characteristics is termed as a linear phase filter, in which all input signal frequency components are delayed by the same amount of time. This is an important characteristic of a distortionless system in which the signal waveshape in the passband is unaltered.

1.1 Group Delay of Continuous-Time Filters

Practically, different types of filters exhibit different gain and group delay characteristics. A 1st-order RC-LPF shown in Figure 1.1 has a transfer function H(s) which can be expressed as,

$$H(s) = \frac{V_o(s)}{V_{IN}(s)} = \frac{1}{1 + sRC} \tag{1.2}$$

The phase of the filter $\measuredangle H(j\omega) = -\tan^{-1}(\omega RC)$ and the group delay τ of the circuit is expressed as,

$$\tau = \frac{d(\measuredangle H(j\omega))}{d\omega} = \frac{RC}{1 + \omega^2 R^2 C^2}. \tag{1.3}$$

The expression for τ with $\omega^2 R^2 C^2 << 1$ gives the low-frequency group delay and is expressed as $\tau \approx RC$. Figure 1.2(a) and Figure 1.2(b) shows the corresponding gain response and the group delay plot of H(s).

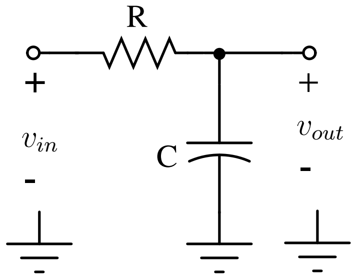
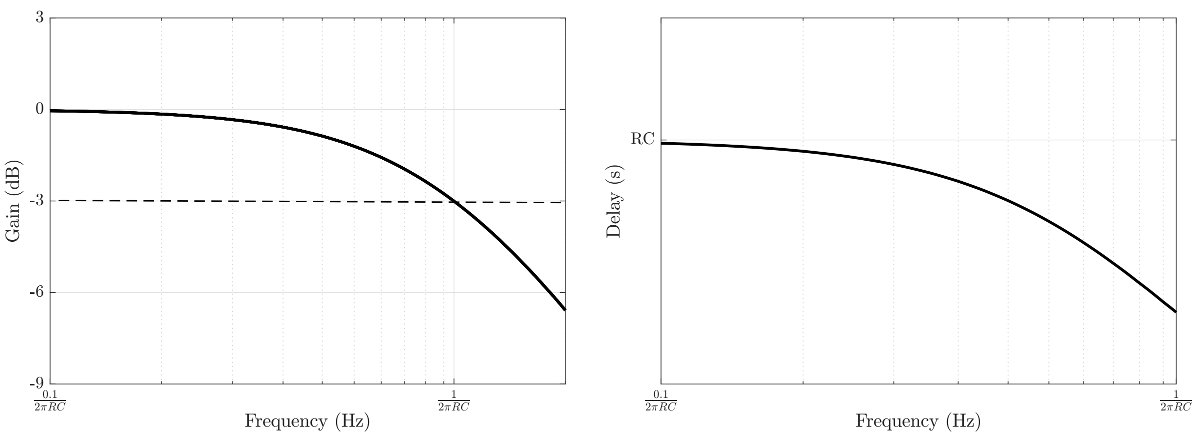


Figure 1.1: 1st order RC LPF.



(a) (b)

Figure 1.2: (a) Gain and (b) group delay plots of 1st order RC LPF.

A 1st order active all-pass filter (APF) shown in Figure 1.3 has a transfer function $H(s) = (sRC - 1)/(sRC$ The APF shows unity gain across all the frequencies. The phase of the filter $\measuredangle H(j\omega) =$

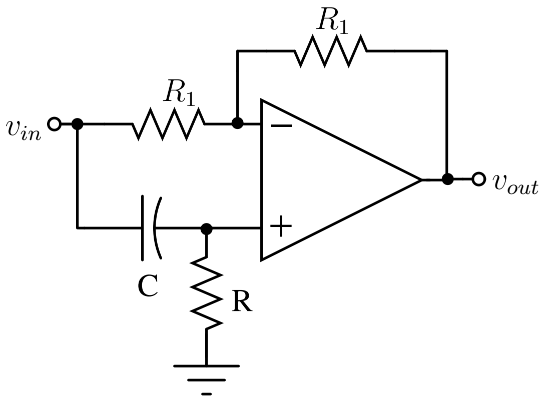


Figure 1.3: 1st order active APF.

$-2\tan^{-1}(\omega RC)$ and the group delay $\tau(\omega)$ of the circuit is expressed as,

$$\tau(\omega) = \frac{d(\measuredangle H(j\omega))}{d\omega} = \frac{2RC}{1+\omega^2 R^2 C^2}. \tag{1.4}$$

The expression for $\tau(\omega)$ with $\omega^2 R^2 C^2 << 1$ gives the low-frequency group delay and is expressed as $\tau \approx 2RC$. Figure 1.4 shows the group delay plot of the APF. Hence, the group delay

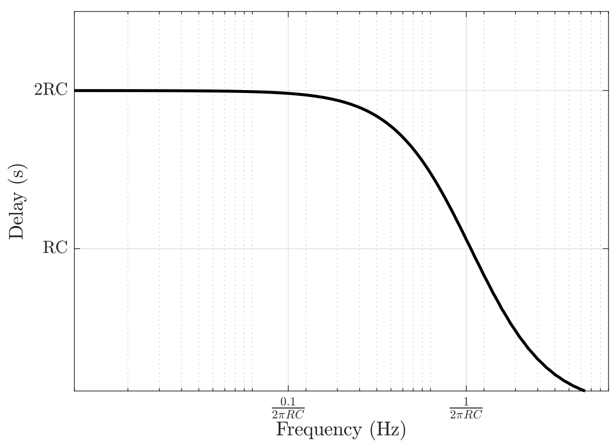


Figure 1.4: Group delay plot of 1 st order APF.

of these filters remains constant up to a certain range of frequencies.

1.1.1 Variable-delay filters and their applications

Two main techniques for changing a filter's delay are (a) Frequency scaling and (b) Variable order methods. The delay of a filter with fixed order can be varied using the frequency scaling technique, in which the filter's bandwidth is varied. In this method, the resistance R of the filter is unchanged while the inductive and capacitive reactances of the filter jX_L, jX_C are modified. Consider a 2nd order RLC-LPF shown in Figure 1.5.

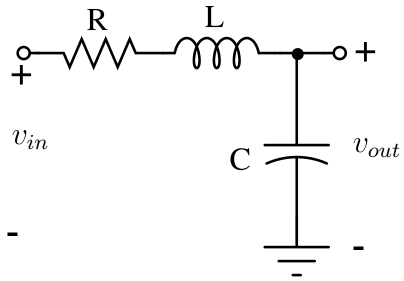


Figure 1.5: 2 nd order RLC-LPF.

The filter has a 3-dB bandwidth of $f_0 = 1/(2\pi\sqrt{LC})$ with low-frequency group delay $\tau \approx \sqrt{LC}$. On decreasing the values of L and C by a factor of 2 the bandwidth is increased by a factor of 2 while the group delay is reduced by a factor of 2 as seen in Figure 1.6. In this way, the frequency-scaling method can be used for changing the delay of the filter. For active G_m-C analog filters, the frequency scaling is achieved with variable Gm or variable C scaling methods [8].

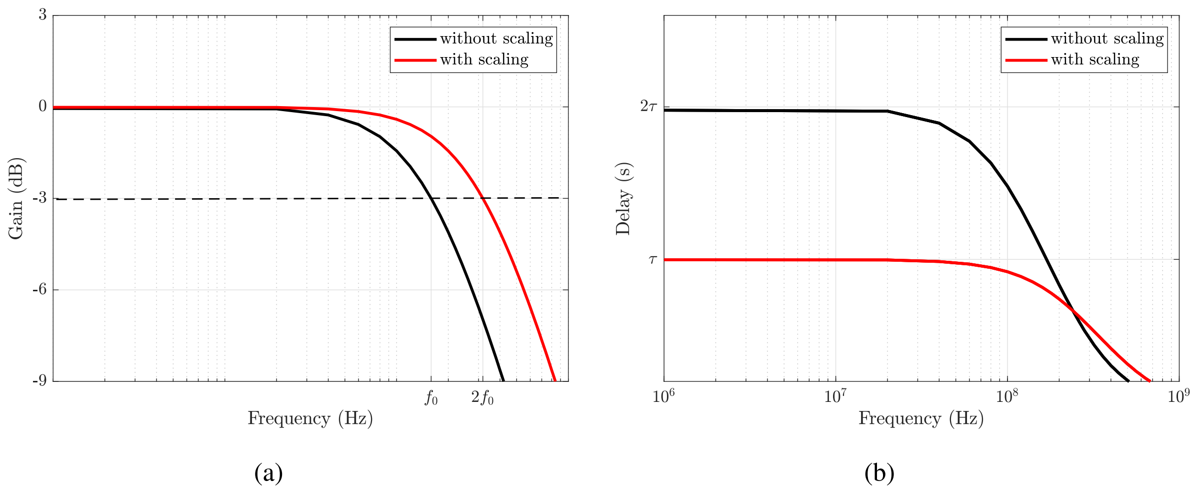
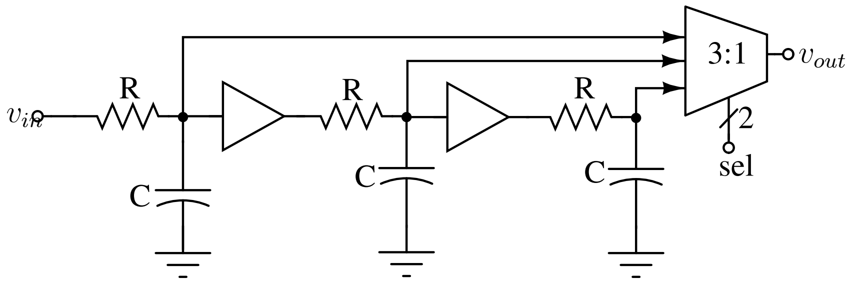


Figure 1.6: (a) Gain and (b) group delay plots of 2^{nd} order RLC LPF without and with frequency scaling.

The delay of a continuous-time filter can also be varied by changing its order. Consider a 3rd order RC-LPF realized by cascading 3 identical stages of 1st order RC-LPFs. Different delays are obtained by switching the order of the filter as shown in Figure 1.7.

APF with variable-delay are also known as true time-delay (TTD) circuits. These are used to offer variable-delays for the signals in timed array transceivers for beamforming applications



Order	τ
1	RC
2	2RC
3	3RC

Figure 1.7: Variable order RC-LPF with different τ.

[9], in design of equalization circuits [10], finite impulse response (FIR) filters [11] etc. Tunable delay LPF are also used as variable-delay cells used in the realization of N-stage ring VCO [12]. variable-delay LPF and APF are also used for time stretching of narrow sporadic continuous time pulses [13].

1.2 Motivation

Most of the reported TTD circuits and pulse expansion circuits are realized with variable-delay APF. APF is preferred over LPF for these applications as they maintain unity gain with linear phase across the desired bandwidth. But on-chip design and implementation of higher-order variable-delay APFs is complex with the disadvantage of large power dissipation. Hence, there is a need to explore simpler yet power-efficient integrated circuit techniques for realizing these applications.

TTD circuits are preferred in wideband beamforming techniques used in timed array receivers. They are supposed to have frequency-independent gain and delay characteristics which yield good receiver performance. But practical TTD elements have a certain amount of non-idealities in the gain and delay characteristics. A study on the effect of these practical implications and quantifying these non-idealities is important.

1.3 Contribution of the Thesis

Contribution-1

This work shows that a variable-delay LPF can also be used to implement programmable delay lines resulting in much simpler and lower power circuits. The variable-delay LPF presented in this work is a 1^{st}-order RC circuit whose effective impedance is varied using the principle of the Miller effect. Further, the other applications of the proposed variable-delay LPF in expansion, compression of continuous-time pulses, and voltage-controlled oscillator (VCO) circuits are explored.

Contribution-2

A variable-delay 3^{rd}-order G_m-C APF is presented in this work. The delay tuning is achieved with the variable-Gm frequency scaling technique. The variable-delay APF is used to realize TTD, continuous time pulse expansion, and compression circuits. An inline calibration method for varying negative conductance in high-gain operational transconductance amplifiers (OTAs) is discussed in this work.

Contribution-3

In practice, TTD circuits exhibit frequency-dependent delay and gain characteristics. This work analyses the effect of these non-idealities on different types of beamformings used in a timed array receiver. In addition, quantifying these non-idealities for the practical TTD elements is also presented in this work.

1.4 Dissertation outline

The thesis is organized as follows.

- Chapter 2 reviews the literature for reported TTD based beamforming circuits and continuous time pulse expansion techniques.

- Chapter 3 discusses the technique of changing the delay of RC LPF using the Miller effect and application of the variable-delay LPF in the design of a 10-stage TTD element and a 15-stage TTD circuit with delay correction technique. The chapter also discusses the design of 3-stage ring VCO using the variable-delay LPF as a variable-delay cell and presents simulation results of the designed VCO.

- Chapter 4 discusses the design of 10, 15, and 25 stage expansion and 15 stage compression circuits using variable-delay LPF. The design of various circuits required for testing

N-stage expansion circuits, and post-layout simulation results of expansion and compression circuits are discussed in this chapter.

- Chapter 5 focuses on explaining the delay tuning technique of 3^{rd}-order LC APF. The design of variable-delay active APF based TTD circuit is discussed and simulation results of the implemented design are reported in this chapter.

- Chapter 6 discusses the design of variable-delay APF based expansion and compression circuits. In addition to discussing simulation results of the implemented designs, a qualitative comparison between LPF and APF based expansion circuits is performed in this chapter.

- Chapter 7 discusses some of the practical implications of TTD based beamforming and quantification of non-idealities associated with practical TTD is also presented.

- Chapter 9 discusses the conclusion and future direction of this work.

- The implementation details of the on-chip ramp generator circuit for expansion and compression operation are discussed in Appendix A.

- Details of inline calibration technique for tuning negative impedance in high-gain OTAs are explained in Appendix B.

Chapter 2

Literature survey

The major focus of the thesis is on different delay tuning techniques for 1st order LPF, 3rd order APF and their applications in wideband beamforming applications, expansion or compression of continuous time pulses and in the design of 3-stage ring VCO. A literature survey on existing architectures for these applications is reported in this chapter. The first section of the chapter discusses various reported TTD circuits in the literature. The second section discusses the motivation for pulse expansion and explains various reported pulse expansion techniques.

2.1 TTD circuits for beamforming applications

TTD circuits are used as programmable delay blocks in beamforming techniques used in timed array receivers. Different reported TTD architectures are discussed in the following subsections. A detailed survey of various beamforming techniques along with analysis of different implementation styles is defered to Chapter 7.

2.1.1 TTD using tuned LC delay lines

TTD elements can be realized using a transmission line model approximation with passive inductors and capacitors. The total delay τ offered by the artificial transmission line of length ℓ with inductor L_0 and capacitor C_0 per unit length is expressed as $\tau = (\sqrt{L_0 C_0})\ \ell$. Hence, by varying L_0, C_0 and ℓ values, the delay offered by the artificial transmission line can be varied.

In tuned LC model (which is shown in Figure 2.1), the TTD consists of N sections of tunable LC delay units. The values of both L and C are varied without affecting the characteristic impedance Z_o and length ℓ of the transmission line. In [14], the inductance value is varied using

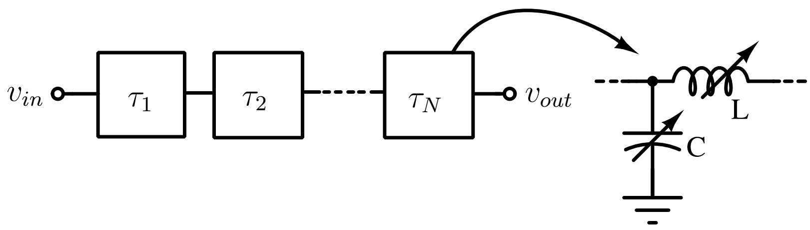


Figure 2.1: TTD realized from tuned LC transmiison line model.

mutual inductance along with varactors for changing capacitance value. In [15] wideband delay lines are realized using varibale L and C elements to form a transmission line structure. The values of L and C for different delay settings are tuned using the Miller effect.

Variable delay can also be achieved by realizing varibale length transmission lines. The variable length transmission line architecture can be implemented using a switched line model, where the input signal is passed through serially connected time delay elements using single pole double throw (SPDT) or double pole double throw (DPDT) switches as shown in Figure 2.2. The controlled switches between the delay elements along with the reference path, vary

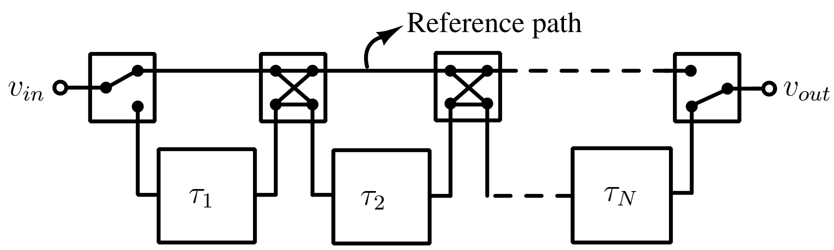


Figure 2.2: Variable length transmission line with N stage switched delays.

the effective length of the transmission line and hence control the delay of TTD. The delay elements $\tau_1, \tau_2 \ldots \tau_N$ could be of identical or nonidentical values. A 8-bit TTD with the switched line topology is reported in [16], where delay cells with a minimum value of 1 ps to maximum value of 128 ps are connected through SPDT switches. The position of the switches are controlled via external digital inputs to achieve a maximum delay of 255 ps with a resolution of 1 ps. In [17] variable delay lines are realized with the same switched line topology to provide controlled delays with 7 delay units. These delay elements are realized using the CLC-π network and are connected in cascade using SPDT and DPDT switches. Bidirectional distributed gain amplifiers are added after each delay element in the chain to compensate for the losses from switches. A maximum group delay of 198.4 ps is achieved with a delay resolution of 1.56 ps. A 5-bit TTD with variable length transmission topology using phase change material (PCM) based SPDT switches is implemented in 0.18 μm SiGe BiCMOS process in [18]. These PCM based switches with low ON resistance offer good matching between cascaded sections and also provide good isolation between active and inactive RF paths. This results in a minimum group delay variation of about 1.6 $\%$ across the bandwidth of 25 to 50 GHz.

Another architecture using the variable length transmission line approximation with trombone topology is shown in Figure 2.3 [19, 20]. It consists of two parallel transmission lines

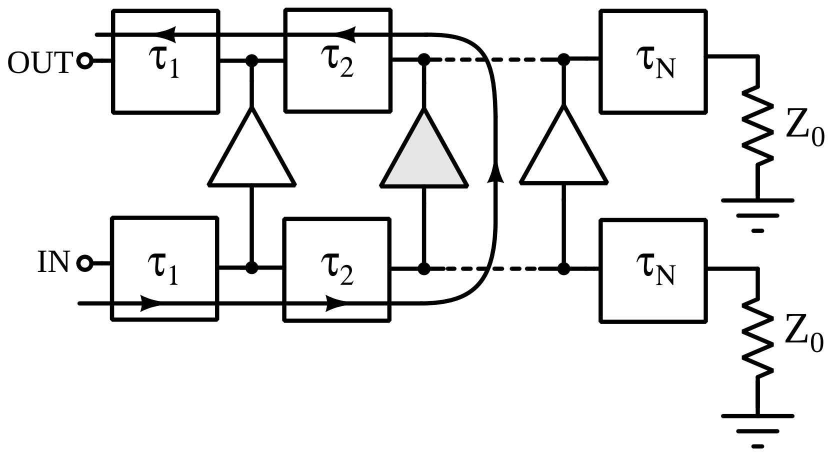


Figure 2.3: Tunable length TTD realized with trombone topology.

terminated with impedance Z_0 and each of these transmission lines is realized with N sections of LC delay elements. The connection between these two lines is programmed using path-select amplifiers to achieve the desired delay value. The path-select amplifiers also amplify the signal through the delay chain in addition to delay tuning. Wideband beamforming with a 4 channel timed array receiver using path-sharing TTD elements realized with trombone architecture is reported in [19]. Each identical LC section of the transmission lines is designed with a delay of 7.5 ps to have a delay resolution of 15 ps. The path-sharing architecture used in this work reduces the maximum delay required on each element and hence minimizes the chip area considerably.

On-chip spiral inductors can be used in the realization of delay units in tuned LC delay lines. The finite Q-factor of the inductors introduces non-idealities in the delay vs frequency characteristics. Hence, it adds a constraint on the maximum achievable delay by limiting the number of delay stages that can be cascaded. With the Q-factor of spiral inductor varying from 3 to 13 across a bandwidth of 1 to 13 GHz, the 5-bit (32 stages) trombone topology exhibits more delay variations compared to that of 4-bit (16 stages) architecture [21], primarily due to finite Q of inductors. Hence, the maximum achievable delay is around 64 ps with a 4-ps delay resolution. Variable gain amplifiers are also added in this architecture, to compensate for the propagation losses that occurred due to the limited Q-factor of the inductor.

Table 2.1 summarizes the performance of various LC delay-based TTD elements in terms of the frequency range, delay resolution, maximum, delay, gain and delay variations.

Table 2.1: Performance summary of different LC based TTDs.

	Process-Technology	Topology	Bandwidth (GHz)	Delay resolution (ps)	Maximum delay (ps)	Delay variation(%)	Gain variation(dB)
[14]	CMOS 90-nm	Tuned LC	DC-8	14	40	10	±0.5
[15]	CMOS 45-nm	Tuned LC	11-24	10	50	11	–
[16]	GaAs 250-nm	Switched Line	6-18	1	255	2.43	±2
[17]	CMOS 130-nm	Switched Line	8-16	1.56	198.4	2.16	±1
[18]	SiGe BiCMOS 180-nm	Switched Line	25-50	4	124	1.6	±4.4
[19]	CMOS 130-nm	Trombone	1-15	15	225	14	±3
[20]	SOI 45-nm	Trombone	135-145	0.3*	11.57*	1.4*	±0.57

* The delay resolution, maximum delay and delay variation of [20] are calculated from the reported phase vs frequency characteristics.

2.1.2 TTD from passive and active APF

The ideal APF is the best candidate for realizing delay elements as they maintain unity gain and linear phase response across all the frequencies. APF can be realized using passive or active components depending on the various parameters like operating frequency range, delay-bandwidth product and power consumption in TTD.

Passive APFs can be used as delay elements in the switched line and trombone architectures of TTD implementation. A second-order passive APF (shown in Figure 2.4(a)) is used as delay units in the combined 10-stage trombone and 3-bit switched line configurations in [22]. Here the authors use coarse delay tuning which is done by switching path-select amplifiers, and fine delay tuning which is done with the cascaded switched line delay unit. There is an additional continuous delay tuning stage for compensating delay errors introduced in the preceding delay stages. In the modified trombone architecture as reported in [23], two types of delay cells, namely type-1 and type-2 delay cells are realized using passive 2nd-order and 4th-order APFs respectively. The input line uses type-1 delay cells and the output line uses type-2 delay cells with the controlled amplifier blocks placed between the input and output lines for delay tuning. In another passive APF based TTD reported in [24], a 4-bit switched line TTD architecture uses a coupled APF (C-APF) and non-coupled APF (NC-APF) for realizing delay units. In the presence of parasitics, the delay vs frequency characteristics of C-APF and NC-APF are not flat with frequency. C-APF exhibits a mildly expansive delay profile with increasing frequency, while NC-APF exhibits a declining delay with increasing frequency. In order to compensate for the delay variation, the authors use alternating C-APF and NC-APF stages for realizing the TTD. The authors report an example of a 4$^{\text{th}}$-order passive APF by cascading a 2$^{\text{nd}}$-order C-APF and NC-APF which is shown in Figure 2.4(b). The APFs are designed to have four different delay values of 4.6 ps, 6 ps, 12 ps and 23 ps. These different delay sections are cascaded using SPDT and DPDT switches for controlling delay values. An integrated 5-bit controlled delay line using internal switched topology is reported in [25]. The delay elements are realized using resistor-incorporated passive APFs with an internal switch feature. A 2$^{\text{nd}}$-order filter used in this work is shown in Figure 2.4(c), where the control signal V_c switches the filter between bypass mode and delay mode. The internal switched delay cells with values varying from 3.9 ps to 62.5 ps are designed using 2$^{\text{nd}}$, 4$^{\text{th}}$ and 6$^{\text{th}}$-order APFs and are cascaded to provide a maximum delay upto 125 ps.

Delay lines can also be realized using active APFs, which include transistors in combination with resistors and capacitors to have Gm-RC and G_m-C architectures as reported in [26] and [27] respectively. An example design of TTD element implemented using six cascadable 1$^{\text{st}}$-order G_m-C filters with 3-bit switched capacitors for delay tuning is shown in Figure 2.5(a) [28]. The proposed APF has phase linearization and bandwidth extension techniques to maintain flat group delays from 1 GHz to 2.5 GHz. This variable delay element has an additional feature of fine delay tuning with 3 cascaded adjustable delay cells. A second-order voltage mode active APF with variable delay is proposed in [29]. Figure 2.5(b) shows the transistor level schematic of variable delay APF reported in this work. Transistor M1 is biased in common source configuration with source degenerating capacitive feedback from C_s and parallel combination of R_1 and C_1 in the feedforward path. This results in 2$^{\text{nd}}$-order APF with two poles and two zeros and continuous delay tuning is achieved by implementing Cs with varactor. The delay of APF is varied from 30 ps to 57 ps across the wideband range of 0.4 GHz to 2 GHz. Figure 2.5(c) shows a wideband 3-bit TTD element implemented with variable order G_m-C filter reported in [30]. Here, the order of the active Bessel APF is varied from 2 to 9 to obtain delay values ranging from 250 ps to 1700 ps. The technique maintains a constant bandwidth of 2 GHz across all the delay settings by changing the capacitance values for each delay setting.

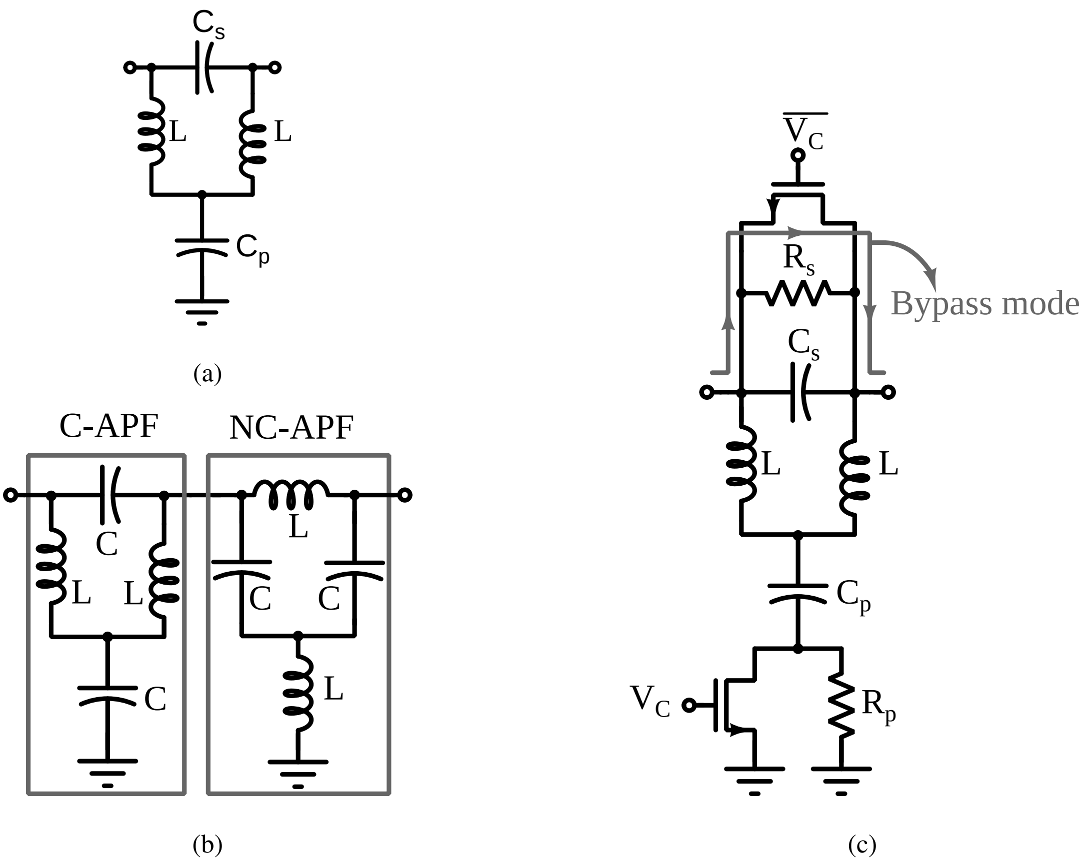


Figure 2.4: Different passive APFs used in TTD implementation: (a) 2nd-order APF (b) 4th-order APF using coupled and non coupled APFs and (c) 2nd-order internal switched APF.

Table 2.2 summarizes the performance of various active and passive APF based TTD elements in terms of the frequency range, delay resolution, maximum, delay, gain and delay variations.

Table 2.2: Performance summary of APF based TTDs.

	Process-Technology	Topology	Bandwidth (GHz)	Delay resolution (ps)	Maximum delay (ps)	Delay variation(%)	Gain variation(dB)
[22]	CMOS 130-nm	Passive-Trombone	1-20	5.6	400	17.8	±15
[23]	CMOS 180-nm	Passive-Trombone	8-18	15.6	109.3	19	±1
[24]	CMOS 28-nm	Passive-Switched Line	3-30	4.6	68.5	4.3	±1.6
[25]	CMOS 180-nm	Passive-Internal switched	8-18	3.9	125	21	±2
[30]	CMOS 130-nm	Active G_m-C	DC-2	250	1700	24	±0.7
[28]	CMOS 140-nm	Active G_m-C	1-2.5	13	550	1.8	±1.4
[29]	CMOS 65-nm	Active Gm-RC	0.4-2.5	2	50	2.49	±0.375

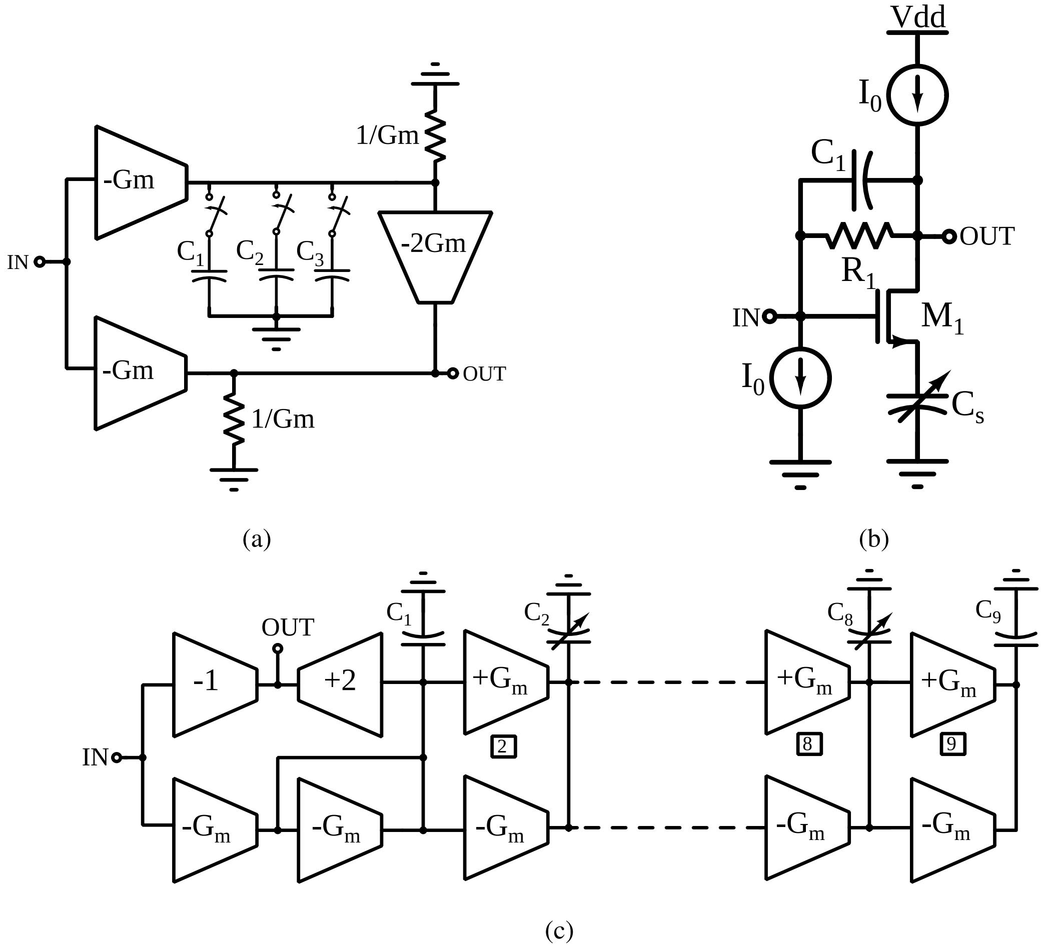


Figure 2.5: Various active APFs used in TTD implementation: (a) 1st-order switched C G_m-C APF (b) Variable-order APF and, (c) 2nd-order Gm-RC APF

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2.2 Expansion of Continuous-Time Narrow Pulses

Data acquisition systems such as ground penetrating radars [31] or radars used in through wall imaging [32] wait for echo signal from target, after transmitting short nano-second pulses. The reflected signal is then processed to study the image of desired object. In case of some gaming devices [33], train of high frequency pulses are transmitted. The reflected input signal to system is used to detect the player position and movement. Features like shape and size of pulses can be used to identify particles in Neutrino and other particle detectors [34, 35]. In these applications the pulses are sporadic in nature, which occur infrequently in time.

Directly digitizing these narrow pulses in such applications require wide bandwidth (multi-GHz) ADCs. Various attempts have been made to avoid full fledged high speed ADCs. One such method involves stretching the pulses in time domain and then digitize using low speed

ADCs. To illustrate this, consider a narrow pulse of width 2 ns as shown in Figure 2.6.(a), to be sampled with 40 samples. The required sampling rate is 20 Gsamples/s, which requires

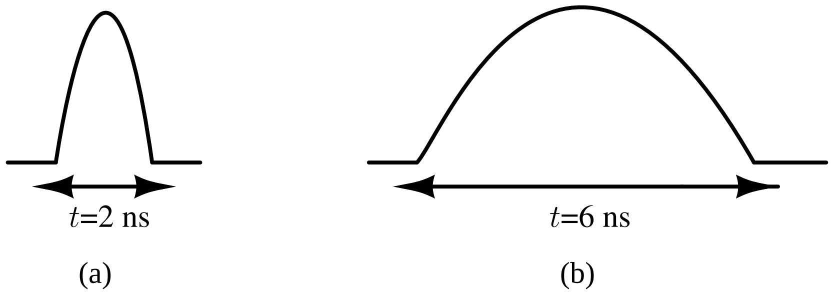


Figure 2.6: (a) Narrow pulse of 2 ns, (b) Pulse stretch to 6 ns

very high speed and fine resolution ADC. Stretching the same narrow pulse by factor of 3, shown in Figure 2.6(b), increases the pulse duration to 6 ns. The required sampling rate with the same number of samples for the stretched pulse is 6.33 Gsamples/s. Therefore the pulse can be digitized using low speed ADCs. There are various time stretching methods reported to slow down the incoming continuous time sporadic pulses. These methods include expansion using optical signals and dispersive filters, sampling based time stretching and expansion using continuous time filters. This section reviews various reported methods for time stretching of analog pulses.

2.2.1 Photonic Time Stretching

The basic principle of photonic time stretching [36], [37] is the dispersion of intensity modulated optical chirp signal. A chirp signal is a signal whose frequency increases (decreases) linearly with time. An optical chirp signal is intensity modulated with the narrow pulse to be stretched. The modulated signal is transmitted through a linear dispersive optical fiber. The dispersed modulated optical signal is then converted to electrical signal using a photodetector for further processing. Figure 2.7 shows the block diagram of photonic time stretching system. Direct generation of optical chirped carriers has its own challenges. A linearly chirped optical pulse can be generated when a narrow optical pulse from an optical source such as Erbium doped fiber laser (EDFL) [36], propagates through a optically dispersive device such as a spool of single mode optical fiber of length L_1. The time duration of the chirped carrier signal has to be longer than the duration of pulse to be stretched. This requires L_1 to be in the order of kilometers. The electrical signal to be stretched is intensity modulated with this chirped optical

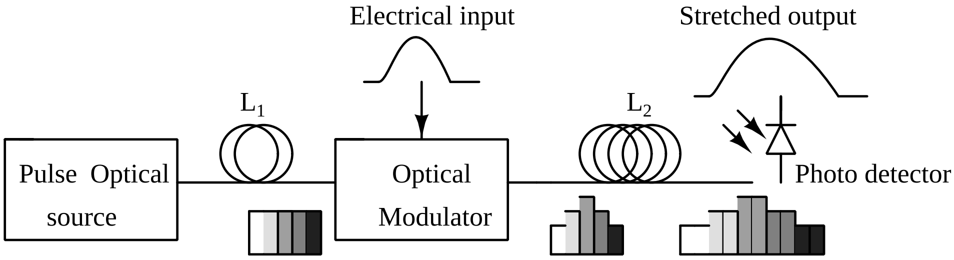


Figure 2.7: Photonic Time Stretching System

carrier using an electro-optic modulator such as Mach-Zehnder intensity modulator [36]. The intensity modulated signal is then passed through a second linearly dispersive fiber of length L_2. The output electrical signal from the photo detector represents the time-stretched version of the input narrow pulse waveform. The stretch factor, S for the photonic time stretching is given by

$$S = \frac{L_1 + L_2}{L_1} \tag{2.1}$$

The stretch factors for various fiber lengths L_1, L_2 are tabulated in Table. 2.3. As observed from Table. 2.3, the stretch factor increases with the increase in optical fiber length L_2.

Table 2.3: Stretch factor for different lengths of fiber.

Optical fiber Lengths L_1,L_2	Stretch factor M
1.1 km, 2.2 km	3
1.1 km, 5.5 km	6
1.1 km, 7.6 km	8

2.2.2 Stretching using Dispersive Filter with Distributed Amplifier

The method is similar to that of photonic time stretching, but operates in the microwave region instead of visible spectrum. This technique of time stretching is based on the application of dispersive filter such as a parallel LC filter that exhibits frequency dependent group delay near resonance. Figure 2.8 shows the block diagram of time stretching system using a dispersive filter [38], [39]. The electrical input signal to be expanded is amplitude modulated with respect to chirped carrier signal. The chirped carrier signal is generated from a VCO, as shown in Figure 2.9(a). The ramping VCO is an LC-tank type VCO with a control signal applied to the

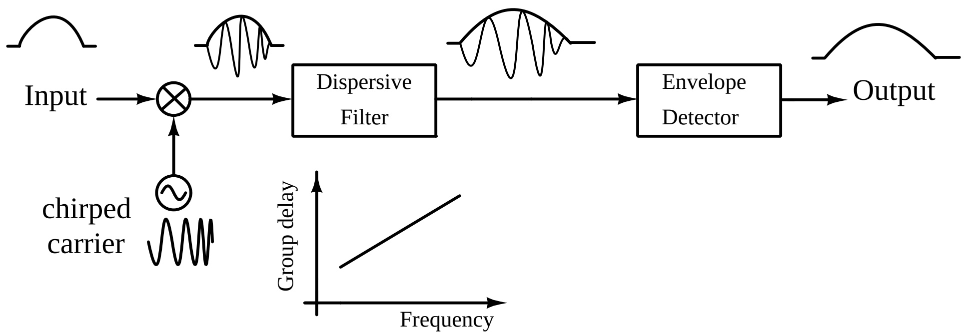


Figure 2.8: Time Stretching System with DDL

varactor. With the increase in control voltage, V_{ctrl}, the instantaneous frequency of LC oscillator is increased, yielding an up-chirped carrier. The ramping VCO is followed by a two-stage amplitude modulator as shown in Figure 2.9(b). The input signal applied at the gate varies the current through the differential amplifier, which varies the gain with respect the input voltage. Finally an amplitude modulated chirped carrier is available for the dispersive filter.

Figure 2.10 shows the dispersion enhanced distributed amplifier system [39], [38]. The dis-

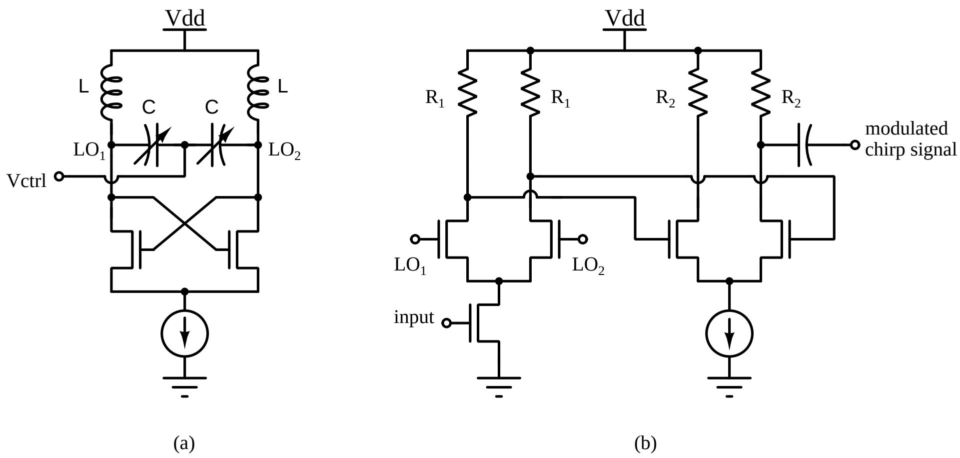


Figure 2.9: (a) Ramping VCO, (b) Two stage amplitude modulator for modulated chirp carrier.

persive parallel LC filter is operated at resonance which exhibits strong dispersion. The gate and drain lines with the common source amplifier form the distributed amplifier system. The distributed amplifier configuration compensates for any loss during dispersion, which provides amplified stretched signal. Finally, the stretched signal is obtained by demodulating the output signal.

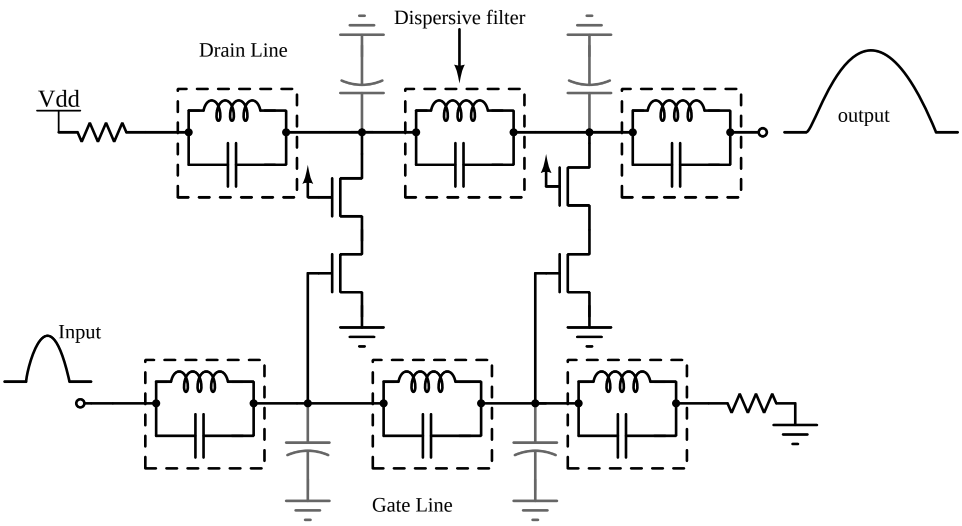


Figure 2.10: Dispersive enhanced distributed amplifier system

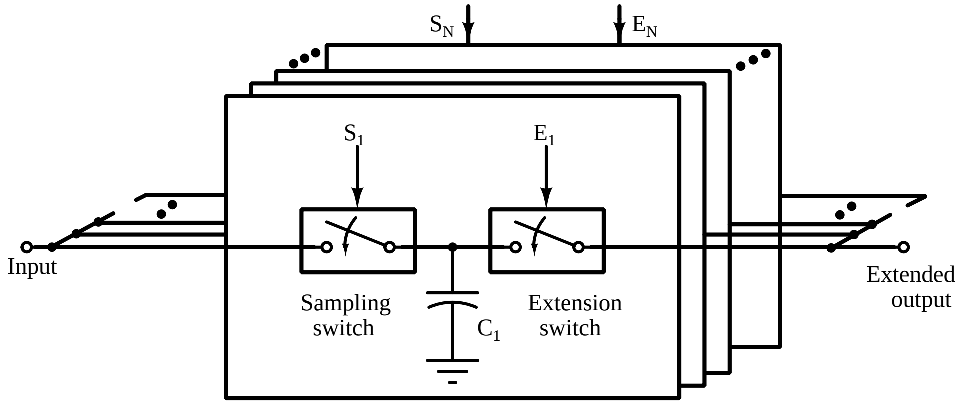
2.2.3 Sampling Based Time Stretching

In this method of time extension, a narrow pulse to be stretched is sampled at a high rate. The sampled signal values are stored in an analog memory device. These sampled values are then read at a slow speed, which results in the extension of the input narrow pulse [33]. Figure 2.11(a),(b) shows the schematic and timing diagram of high-speed sampling and time extension block. The time interleaving method is used for high-speed sampling. With the high speed sampling clock, S_N the time-interleaved sampler samples the input signal sequentially with the interval of δ_S. These sampled values are held on to the capacitors during the sampling period. In the extension period, the stored signal values are extended by the interval of δ_e, with the extension clock, S_E. The stretch factor, S for the stretched signal is given by

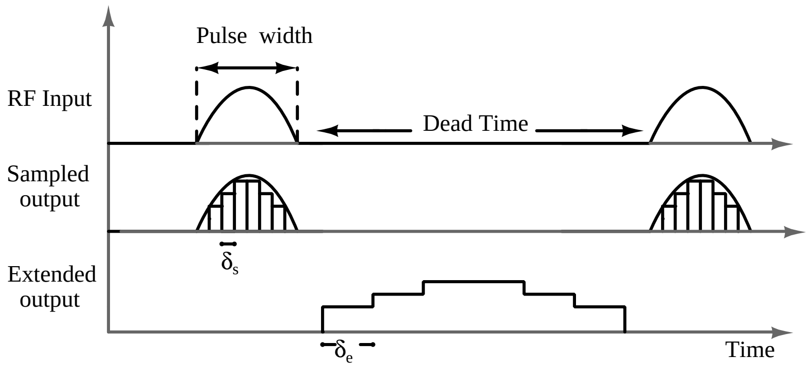
$$S = \delta_e/\delta_s \tag{2.2}$$

2.2.4 Stretching by Continuous Time Filters

Pulses of short duration can be stretched in time by driving them into a filter, whose group delay is more than the pulse duration. Once the pulse is completely inside the filter, the pulse is stored in the form of filter state variables [8], [13]. By instantaneously varying the filter bandwidth, the output of the filter can be expanded or compressed. Consider a narrow pulse input $x(t)$ with pulse width w applied as input to a N^{th} order continuous time filter. The filter has flat group delay τ with $\tau > w$, and unit magnitude over the bandwidth of input pulse. The output of the filter can be analyzed for different values of the control signal $\alpha(t)$.



(a)



(b)

Figure 2.11: (a) Sampling Based Time Stretching Architecture, (b) Timing diagram

1. Applying the control signal $\alpha(t) = 1$ throughout the time, the output $y(t) = x(t - \tau)$, as seen Figure 2.12(a).

2. The signal $\alpha(t)$ varied from 1 to 0.5 at time t_o, where the output is not changed and input is reduced to zero. The output $y(t) = x(t/2 - \tau)$ is the stretched version of the input signal as shown in Figure 2.12(b).

3. The signal $\alpha(t)$ varied from 1 to 2 at time t_o. The output $y(t) = x(2t - \tau)$ is the compressed input signal as shown in Figure 2.12(c).

Thus, variation in α varies the filter time constants and in turn varies the filter bandwidth. This is called frequency scaling. Figure 2.13(a) shows the G_m-C realization of second-order state variable filter with multiplier α.

Frequency scaling of this filter is achieved by multiplying unit transconductance and conductance by α with C unchanged. This method is called constant C frequency scaling. Fig-

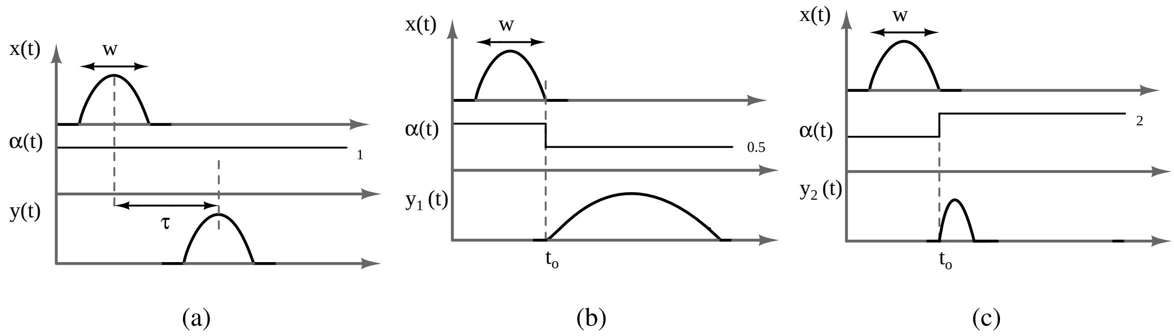


Figure 2.12: (a) Filter output with constant α. (b) Pulse stretching by dynamically reducing α. (c) Pulse compression by dynamically increasing α.

ure 2.13(b) shows the block level architecture of expansion and compression circuit realized using cascades of 7^{th} order APF and 5^{th} order LPF reported in [13]. In this work, the frequency scaling of an N^{th} order filter is achieved with constant C scaling in which the value of all the transconductors of the filter is varied during expansion or compression operation. For instance, the filter is switched to low bandwidth mode by turning off the highlighted transconductors during expansion operation as shown in Figure 2.14.

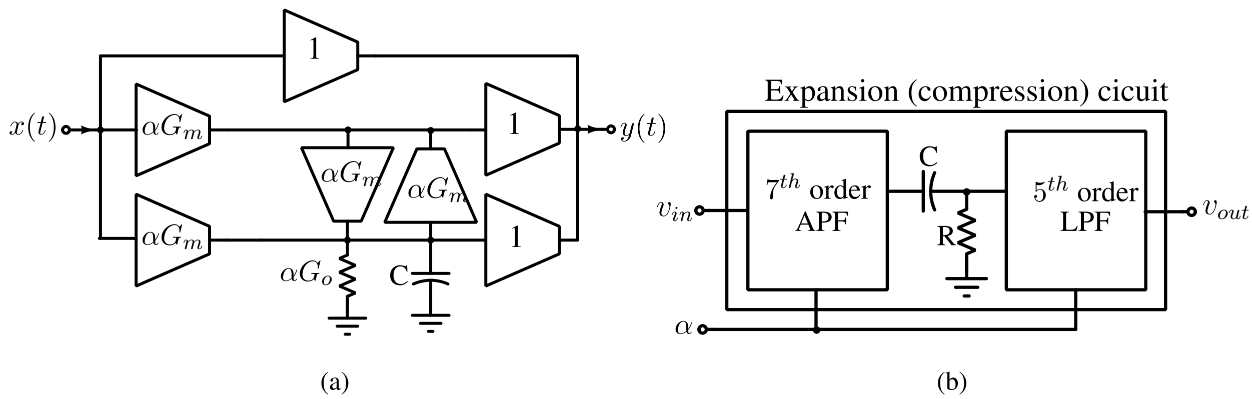


Figure 2.13: (a) G_m-C implementation of second order filter with multiplier α, (b) Block diagram of expansion (compression) circuit.

2.3 Summary

The value of inductors and capacitors used in realization of wideband delay lines are large and have low quality factor making them unsuitable for on-chip implementation. A true time delay line is implemented using second order all-pass networks with a 10-state trombone topology in [22]. But the architecture suffers from huge losses in gain due to the low quality factor of

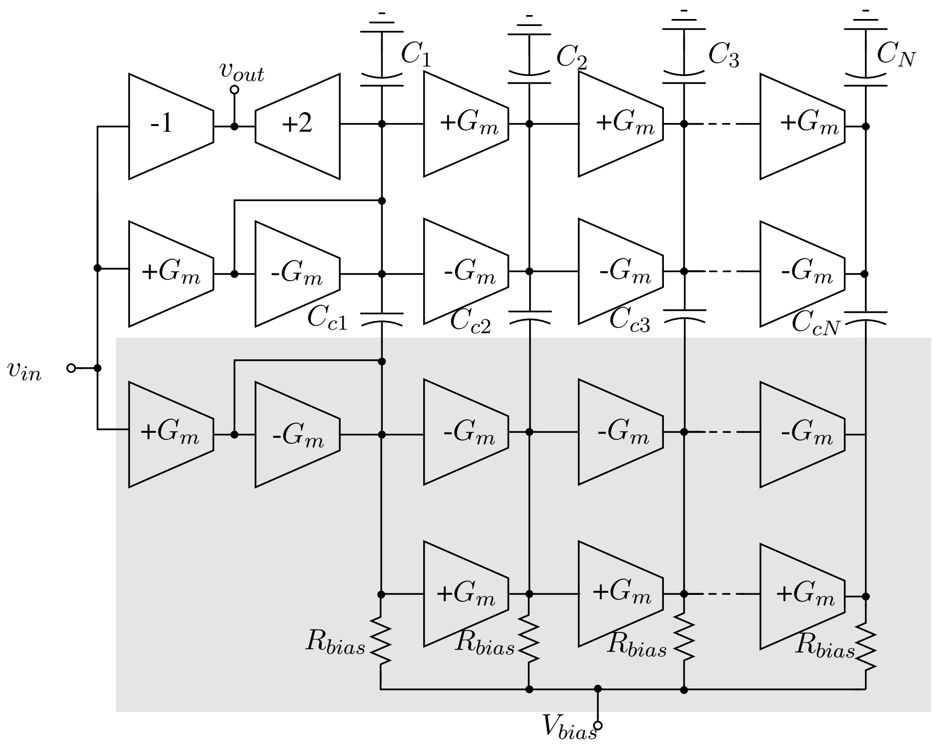


Figure 2.14: Frequency scaling by switching the transconductance values in N^{th} order G_m-C filter.

capacitors. APF, which maintain unit magnitude response and linear phase are the best option for realizing variable delay lines. Inductorless-active Gm-RC [26] and G_m-C [28] APF configurations are more compatible for on-chip active true time delay implementations. The Gm-RC based APF implementation has degraded performance because of high parasitic capacitance. In addition, they need intermediate buffer stages or DC blocking capacitors to realize cascadability [26]. First order G_m-C APFs are cascaded to realize delay line with switched capacitor architecture for delay tuning in [28]. This single ended architecture requires an additional gain enhancement circuit to compensate for the losses due to finite output impedance. Higher order G_m-C APF are used to obtain larger delay range with reduced gain and delay variation in [30]. In this method, delay tuning is done by changing the order of the filter, while varying C to maintain the constant bandwidth. The technique of switching the capacitors demands high quality factor to be maintained in the capacitors for all delay settings. Also changing the C values with varactors requires additional control voltages for delay tuning. As the order of G_m-C APF increases, the range of capacitance values and the resulting Gm also increases. As a result, this imposes the constraint of high bias currents, and hence, resulting in increased power consumption and large area. Hence, there is a need for simpler design for on-chip baseband beamforming TTD circuits with lower power consumption compared to the existing architectures.

From various reported continuous time pulse expansion methods following observations are done. The photonic method requires lengthy optical fibres in the range of kilometers to obtain a minimum stretch factor. Expansion using dispersive filters and optical time expansion methods require the generation of highly precise chirp carrier. The sampling based method requires external high frequency clock distribution network. Time expansion using continuous time filter requires the design of higher order filters such as cascade of 7^{th}-order APF and 5^{th}-order LPF. Also in its G_m-C implementation, the OTA blocks have to be switched instantaneously and precisely with a period much less than the incoming short duration pulse, which is challenging. Thus, there is a need to find a simple and efficient on-chip implementation of expansion and compression of continuous time pulses.

Chapter 3

Variable-Delay LPF-TTD Circuit and 3-stage VCO

This chapter explains the principle of delay tuning used in the proposed variable-delay LPF. The rest of the chapter then focuses on the realization of TTD circuits for wideband beamforming applications and 3-stage ring VCO based on this variable delay LPF.

3.1 Proposed Variable Delay LPF

A 1^{st} order LPF maintains fixed gain and constant low-frequency group delay of value RC up to a certain range of frequencies. The group delay τ of 1^{st}-order RC LPF can be varied by varying the circuit's time constant RC. In the proposed circuit, the impedance of the LPF is varied using the principle of the Miller effect. Figure 3.1(a) shows a proposed 1^{st}-order variable delay LPF, in which an input voltage v_{in} and its scaled version αv_{in} are applied to input terminals of the circuit. The principle of delay tuning can be understood intuitively for $\alpha = \pm 1$ by evaluating the effective impedance from the node v_{in}. The circuit offers minimum impedance for $\alpha = +1$, and maximum for $\alpha = -1$. Thus, with the principle of the Miller effect, effective impedance and the resulting delay of the circuit can be varied.

As the circuit is linear, the output voltage $V_{out}(s)$ of the circuit can be evaluated using the superposition theorem. With $\alpha V_{in}(s)$=0 (as seen from Figure 3.1(b).) the output voltage $V_{o1}(s)$ is given as,

$$V_{o1}(s) = \frac{(1 + sRC)}{(1 + s2RC)} V_{in}(s) \tag{3.1}$$

For $V_{in}(s)$ =0, (as seen from Figure 3.1(c) the output voltage $V_{o2}(s)$ is given as,

$$V_{o2}(s) = \frac{(\alpha sRC)}{(1+s2RC)} V_{in}(s) \tag{3.2}$$

The resulting output voltage $V_{out}(s)$ and transfer function of the circuit $H_\alpha(s)$ is expressed as,

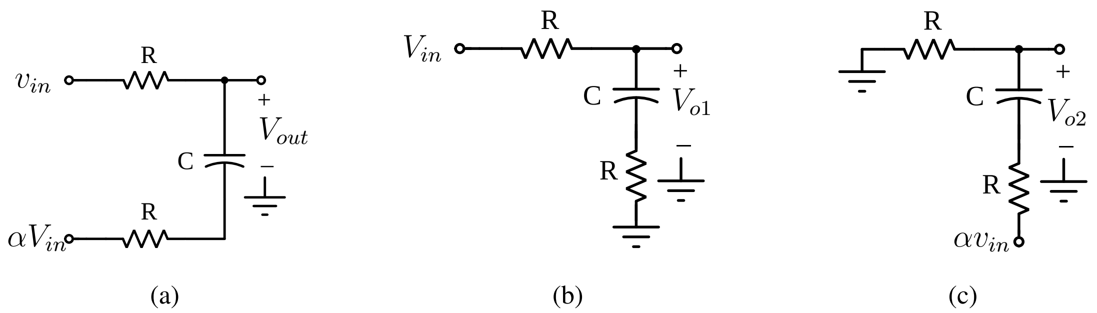


Figure 3.1: (a) Proposed variable delay circuit, (b) Circuit with αv_{in} is grounded, (c) Circuit with V_{in} is grounded.

$$V_o(s) = V_{o1}(s) + V_{o2}(s) \tag{3.3}$$

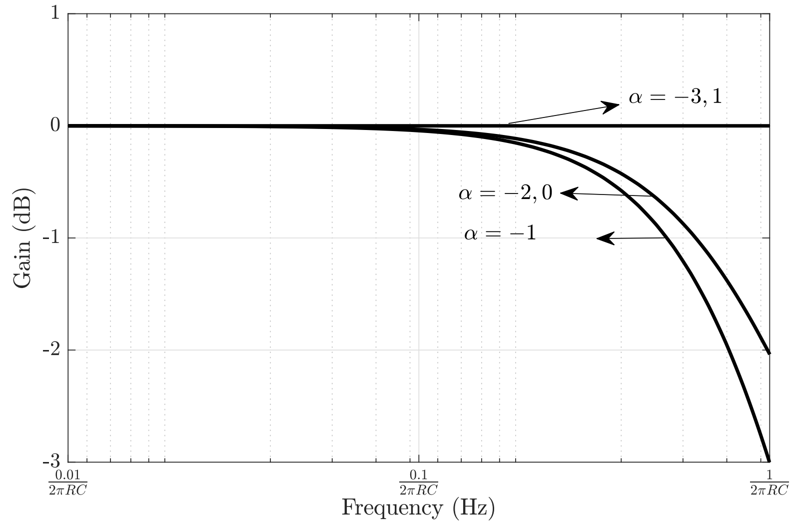
$$H_\alpha(s) = \frac{1+s[1+\alpha]RC}{(1+s2RC)} \tag{3.4}$$

From Eq. 3.4 it is observed that $H_\alpha(s)$ is 1st order system with a pole and a controlled zero frequencies and has low-frequency group delay $\tau_\alpha \approx (1-\alpha)RC$. $H_\alpha(s)$ has positive delay values for $\alpha < 1$ and exhibits low-pass behavior for $\alpha > -3$ as seen from the gain, phase plots (Figure 3.2(a),(b)) of the system. Table 3.1 tabulates $H_\alpha(s)$ and pole-zero plots for different values of α. $H_\alpha(s)$ shows the same gain response for $\alpha = 1$ and $\alpha = -3$ because of zero in RH plane. A similar explanation is valid for $\alpha = 0$ and $\alpha = -2$.

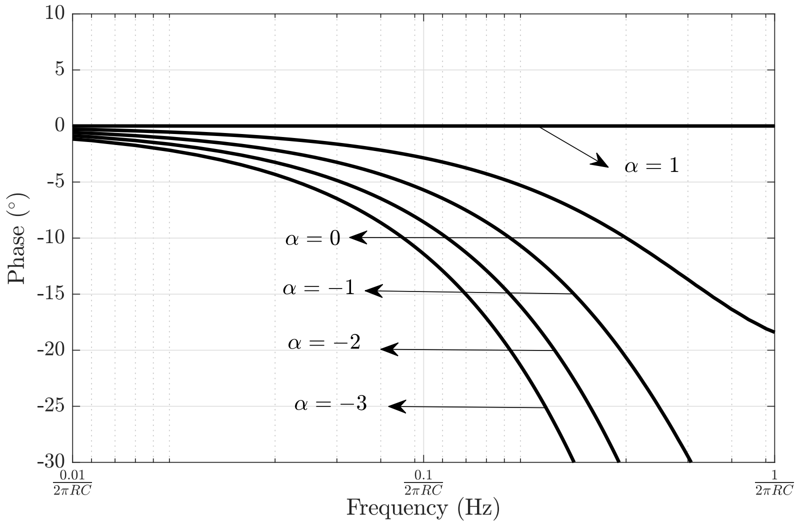
3.2 Variable delay LPF as TTD circuit

The variable delay LPF can be used as a TTD circuit for beamforming applications with different fixed control signal values α. Figure 3.3 shows delay plots for different values of α with -3 < α ≤1. It is observed that the LPF up to the frequency of $\frac{0.1}{2\pi RC}$ with a constant gain and flat delays can be used in the realization of programmable delay lines.

The TTD realized from a single variable delay LPF has a limited delay range of 4RC. The delay range of the TTD can be increased by cascading N identical stages of tunable delay LPFs as shown in Figure 3.4(a). Figure 3.4(b) shows a n stage variable delay element in differential architecture.



(a)



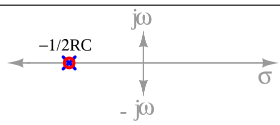
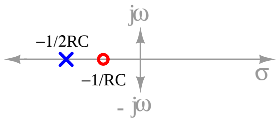
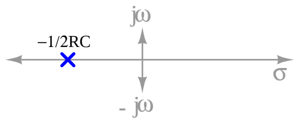
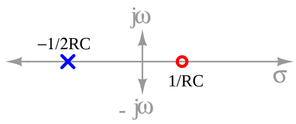
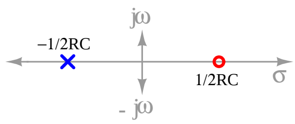
(b)

Figure 3.2: (a) Gain and (b) phase plots of variable delay LPF for different values of α.

3.2.1 Implementation Details and Effect of Circuit Non-Idealities

A block-level implementation of the circuit using a transconductor G_B and Gilbert multiplier G_{MUL} is shown in Figure 3.5. The multiplier G_{MUL} block generates signal αv_{in} for controlling delay. The transcondutcor G_B with R is a unity gain buffer that isolates two stages. The transistor level schematics of G_B and G_{MUL} are shown in Figure 3.6(a) and Figure 3.6(b) respectively. The G_{MUL} consists of an active load Gilbert multiplier followed by G_B to increase the multiplier gain and keep the output resistance of the multiplier to R. The input signal α to multiplier is an external control signal which is differential in nature. The control voltage range is limited

Table 3.1: Transfer function $H_\alpha(s)$ for different values of α with pole-zero plots.

α	$H_\alpha(s)$	Pole-Zero Plot
1	$\frac{(1+s2RC)}{(1+s2RC)}$	
0	$\frac{(1+sRC)}{(1+s2RC)}$	
-1	$\frac{1}{(1+s2RC)}$	
-2	$\frac{(1-sRC)}{(1+s2RC)}$	
-3	$\frac{(1-s2RC)}{(1+s2RC)}$	

by the minimum and maximum voltages required to keep all the transistors in the saturation region.

The parasitic capacitances of G_B and G_{MUL} effect the gain and delay characteristics of each variable delay cell of N-stage TTD. Consider 2 cascaded stages, the n^{th} and $(n+1)^{th}$ delay cell with parasitic capacitances at different nodes as shown in Figure 3.7(a). The capacitances $C_{dB(n)}$ and $C_{dB(n+1)}$ refers to total drain capacitance of the buffer at n^{th} and $(n+1)^{th}$ stage respectively. $C_{dM(n)}$ refers to total drain capacitance of the multiplier at n^{th} stage. $C_{gB(n+1)}$ refers to total gate capacitance of the multiplier at $(n+1)^{th}$ stage. The small signal equivalent representation of n^{th} stage variable delay element with the parasitic capacitances C_{p1} and C_{p2} is shown in Figure 3.7(b).

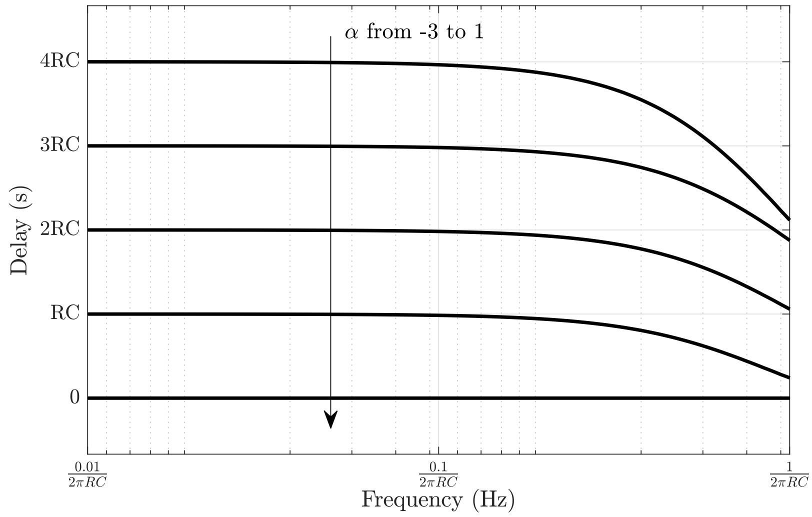
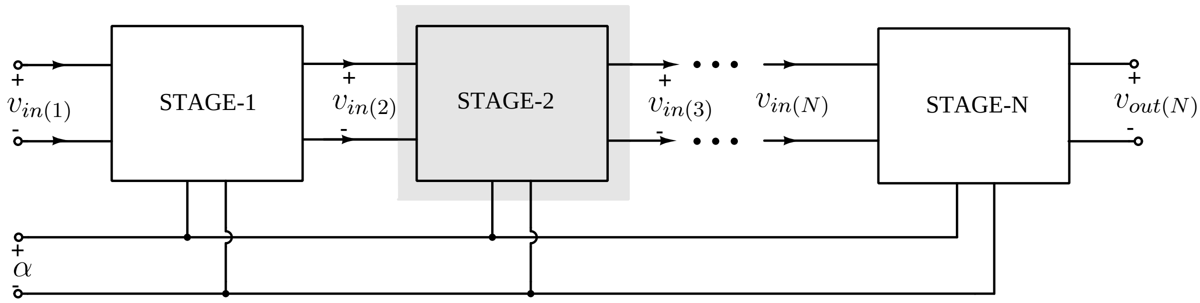
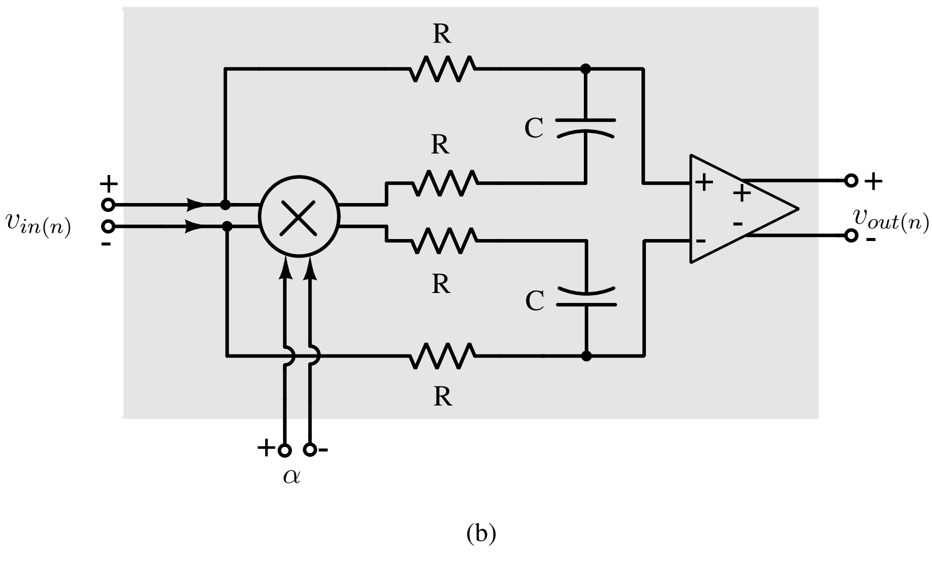


Figure 3.3: Delay plots of $H_\alpha(s)$ for α increased from -3 to 1.



(a)



(b)

Figure 3.4: (a) Cascaded N stages of varibale delay LPFs for delay enhancement, (b) Varibale delay LPF with differential architecture.

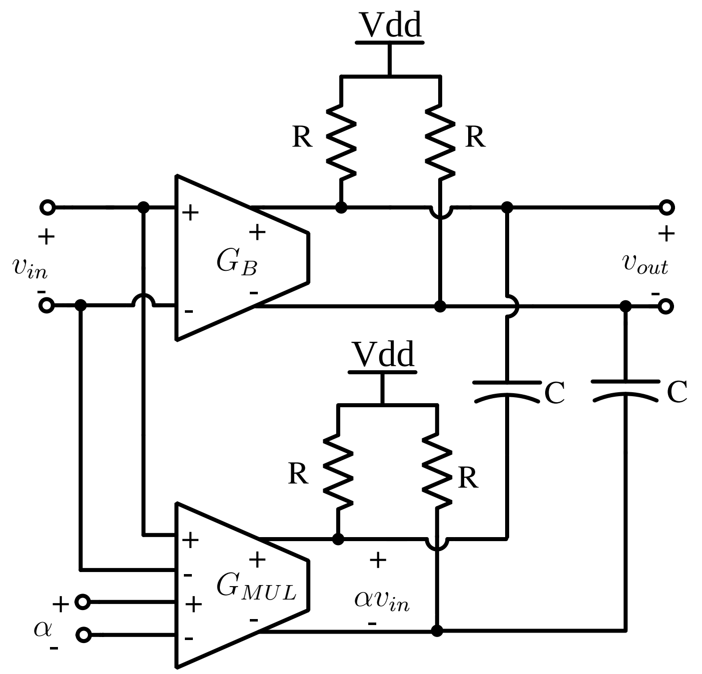


Figure 3.5: Block level representation of variable delay circuit using transconductors: G_B and G_{MUL}.

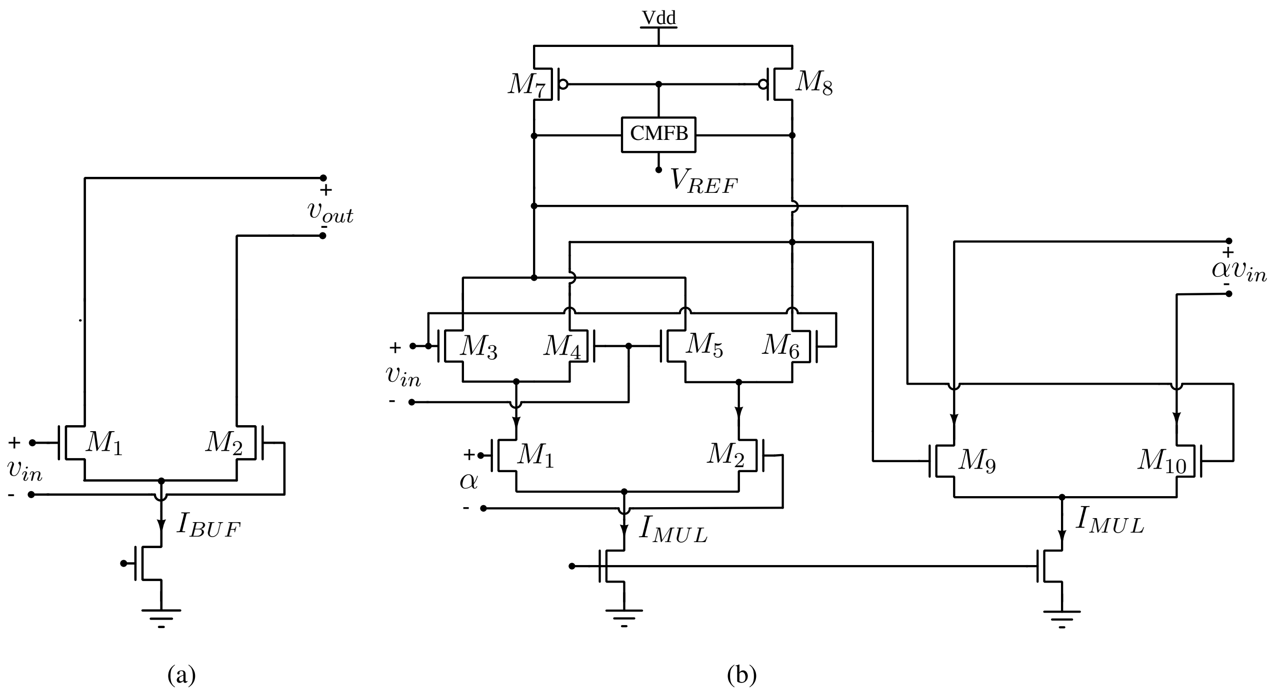


Figure 3.6: (a) Schematic representation of G_B, (b) Schematic representation of G_{MUL}.

The parasitic capacitances C_{p1} and C_{p2} are expressed as

$$C_{p1} = C_{dB(n)} + C_{gB(n+1)} + C_{gM(n+1)}, \tag{3.5}$$

$$C_{p2} = C_{dM(n)}. \tag{3.6}$$

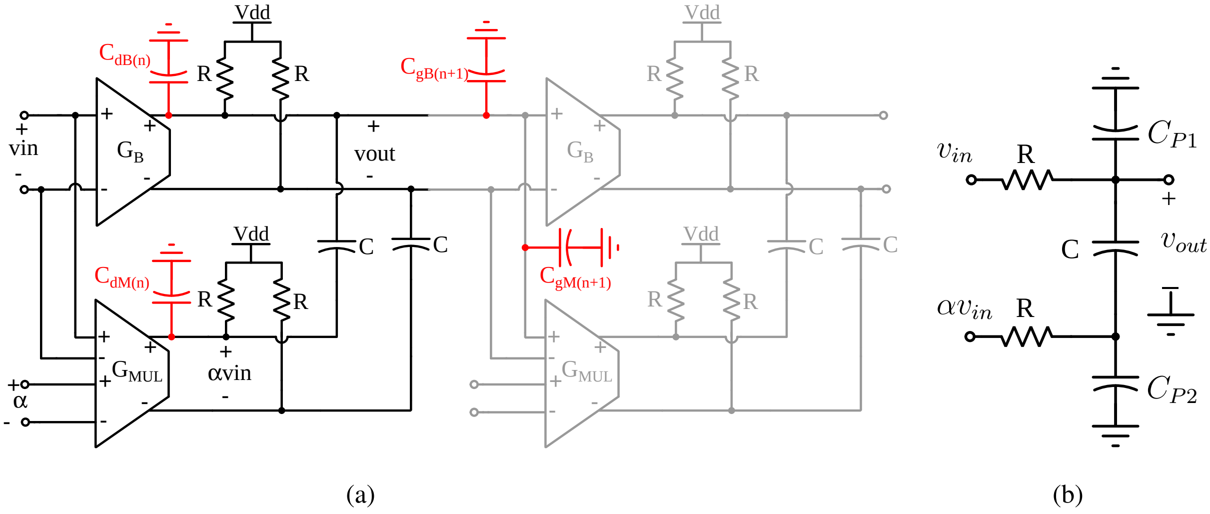


Figure 3.7: (a) Cascaded n^{th} and $(n+1)^{th}$ delay cells with various parasitic capacitances, (b) small signal representation of (a).

The transfer function $H_\alpha(s)$ of n^{th} stage time delay LPF in presence of C_{p1} and C_{p2} is evaluated to

$$H_\alpha(s) = \frac{(1 + s[(1+\alpha)RC + RC_{p2}])}{1 + s[2RC + RC_{p1} + RC_{p2}] + s^2[R^2CC_{p1} + R^2CC_{p2} + R^2C_{p1}C_{p2}])}. \tag{3.7}$$

From Eq. 3.7 it is clear that $H_\alpha(s)$ is 2^{nd} order system with 2 poles and 1 zero. Hence, the function $H_\alpha(s)$ can be approximated as,

$$H_\alpha(s) \approx \frac{(1 + s[(1+\alpha)RC + RC_{p2}])}{\Big(1 + s[2RC + R(C_{p1}+C_{p2})/2]\Big)\Big(1 + s[R(C_{p1}+C_{p2})/2]\Big)} = \frac{(1+s/z_1)}{(1+s/p_1)(1+s/p_2)}. \tag{3.8}$$

where

$z_1 = 1/[(1+\alpha)RC + RC_{p2}]$ is the variable zero frequency,

$p_1 = 1/[2RC + R(C_{p1}+C_{p2})/2]$ and $p_2 = 1/[R(C_{p1}+C_{p2})/2]$ are the pole frequencies.

If the parasitic capacitances C_{p1} and C_{p2} are related to the capacitor C as $C_{p1} = xC$ and $C_{p2} = yC$, then $H_\alpha(s)$ can be written as

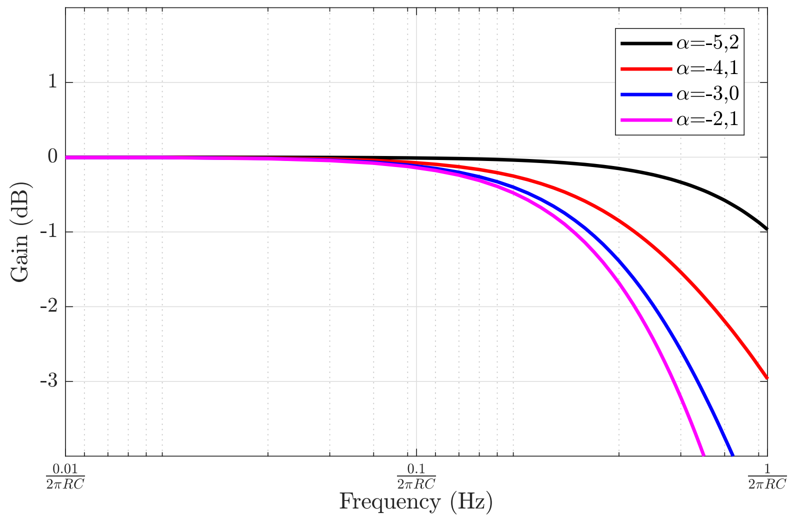
$$H_\alpha(s) \approx \frac{(1 + sRC[(1+\alpha+y])}{\Big(1 + sRC[2 + (x+y)/2]\Big)\Big(1 + sRC[(x+y)/2]\Big)} \tag{3.9}$$

The resulting low frequency delay τ_α is approximated to,

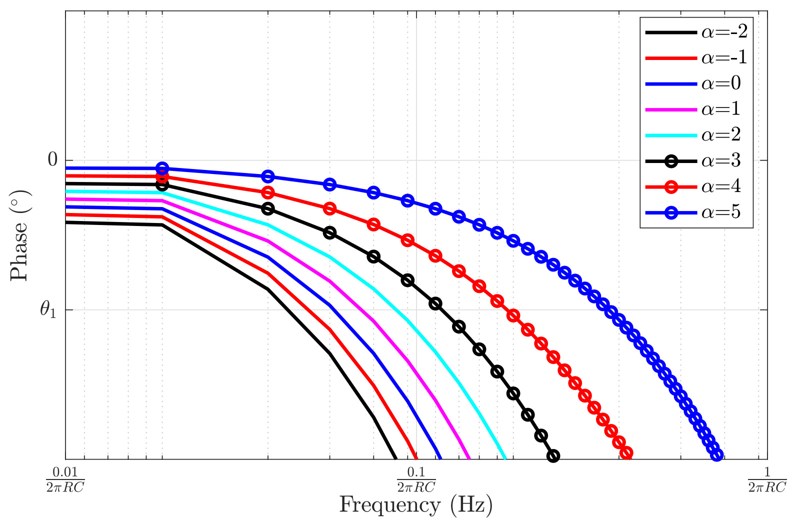
$$\tau_\alpha \approx \big[(1 + x - \alpha)\big]RC. \tag{3.10}$$

From Eq. 3.9 and Eq. 3.10, it can be observed that parasitic capacitances effect the gain and delay characteristics.

Transistor-level implementation shows $x \approx 2$ and $y \approx 0.5$ in the presence of parasitic capacitances. $H_\alpha(s)$ with its pole-zero plots for different values of α are tabulated in Table. 3.2. It is observed that the proposed variable delay LPF can act as TTD for 3 $< \alpha \leq$-5. From gain and phase plots shown in Figure 3.8(a) and Figure 3.8(b), it is observed that $H_\alpha(s)$ has same gain response for $\alpha = -5$ and $\alpha = 2$, because of the zero appearing in RH plane. For the same reason, $H_\alpha(s)$ shows a higher phase lag for $\alpha = -5$ compared to that of $\alpha = -2$. A similar explanation is valid for other values of α. The group delay τ_α decreases linearly from 8RC to



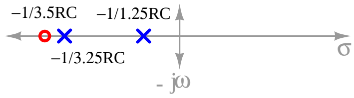
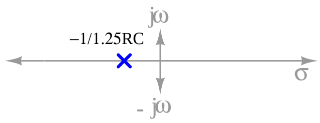
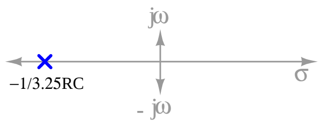
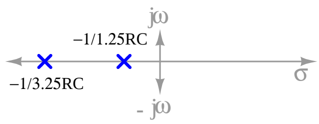
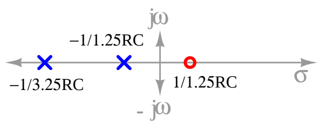
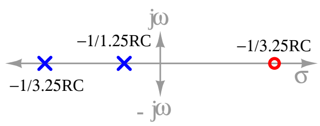
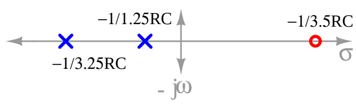
(a)



(b)

Figure 3.8: (a) Gain and (b) phase plots of variable delay LPF for different values of α in the presence of parasitic capacitances.

Table 3.2: Transfer function $H_\alpha(s)$ for different values of α with pole-zero plots.

α	$H_\alpha(s)$	Pole-Zero Plot
2	$\frac{(1+s3.5sRC)}{(1+s3.25RC)(1+s1.25RC)}$	
1.75	$\frac{1}{(1+s1.25RC)}$	
-0.25	$\frac{1}{(1+s3.25RC)}$	
-1.5	$\frac{1}{(1+s3.25RC)(1+s1.25RC)}$	
-2.75	$\frac{(1-s1.25RC)}{(1+s3.25RC)(1+s1.25RC)}$	
-4.75	$\frac{(1-s3.25RC)}{(1+s3.25RC)(1+s1.25RC)}$	
-5	$\frac{(1-s3.5sRC)}{(1+s3.25RC)(1+s1.25RC)}$	

RC when α is varied from -5 to 2 as seen in Figure 3.9. For large delay setting of $\alpha = -5$ with 2 poles and 1 RH zero, the frequency-dependent delay $\tau_g(\omega)$ starts reducing at the frequency $f = 0.07/(2\pi RC)$. The 3-dB frequency reduces for $\alpha = -1.5$ in which the LPF has only 2 poles. It is observed that in the presence of parasitic capacitances, there is an increase in delay

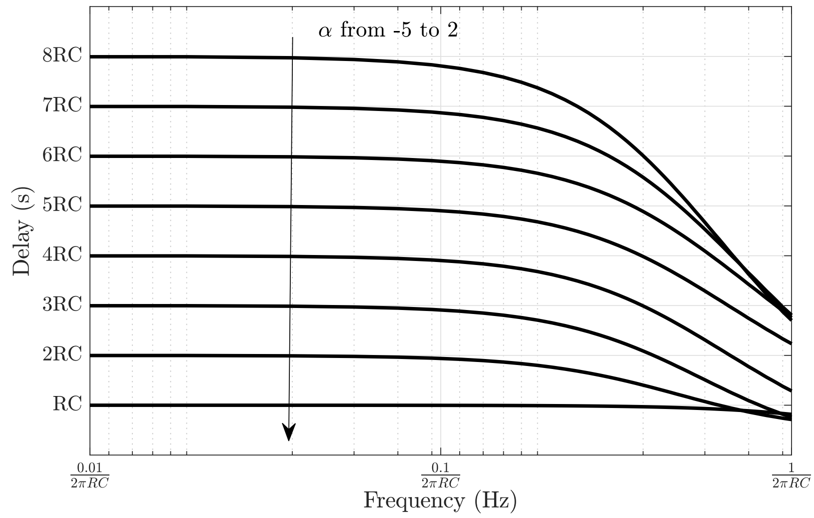


Figure 3.9: (a) Gain, (b) phase, and (c) group delay plots of variable delay LPF for different values of α in the presence of parasitic capacitances.

range from a variable delay cell. However, the maximum bandwidth at which delay remains constant is reduced. Hence, this would effect the (delay-range $\times$ maximum bandwidth) factor of the N-stage TTD circuit.

3.3 10-stage TTD Circuit

Figure 3.10 shows the block-level architecture of the test setup for measuring the performance of a 10-stage TTD circuit implemented with variable delay LPFs. The control signal required for delay tuning is generated by a 5-bit DAC. The output signal is taken out for measurement using a buffer BUF_1 controlled with FLIP digital control bit used to cancel feedthrough parasitics (discussed below) followed by another buffer BUF_2 with 50 Ω output impedance.

The control signal α is varied to have different delay values from the TTD. The different values of α are obtained from N-bit DAC, where an N-bit digital word generates desired control voltages for delay tuning. For better understanding, consider 3-bit DAC which generates 8 different voltage levels from 3-bit digital input. The internal architecture for 3-bit DAC is

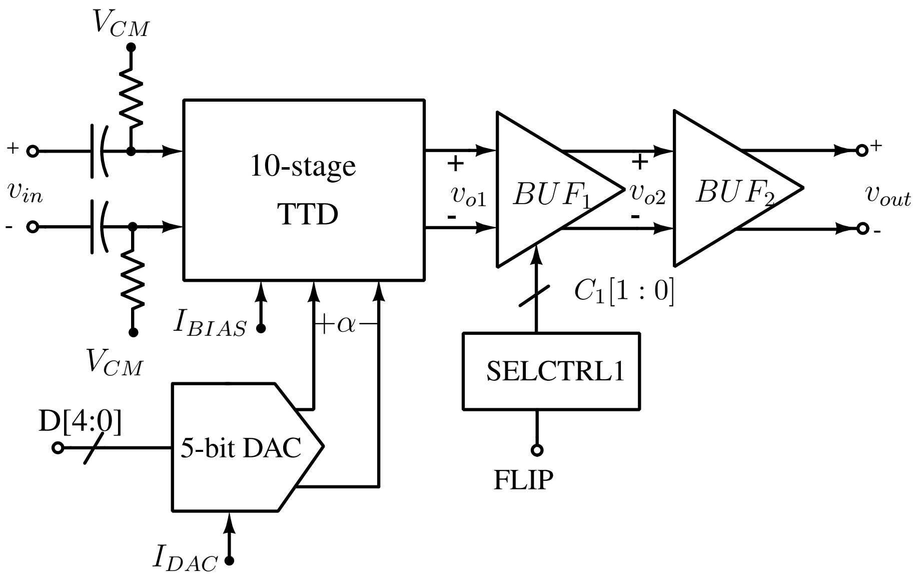


Figure 3.10: Testbench setup for 10-stage TTD circuit.

shown in Figure 3.11. The resistor ladder generates different voltage levels from $V_1 - V_8$. The error amplifier E_A sets voltage V_5 to desired on-chip reference voltage V_{REF} through a negative feedback loop formed by M_P and resistors R. Rest of the voltages are set by V_5 and external bias current I_{DAC}. The transistor level schematic of E_A is shown in Figure 3.14(a). The desired differential voltages from the resistor ladder are selected by a 3-bit digital input using switch matrices and 3-bit adder/subtractor. The N-bit adder/subtractor generates $2N$ switch control bits from N-bit digital input. Hence, for a 3-bit DAC, 6 switch control bits are generated from a 3-bit adder/subtractor. A CMOS implementation of 3-bit adder with inputs a[2:0] and b[2:0] is shown in Figure 3.12. The input b is always connected to digital input "100". The adder generates switch control bits S_{a0}, S_{a1}, S_{a2} and its complementary versions $\overline{S_{a0}}$, $\overline{S_{a1}}$, $\overline{S_{a2}}$. Similarly, the 3-bit subtractor generates the required control bits. The functional table for switch control bits from adder and subtractor for all input digital inputs are tabulated in Table. 3.3. The differential control signals are connected to the input of multiplier G_{MUL} through a buffer A_{BUF}. The buffer A_{BUF} is a 2-stage OTA with open loop gain of 60 dB configured in unity gain configuration. The transistor level schematic of A_{BUF} is shown in Figure 3.14(b).

The undesired effects of input-output feedthrough is nullified by measuring outputs with flipped polarities using control bits $C_1[1:0]$ from control block SELCTRL1 [40]. The output of BUF_1 is passed through buffer BUF_2 with output impedance of 50 Ω to provide impedance matching. The transistor level schematics of BUF_1 and BUF_2 are shown in Figure 3.15(a) and Figure 3.15(b) respectively.

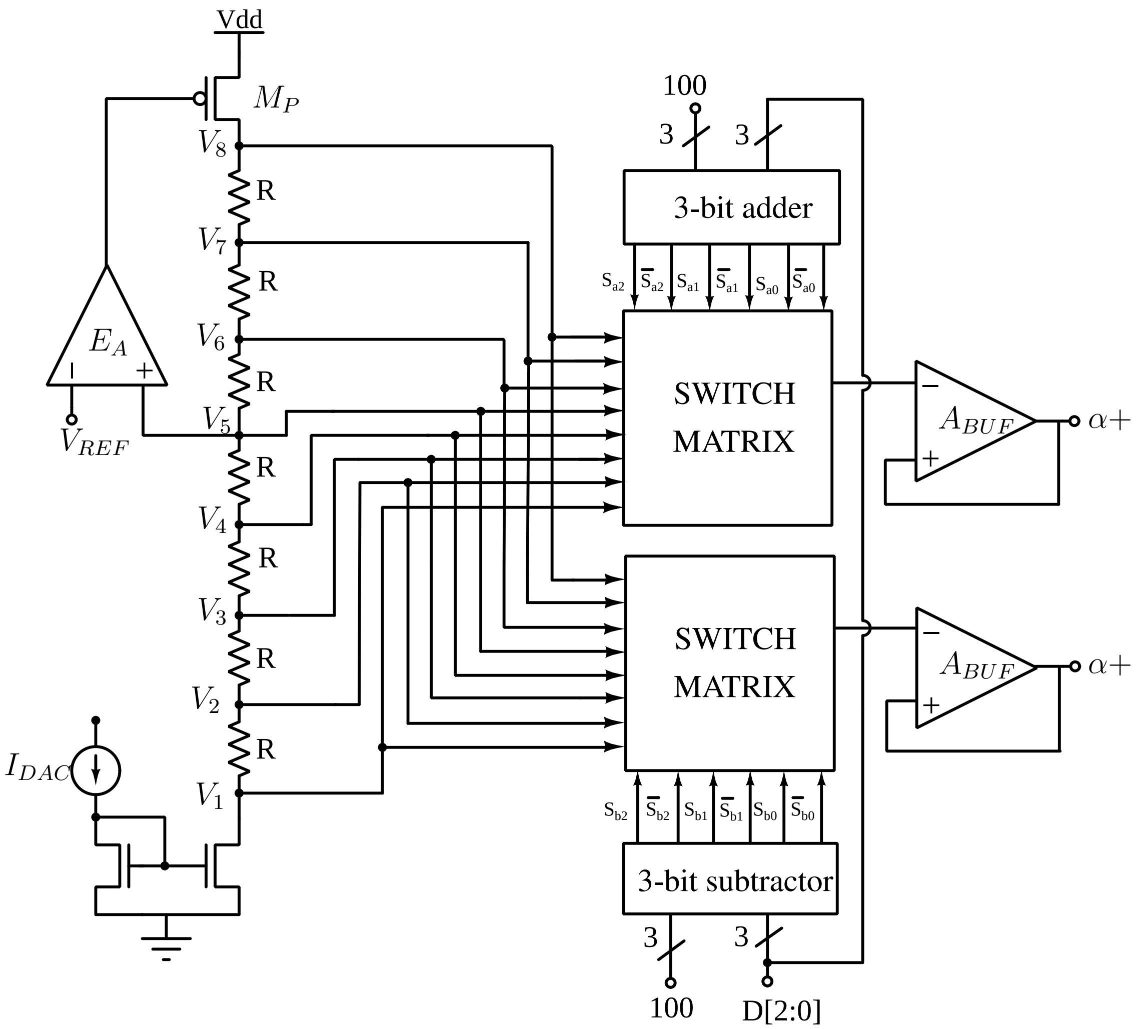


Figure 3.11: Internal architecture of 3-bit DAC for generating differential control signal.

Table 3.3: Functional table of 3-bit adder for 3-bit digital input with other input connected to “100”.

D_3	D_2	D_1	S_{a2}	S_{a1}	S_{a0}	$\overline{S_{a2}}$	$\overline{S_{a1}}$	$\overline{S_{a0}}$	S_{b2}	S_{b1}	S_{b0}	$\overline{S_{b2}}$	$\overline{S_{b1}}$	$\overline{S_{b0}}$
0	0	0	1	0	0	0	1	1	0	1	1	1	0	0
0	0	1	1	0	1	0	1	0	0	1	0	1	0	1
0	1	0	1	1	0	0	0	1	0	0	1	1	1	0
0	1	1	1	1	1	0	0	0	0	1	1	1	0	0
1	0	0	0	0	0	1	1	1	1	1	1	0	0	0
1	0	1	0	0	1	1	1	0	1	1	0	0	0	1
1	1	0	0	1	0	1	0	1	1	0	1	0	1	0
1	1	1	0	1	1	1	0	0	1	0	0	0	1	1

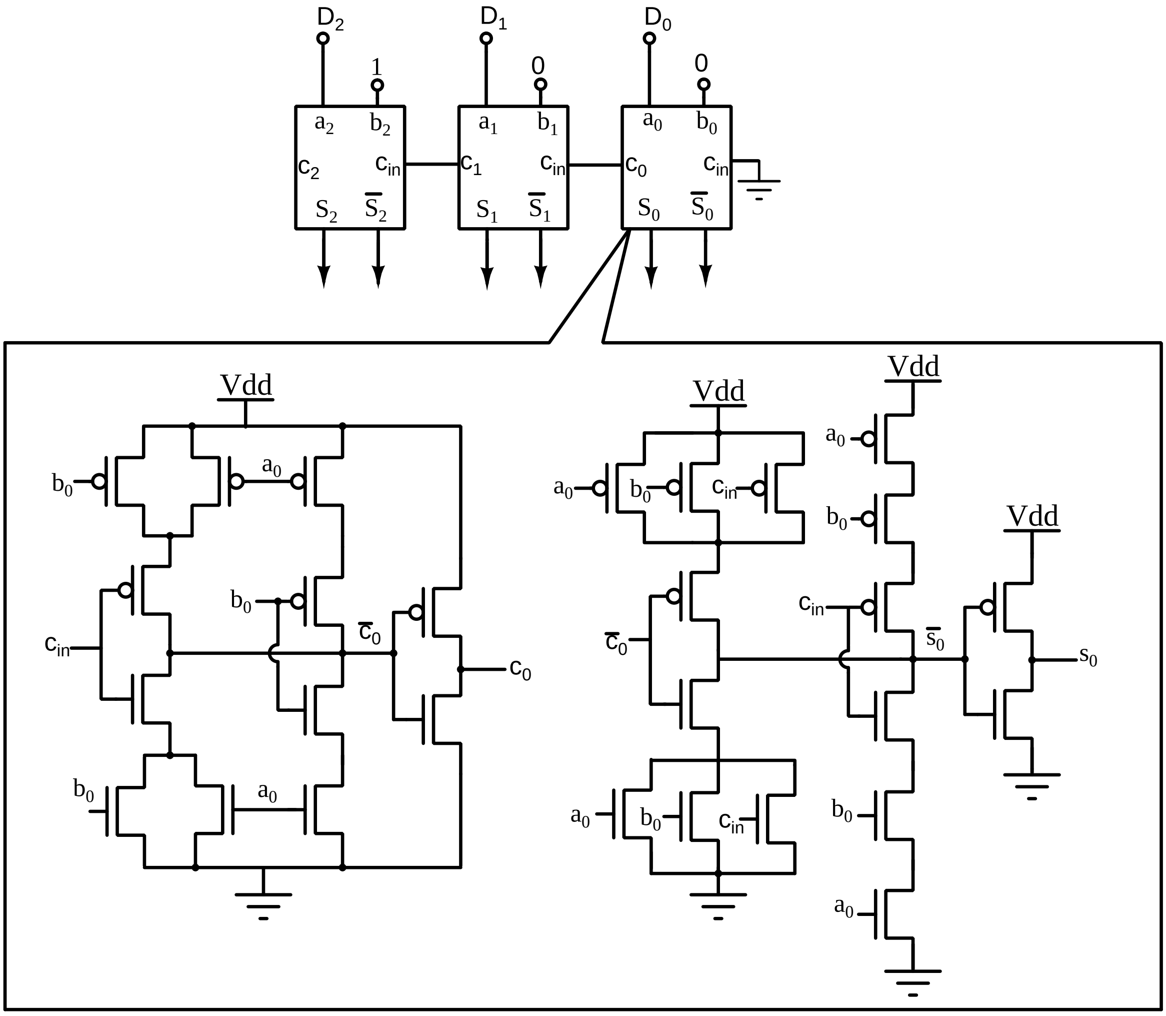
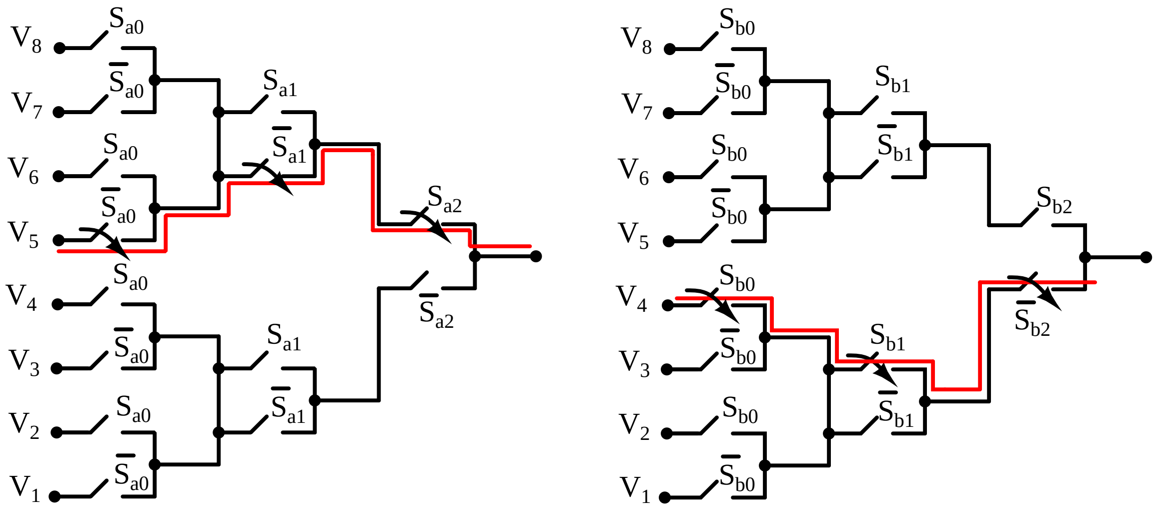


Figure 3.12: Block level representation of 3-bit adder using 1 bit full adders and transistor level implementation of 1-bit full adder is shown.

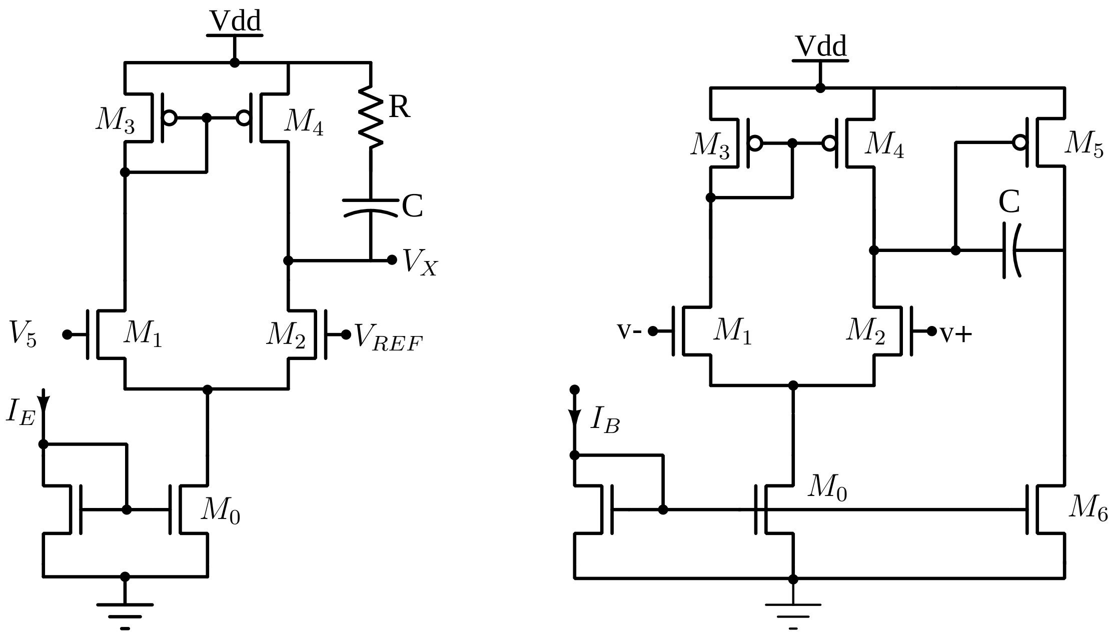
3.3.1 Post-Layout Simulation Results

The proposed delay element is implemented in CMOS UMC 180-nm technology. Figure 3.16 shows the layout of true time delay element with other required blocks. The proposed TTD with cascaded 10-stages of variable delay LPF occupies an area of 0.21 μm $\times$ 0.29 μm. The values of resistors R in G_B and G_{MUL} units of each delay cell are chosen to 700 Ω with C=18 fF. Both the buffers and multipliers are biased with currents I_{BIAS}=1 mA. The differential control signals $\pm\alpha$ are generated from 5-bit DAC which is provided with a reference current $I_{DAC} = 30\ \mu A$. Figure 3.17 shows different values of α for all 5-bit digital input. It can be seen that the multiplier input α varies from 0 to ± 1 V with a common mode of 1.25 V. A high-frequency test input signal v_{in} is AC coupled and is biased at a common mode voltage V_{CM}=1.6 V. The post layout gain and delay responses of tunable delay element for different values of α are shown



(a) (b)

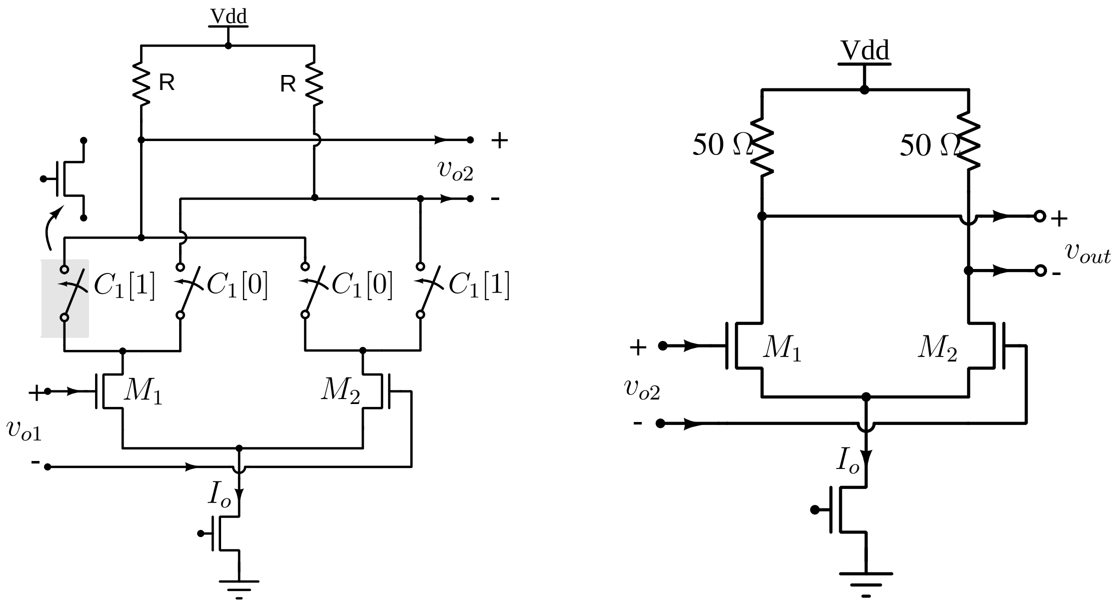
Figure 3.13: Switch matrices passing differential voltages (a) V_5 and (b) V_4 to buffers for digital input="000".



(a) (b)

Figure 3.14: Transistor level of schematic of (a) E_A and (b) A_{BUF}.

in Figure 3.18(a) and Figure 3.18(b) respectively. Results show that the variable delay line has a delay range from 450 ps to 900 ps with a maximum delay variation of ± 60 ps within the frequency range of 1 GHz. The gain response has ± 4 dB variation across a nominal value of 0 dB. The proposed delay element has an average power dissipation of 16 mW.



(a) (b)

Figure 3.15: Tansistor level schematic of (a) BUF_1 and (b) BUF_2

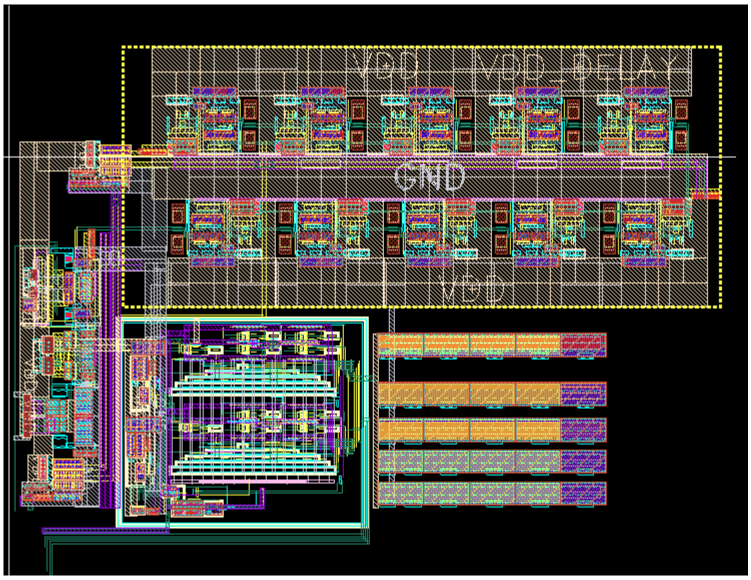


Figure 3.16: Layout of 10-stage TTD circuit with additional circuits for testing.

3.3.2 Measurement Results

The prototype chip is fabricated in UMC 180-nm CMOS technology and is packaged in a QFN40 pin package. The packaged chip is mounted on a 2 layer printed circuit board (PCB) for testing purposes as shown in Figure 3.19. Figure 3.20 shows the measurement setup for testing ten stage TTD. A sinusoidal signal ranging from 100 MHz to 1 GHz with an amplitude of 100 mVpp is applied from RF signal source to BALUN. The BALUN converts this single-

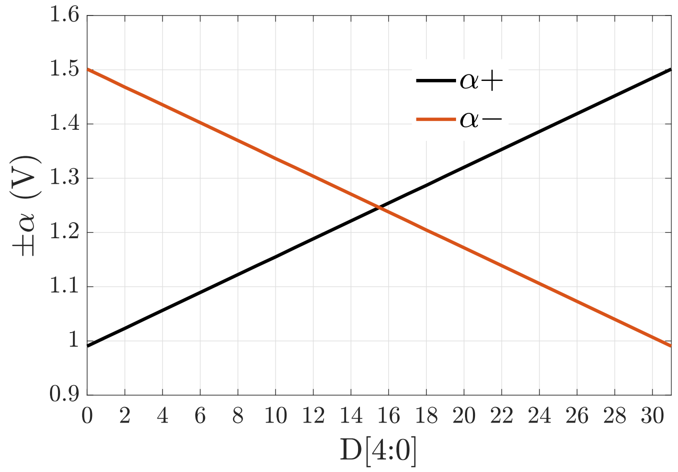


Figure 3.17: 5-bit DAC generating control voltages $\pm\alpha$ from 1.5 V to 1 V.

ended signal to differential signals, out of which one of the signals is fed as an input to the PCB and the other signal is taken as a reference for delay measurements on a digital storage oscilloscope (DSO). The bias currents $I_{BIAS} = 1$ mA and $I_{DAC} = 30\ \mu A$ is applied from the bias current generation setup. The test setup is powered with the supply voltage of 1.8 V applied to the pins $V_{DDdelay}$ and V_{DDTest}. Figure 3.21(a) and Figure 3.21(b) shows the measured gain and group delay response for different delay settings. It is observed that the average delay varies from 350 ps to 740 ps for different 5 bit digital input combinations. The measurement results shows that the maximum deviation in the delay is around $\pm 60\ ps$. The gain vs frequency plot shows that the measured 3-dB frequency is around 400 MHz. The proposed tunable delay circuit consumes current of 10.65 mA with the power dissipation of 19.17 mW from $V_{DDdelay}$

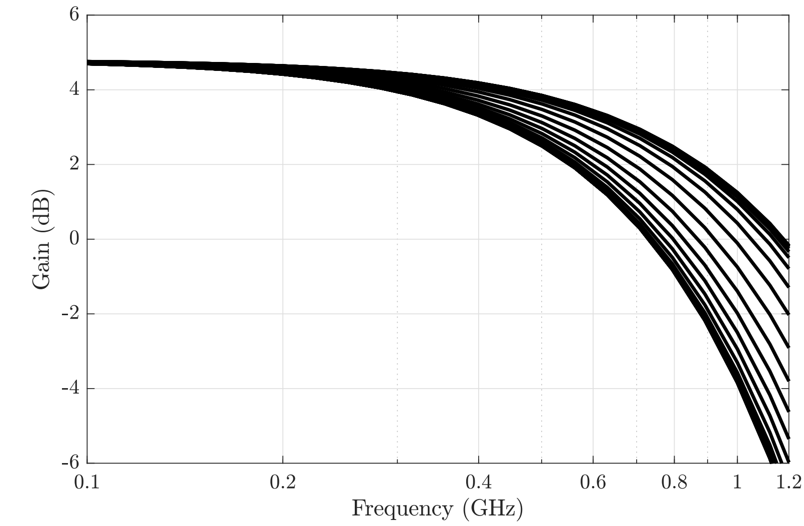
3.4 TTD with Delay Correction and 3-dB bandwidth extension

True time delay cicuits realized using variable delay LPF does not maintain constant delays with frequency. This is due to group delay characteristics of LPF which is a function of frequency. The exact group delay τ_α of variable delay LPF with the controlled signal α is expressed as

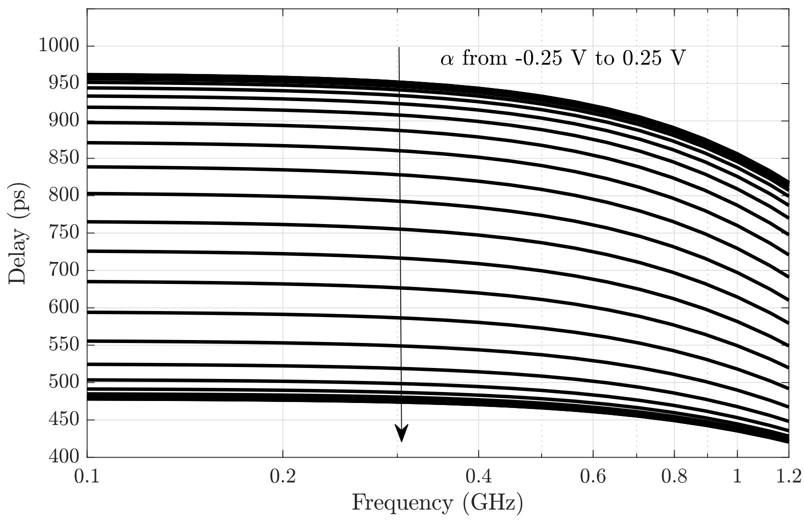
$$\tau_\alpha(\omega) = \frac{2RC}{1+\omega^2 4R^2C^2} - \frac{(1+\alpha)RC}{1+\omega^2(1+\alpha)^2R^2C^2} \tag{3.11}$$

The delay at low frequencies can be approximated to $\tau_\alpha \approx (1-\alpha)RC$.

Consider a TTD circuit that must maintain constant delays at least upto a frequency of



(a)



(b)

Figure 3.18: Post layout (a) gain and (b) delay plots for different values of α.

interest, ω_B which is the bandwidth of baseband signal. Now consider an LPF based TTD that maintains fixed τ_α upto certain frequency, $\omega_a \leq \omega_b$, and then decreases with frequency as shown in Figure 3.22(a). If the decrease in τ_α from ω_a to ω_B is assumed to be linear function of ω, the slope of $d\tau_\alpha/d\omega$ from ω_a to ω_B is expressed as

$$d\tau_\alpha/d\omega = \left(\frac{2RC}{(1+w_B^2 4R^2C^2)} - \frac{(1+\alpha)RC}{(1+w_B^2(1+\alpha)^2R^2C^2)} - (1-\alpha)RC\right)/(\omega_B - \omega_a) \quad (3.12)$$

The group delay can remain flat upto ω_B if another group delay whose value increases linearly from ω_a to ω_B with the slope of $|\,d\tau_\alpha/d\omega|$ is added to $\tau_\alpha(\omega)$. A similar 1-order variable delay RC circuit with different control signal β is used to obtain delays with positive slope

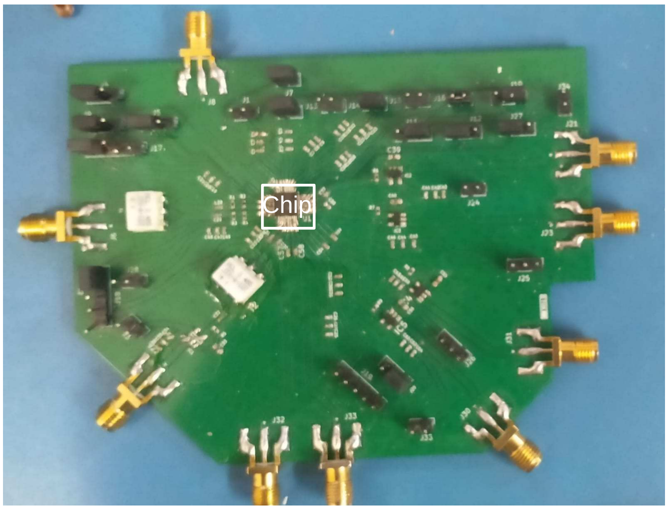


Figure 3.19: Prototype chip mounted on PCB for testing.

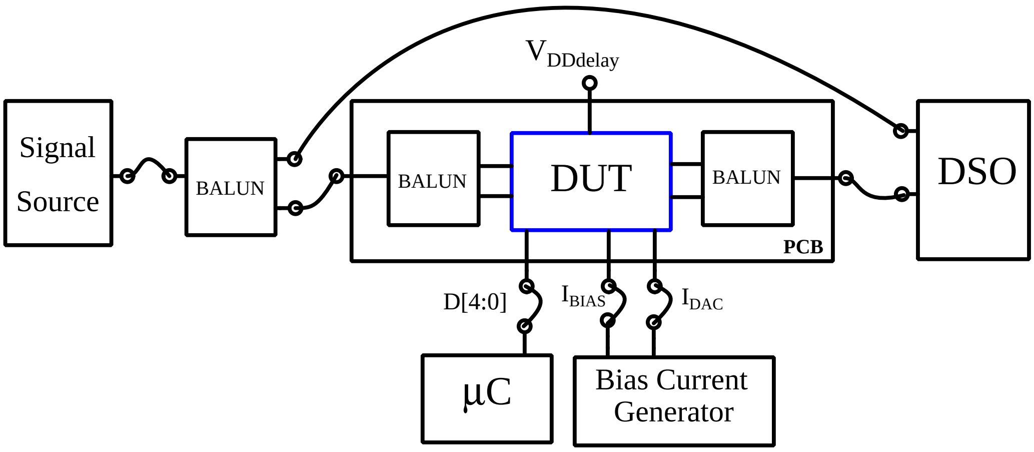
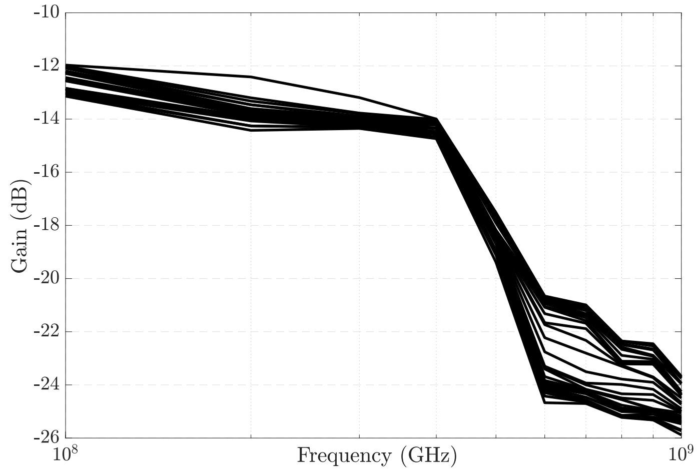


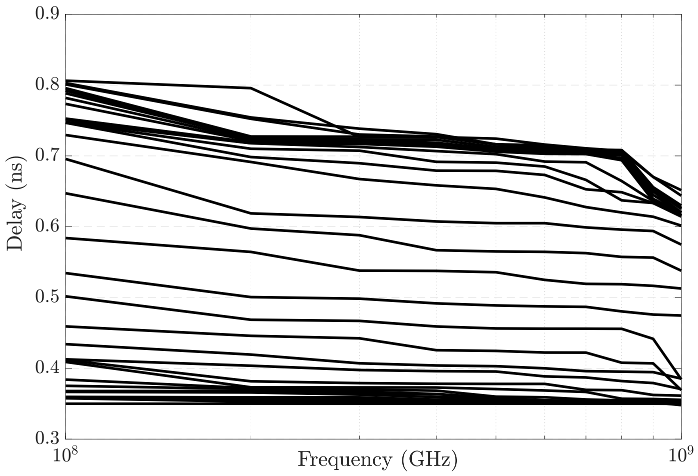
Figure 3.20: Measurement setup for testing ten stage TTD.

within the frequency region of ω_a to ω_B. The transfer function of the circuit $\mathrm{H}_\beta(\mathrm{s})$ and tunable delay τ_β with control signal β are expressed as follows,

$$\mathrm{H}_\beta(\mathrm{s}) = \frac{1 + s(1+\beta)RC}{1 + s2RC} \tag{3.13}$$



(a)

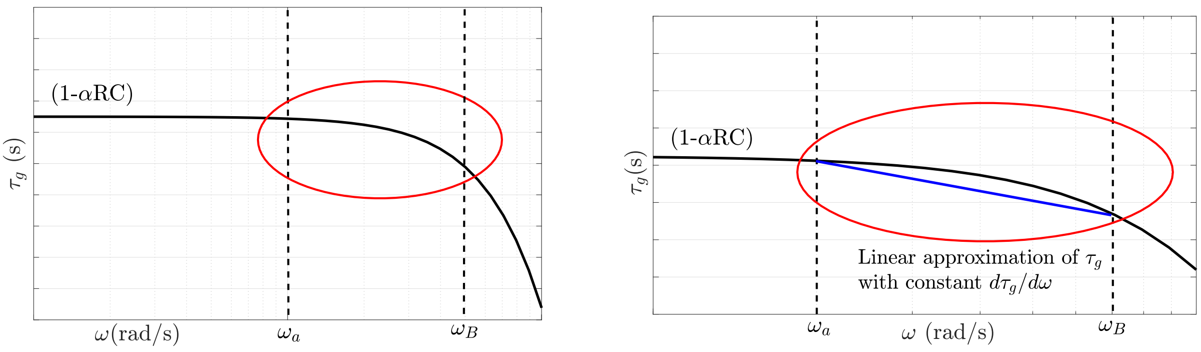


(b)

Figure 3.21: Measured (a) gain and (b) delay response of the TTD for different values of α.

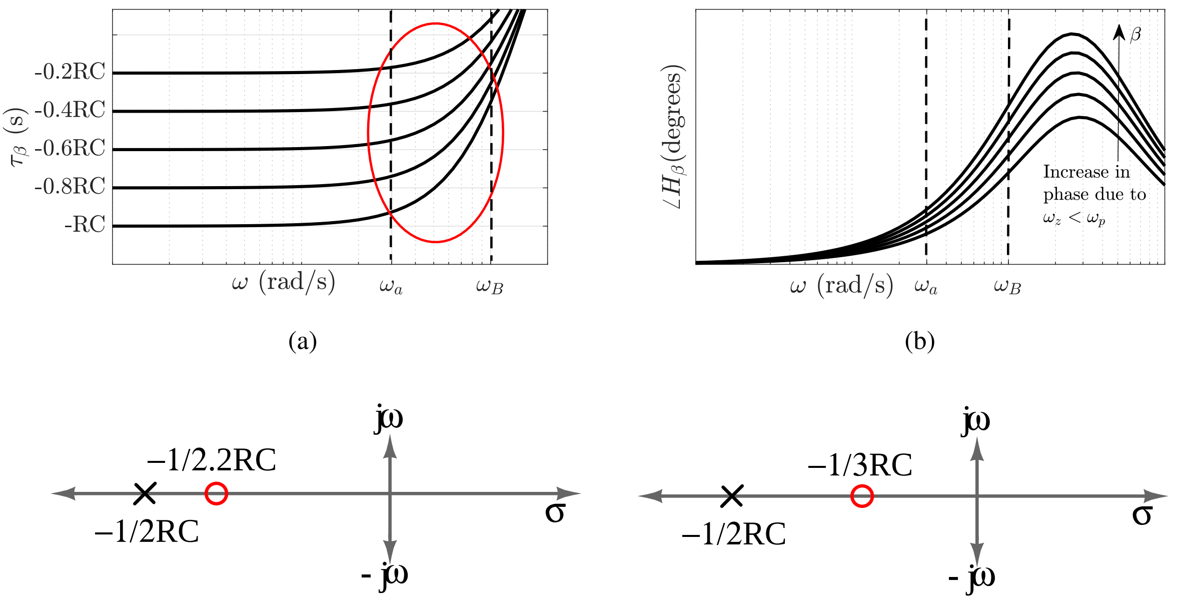
$$\tau_{\beta}(\omega) = \frac{2RC}{1+\omega^2 4R^2C^2} - \frac{(1+\beta)RC}{1+\omega^2(1+\beta)^2R^2C^2} \tag{3.14}$$

Figure 3.23(a) shows the group delay plots of $\mathrm{H}_{\beta}(\mathrm{s})$ for 1.2<β<2 and it can be seen that $\mathrm{H}_{\beta}(\mathrm{s})$ maintains fixed negative delay of $(1-\beta)RC$ at low frequencies (upto ω_a) and then increases. The negative value in the delay τ_β is due to zero appearing at frequency $\omega_z = 1/[(1+\beta)RC]$, which is lower than the pole frequency $\omega_p = 1/2RC$ for 1.2≤β≤2. This can be understood intuitively from phase and pole-zero plots of $\mathrm{H}_{\beta}(\mathrm{s})$ as seen in Figure 3.23(b) and Figure 3.23(c) respectively. If the increase in τ_β within the frequencies ω_a to ω_B is assumed to be linear, the



(a) (b)

Figure 3.22: (a) Delay response of variable delay LPF showing constant delay uptofrequency ω_a, (b) Approximationof variation in τ to a linear function of $\omega$from ω_a to ω_B.



(a) (b)

(c)

Figure 3.23: (a) Group delay τ_β (b) Phase plots of H_β for 1.2<β≤2, and (c) Pole zero plots with β=1.2, 2

slope $d\tau_\beta/d\omega$ in this frequency range is expressed as

$$d\tau_\beta/d\omega = \left(\frac{2RC}{1+\omega_B^2 4R^2C^2} - \frac{(1+\beta)RC}{1+\omega_B^2(1+\beta)^2R^2C^2} - (1-\beta)RC\right)/(\omega_B - \omega_a) \qquad (3.15)$$

The group delay of TTD element realized by variable delay LPF H_α(s) is made flat by cascading correction circuit H_β(s) with it as shown in Figure 3.24. The overall group delay $\tau_{\alpha\beta}$ of the

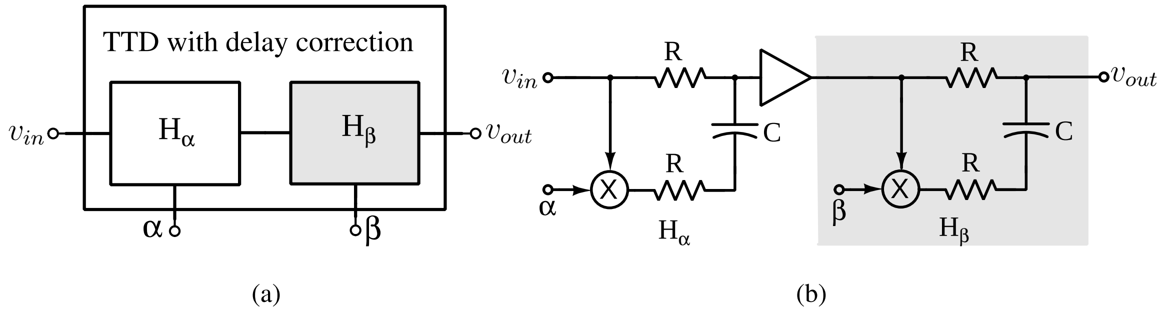


Figure 3.24: Variable delay LPF H_α cascaded with correction circuit H_β.

cascaded system $H_{\alpha\beta}(s)$ for delay corrections is expressed as

$$\tau_{\alpha\beta}(\omega) = \tau_\alpha(\omega) + \tau_\beta(\omega) \tag{3.16}$$

The cascaded system maintains constant delay $\tau_{\alpha\beta} = [2-(\alpha+\beta)]RC$ upto frequency ω_B if the following condition from frequencies ω_a to ω_B is maintained.

$$d\tau_\alpha/d\omega + d\tau_\beta/d\omega = 0 \tag{3.17}$$

Using Eq. 3.11 and Eq. 3.15 the above equation is rewritten as,

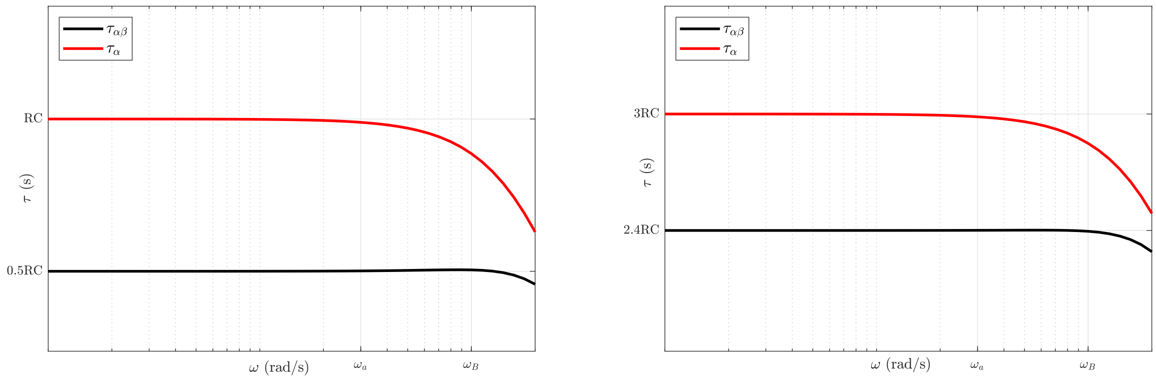
$$\frac{(1+\beta)RC}{(1+\omega_B^2(1+\beta)^2R^2C^2)} + (1-\beta)RC = \frac{4RC}{(1+\omega_B^2 4R^2C^2)} - \frac{(1+\alpha)RC}{(1+\omega_B^2(1+\alpha)^2R^2C^2)} - (1-\alpha)RC \tag{3.18}$$

Hence, for the desired bandwidth ω_B with the designed values of R and C, the required value of β for flat delay is inspected for different delay settings of α. For instance, if the variable delay LPF is designed with R=500 Ω, C=35 fF and ω_B=(0.1/RC) rad/s, then the required β for different values of α is shown in Table. 3.4. Delay plots of the cascaded system $H_{\alpha\beta}$ for 2

Table 3.4: Values of β evaluated from inspection for different α.

α	β	$\tau_\alpha(s)$	$\tau_{\alpha\beta}(s)$
0	1.51	RC	0.5RC
-0.5	1.54	1.5RC	0.96RC
-1	1.56	2RC	1.44RC
-1.5	1.58	2.5RC	1.92RC
-2	1.6	3RC	2.4RC

different values of $[\alpha,\beta]$=$[0, 1.51]$, $[-2, 1.6]$ are shown in Figure 3.25(a) and Figure 3.25(b). It is seen that with the correction circuit H_β the group delay $\tau_{\alpha\beta}$ remains flat upto frequency ω_B compared to τ_α. With the correction circuit, the bandwidth is increased by a factor (ω_b/ω_a) of 2.5. But TTD with the correction circuit consumes twice the power without correction.



(a) (b)

Figure 3.25: Delay plots showing flat group delay upto ω_B with the correction circuit.

3.5 15-stage TTD with delay correction technique

Figure 3.26 shows the block-level architecture of the test set up for measuring the performance of 15-stage TTD circuit with delay correction technique. The proposed TTD circuit consists

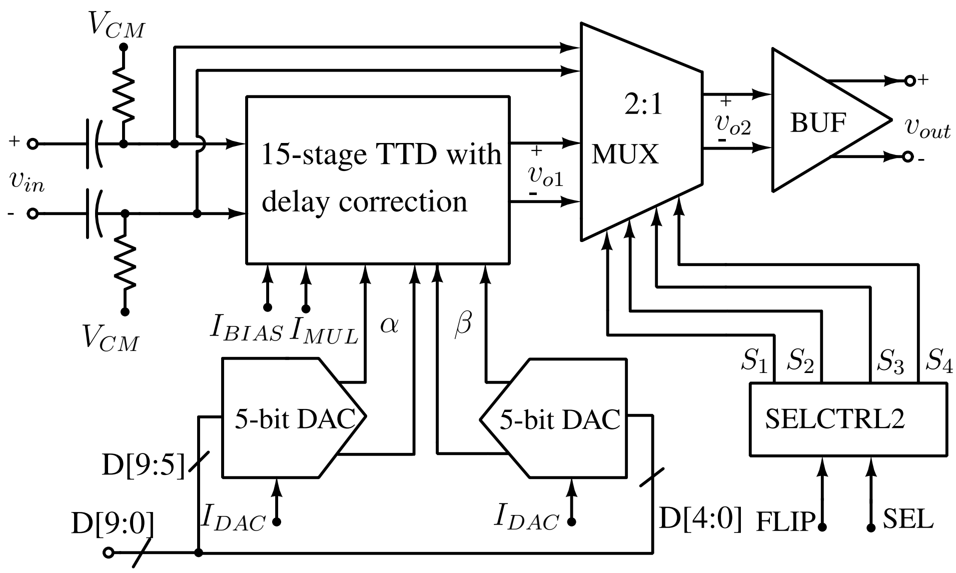


Figure 3.26: Testbench setup for 15-stage TTD circuit with delay correction.

of 15 cascaded stages of variable time delay cell with delay correction technique. The control signals α, β required for delay tuning and delay correction are generated by two 5-bit DACs. The input signal and time delayed output signal are taken out for measurement using a 2:1 MUX controlled with FLIP and SEL digital control bits followed by a buffer BUF with output impedance of 50 Ω.

The required selection lines S_{1-4} of the MUX for measuring input and output signals are obtained from select control unit SELCTRL2 consisting of 2:4 decoder and CMOS TO CML converter. Table 3.5 shows functional table of the select control unit for different input combinations. The internal architecture of the select control unit and transistor level schematic of CMOS TO CML converter are shown in Figure 3.27(a) and Figure 3.27(b) respectively. The

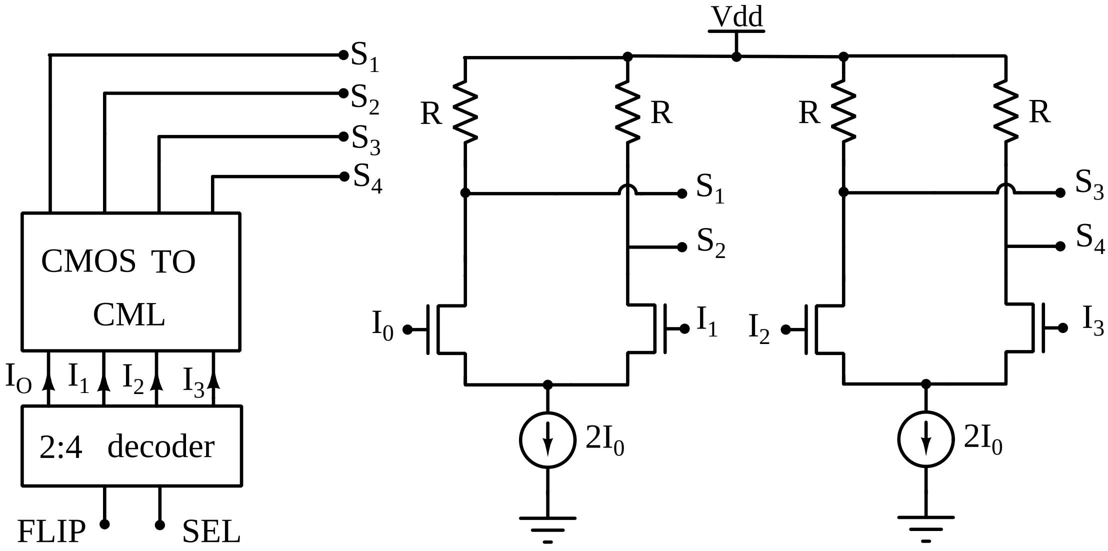
Table 3.5: Functional table of SELCTRL2 for generating signals S_{1-4} .

$FLIP$	SEL	I_3	I_2	I_1	I_0	S_1	S_2	S_3	S_4
0	0	0	0	0	1	V_{DD}	V_{DD}	V_{DD}	$V_{DD} - I_O R$
0	1	0	0	1	0	V_{DD}	V_{DD}	$V_{DD} - I_O R$	V_{DD}
1	0	0	1	0	0	V_{DD}	$V_{DD} - I_0 R$	V_{DD}	V_{DD}
1	1	1	0	0	0	$V_{DD} - I_O R$	V_{DD}	V_{DD}	V_{DD}

transistor level schematic of MUX for selecting input and outputs for measurement is shown in Figure 3.28. The design and transistor level implementation of details of DACs, BUF are similar to that explained in Section 3.3 of Chapter 3.

3.5.1 Post-layout Simulation Results

The 15-stage TTD circuit without and with delay correction technique is designed and implemented in TSMC 65-nm technology. The layouts of these circuits are shown in Figure 3.29 and Figure 3.30 respectively. The 15-stage TTD block without delay correction technique occupies an area of 133 μm $\times$ 159 μm. Post layout gain and delay responses for different values of α are shown in Figure 3.31(a) and Figure 3.31(b) respectively. Results show that the delay values of 15-stage TTD without delay correction technique ranges from 580 ps to 1000 ps with a maximum delay variation of ± 50 ps within the frequency range of 1 GHz. The gain response has ± 6 dB variation across a nominal value of 2 dB. The proposed delay element has an average power dissipation of 18 mW.



(a) (b)

Figure 3.27: (a) Internal architetcure of select control unit, (b) tansistor level schematic of CMOS to CML converter

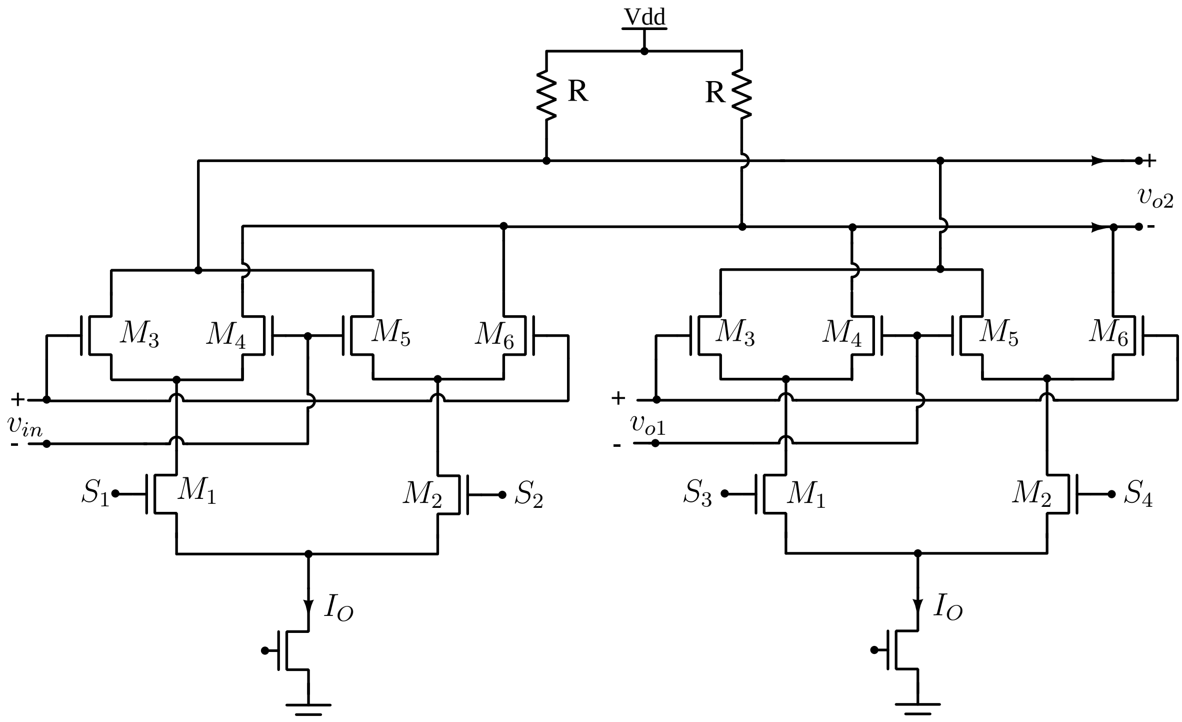


Figure 3.28: Schematic of 2:1 MUX used in the test bench of TTD measurement.

The 15-stage TTD block with delay correction technique occupies an area of 133 μm $\times$319 μm. Post layout gain and delay responses of the TTD circuits for different values of α are shown in Figure 3.32(a) and Figure 3.32(b) respectively. Results show that the 15-stage TTD with delay correction technique has a delay ranges from 750 ps to 1150 ps with a maximum delay variation

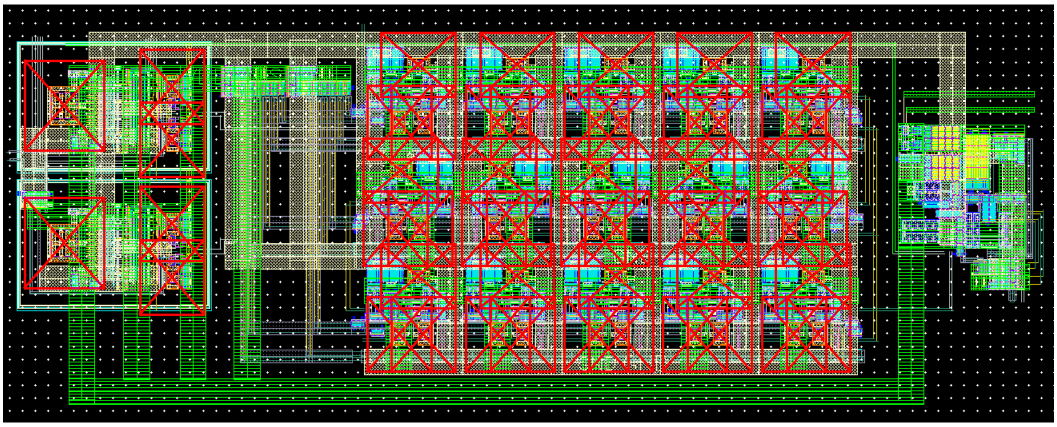


Figure 3.29: Layout of 15-stage TTD circuit without delay correction.

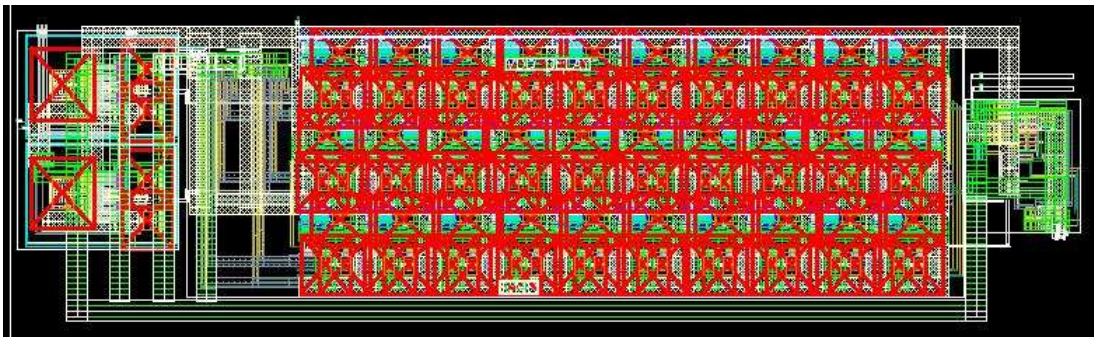


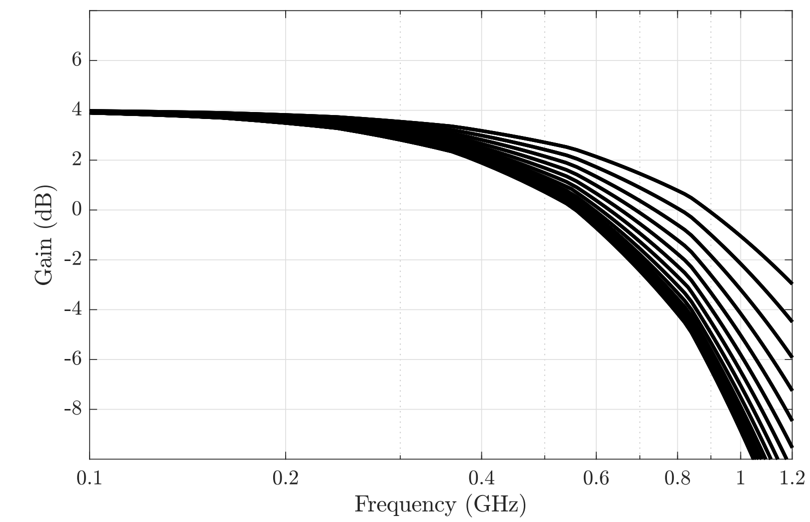
Figure 3.30: Layout of 15-stage TTD circuit with delay correction.

of ± 5 ps within the frequency range of 1 GHz. The gain response has ± 2.5 dB variation across a nominal value of 2.5 dB. The proposed delay element has an average power dissipation of 36 mW.

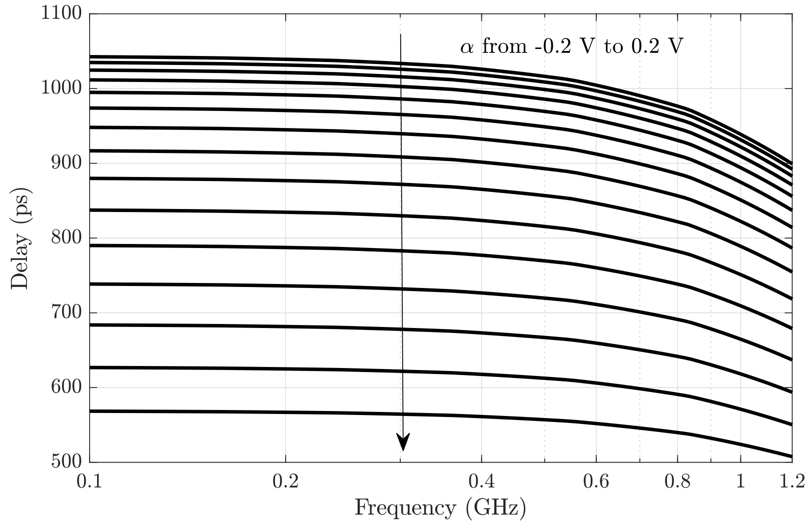
The effect of process corner, supply voltage, and temperature variations on the proposed delay element is studied to check its robustness. Figure 3.33(a), Figure 3.33(b), and Figure 3.33(c) shows the average gain and delay values across all the process corners when temperature varied from $0°C$ to $70°C$ for different supply voltage V_{dd}. Table 3.6 tabulates the maximum gain and delay variations for these conditions.

3.5.2 Noise and Distortion

To quantify the amount of distortion introduced by the proposed variable-delay LPF, 1-dB compression point (P1-dB) and input third-order intercept point (IIP3) have to be determined. The LPF shows a P1-dB $=-28$ dBm as shown in Figure 3.34(a). IIP3 is determined by exciting the filter with 2 tones separated by 1 MHz (990 MHz, 1000 MHz). Figure 3.34(b) shows the



(a)

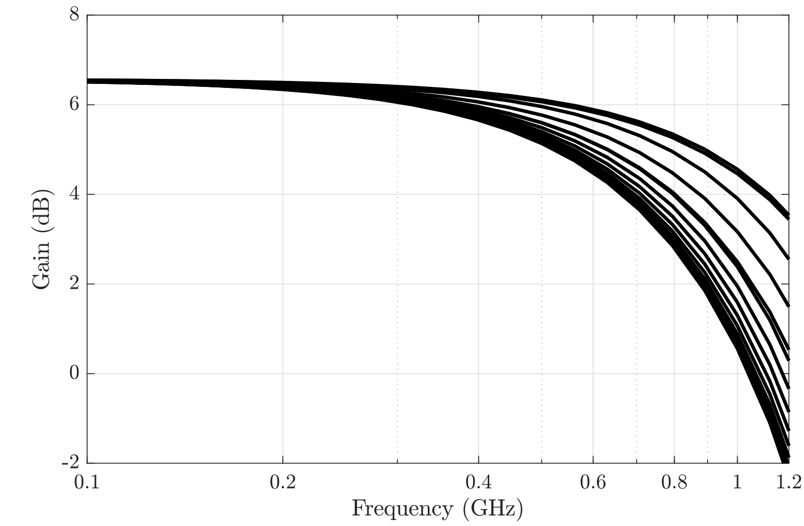


(b)

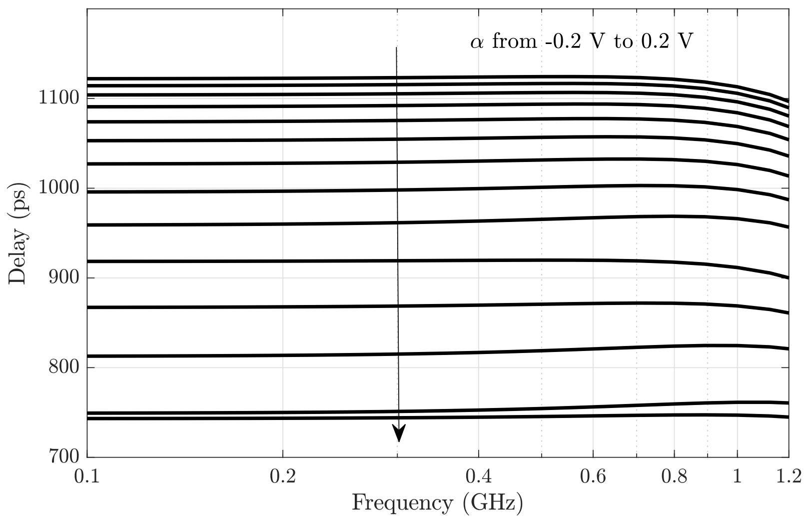
Figure 3.31: Post layout delay and gain plots for different values of α.

LPF has an IIP3 value of -17 dBm. Figure 3.35 shows the input referred noise plot of the LPF. The LPF has a noise voltage density of 13 $nV/\sqrt{Hz}$ at 1 GHz, with a total integrated noise of 422.8 μV.

The dynamic range of the LPF is determined by the maximum and minimum signal levels that the filter can handle. The maximum value is determined from the P1-dB value, which corresponds to a RMS value of 8.83 mV. The minimum signal value is determined from the noise floor of the input-referred noise plot. Considering the signal bandwidth from 10 KHz to 1 GHz, the noise corresponding to 10 KHz is the heighest. This value corresponds to 574 nV/$\sqrt{Hz}$ and



(a)



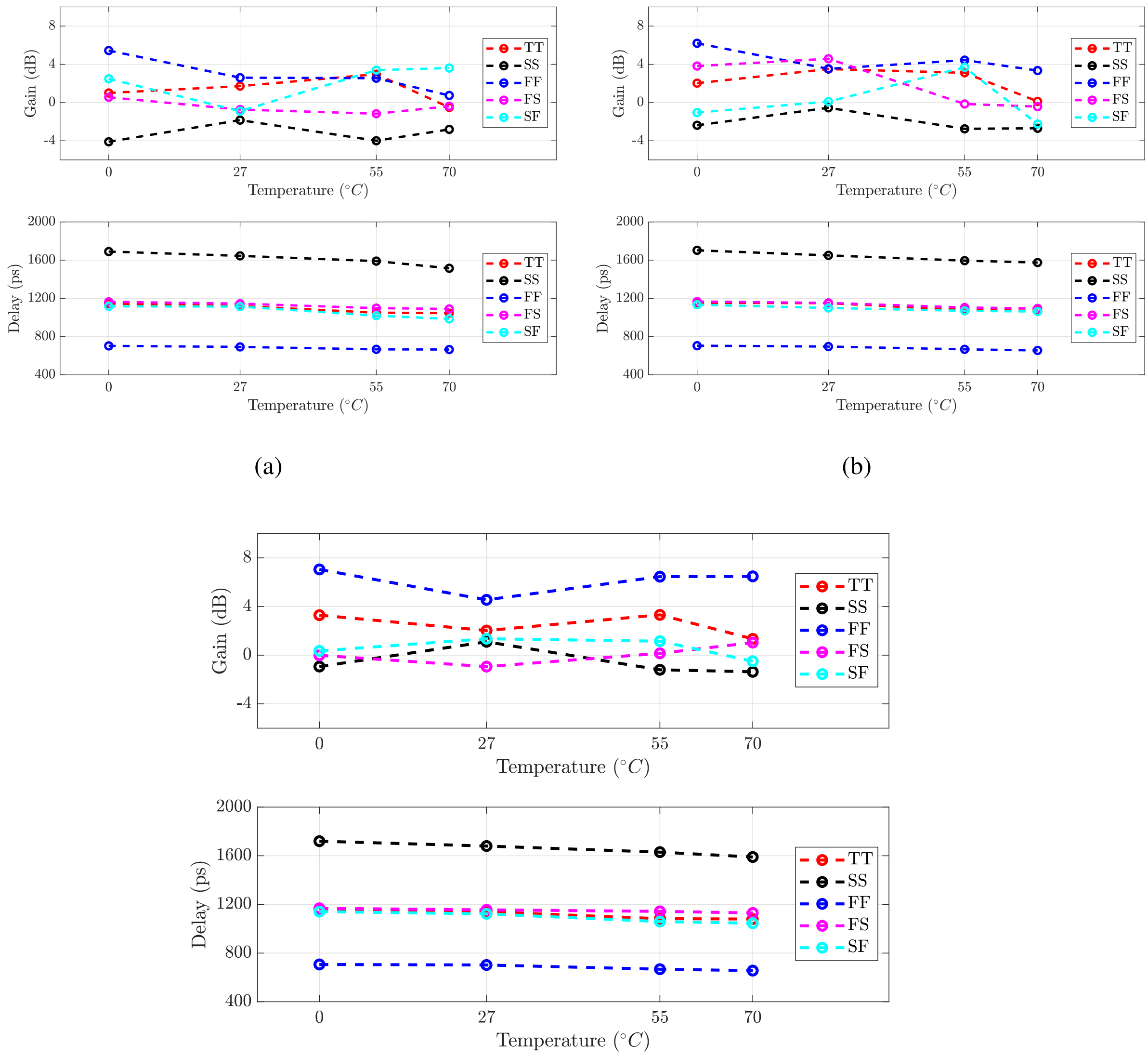
(b)

Figure 3.32: Post layout delay and gain plots for different values of α.

hence, the filter's dynamic range is found to be 83.75 dB.

3.6 Discussions

The post-layout simulation results of 10-stage TTD and 15-stage TTD without and with delay correction technique is compared with different parameters of existing TTD architectures as seen in Table 3.7. To quantify the performance of the proposed TTD architectures and to have a quantitative comparison between the reported TTD circuits, two figure of merit (FOM)s are



(a) (b)

(c)

Figure 3.33: Average gain and delay values across all the process corners when temperature varied from $0°C$ to $70°C$ for (a) V_{dd}=1.16 V, (b) V_{dd}=1.2 V and (c) V_{dd}=1.26 V.

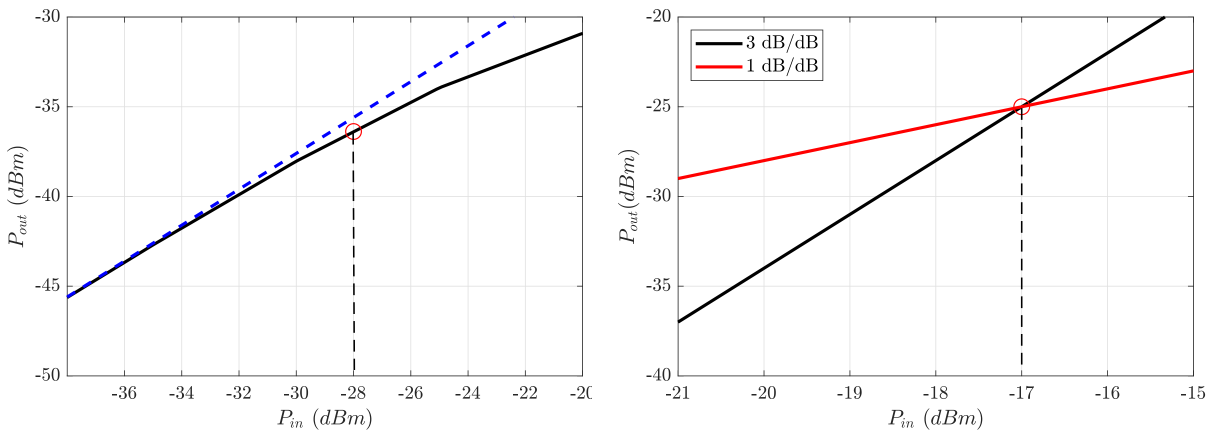
defined for the TTD circuits. Considering commonly reported FOM-1, among all the reported architectures passive-APF based TTD circuit [22] shows the best performance.

$$\text{FOM-1} = 20\log(\text{Power}/(\text{delay range} \times \text{bandwidth})) \tag{3.19}$$

But the circuit shows a larger gain deviation of ± 15 dB. Variable order APF based TTD circuit shows larger delay variations of ± 140 ps compared to other reported works. Hence, to capture the delay and gain variations in the FOM, FOM-2 is defined as

Table 3.6: Maximum gain and delay variations across all the process corners with different supply voltage V_{dd} when temperature varied from $0°C$ to $70°C$.

		Process Corner				
Parameters	V_{dd} (V)	TT	SS	FF	FS	SF
Gain variation (dB)	1.16	±3.8	±7.3	±1.4	±4.1	±3.6
	1.2	±3.8	±7.2	±1.3	±4.7	±3.6
	1.26	±3.8	±7.3	±1.2	±4.1	±3.6
Delay variation (ps)	1.16	±10	±15	±5	±10	±8
	1.2	±5	±15	±5	±5	±12
	1.26	±15	±20	±6	±10	±4



(a) (b)

Figure 3.34: (a) 1 dB compression point of -28 dBm and (b) IIP3 of -17 dBm obtained for variable-delay LPF.

$$\text{FOM-2} = \text{FOM-1} + 20\log\left(\frac{2 \times \text{delay variation}}{\text{delay range}}\right) + 2 \times \text{gain variation}. \tag{3.20}$$

Considering FOM-2 for comparing the performance of LPF-based TTD (with and without delay correction) with other reported architectures, one can note that among all active TTD circuits, the LPF-based TTD with delay correction has the best FOM-2. While the Passive APF based TTD reported in [22] features a better FOM-2, being passive it has an nominal attenuation of 30 dB. In contrast, all the active TTD have nominally zero attenuation.

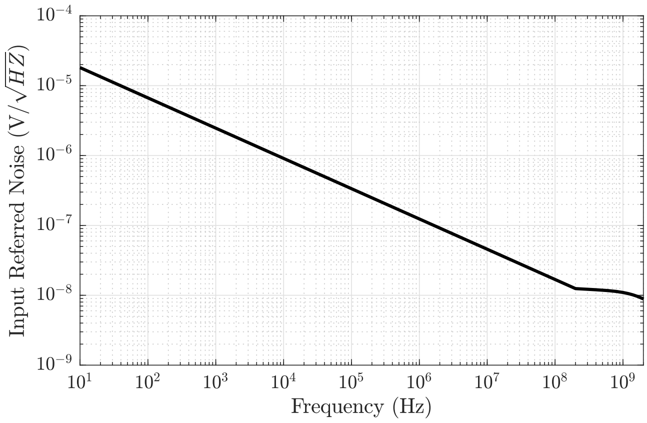


Figure 3.35: Input referred noise plot of the proposed variable-delay LPF.

Table 3.7: Comparison of the proposed TTD without and with delay correction technique with existing architectures.

Parameters	This work	This work	This work	[22]	[28]	[30]	[29]
Technology (nm)	65	65	180	130	140	130	65
Architecture	LPF with corr.	LPF without corr.	LPF	P-APF	A-APF	A-APF	A-APF
Maximum delay(ps)	1150	1000	955	400	550	1700	55
Bandwidth (GHz)	DC-1	DC-1	DC-1	1-20	1-2.5	DC-2	0.4-2
Delay Range (ps)	400	420	450	350	550	1450	27
Delay Variation (ps)	± 5	± 50	± 60	± 8	± 10	± 140	± 2
Gain Variation (dB)	± 2.5	± 6	± 4	± 15	± 1.4	± 0.7	± 0.375
Power (mW)	36	18	16	6	450	364	2.47
FOM-1^{+}	-20.92	-27.35	-28.98	-60.89	-5.26	-18.02	-24.85
FOM-2*	-47.95	-27.82	-32.46	-57.69	-31.25	-30.91	-40.69

FOM-1^{+}=20log(Power/(delay range $\times$ bandwidth))

FOM-2*=FOM1+ 20log ($\frac{2\times\text{delay variation}}{\text{delay range}}$)+2$\times$gain variation

3.7 3-stage Ring VCO

VCO is an important block in clock recovery circuits, PLL and frequency synthesizers. VCOs can be broadly classified into two types: (a) LC-VCO, (b) Ring-VCO. The LC-based VCO exhibits better phase noise performance and higher oscillation frequency compared to ring-VCO. But they have limited tuning range at low frequencies and are not compatible with on-chip integration because of large inductor size at these frequencies. The ring-VCO has wide frequency range and is suitable for integrated circuit designs.

Figure 3.36 shows a block diagram of N-stage VCO with differential architecture, in which identical variable delay cells are interconnected in positive feedback for oscillations. The in-

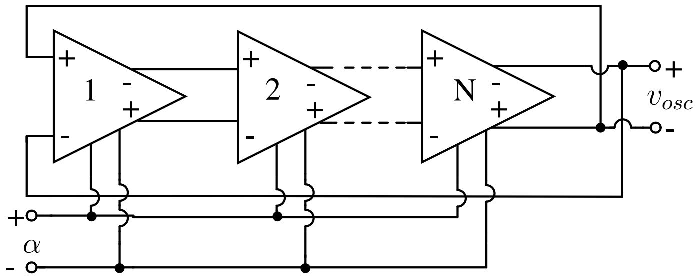


Figure 3.36: Block diagram of N-stage ring-VCO with control signal α.

dividual delay cell is a CML buffer which acts as LPF. A circuit-level schematic of the CML based delay cell is shown in Figure 3.37. The delay offered by a delay cell is $\tau \approx RC$. The

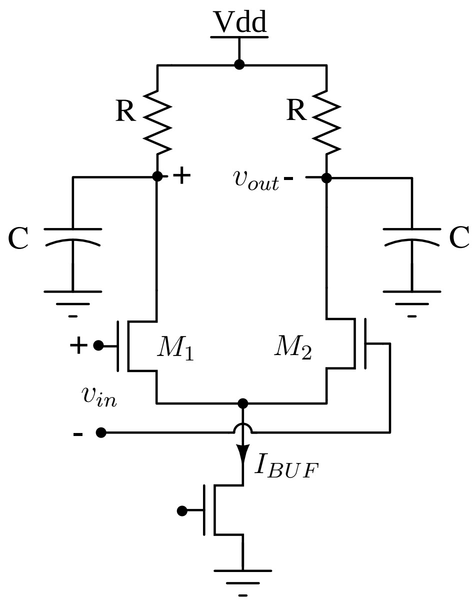


Figure 3.37: Conventional CML buffer as delay cell in ring oscillator

frequency of oscillations f_{osc} of N stage ring oscillator is given by,

$$f_{osc} = \frac{1}{2NRC} \tag{3.21}$$

The frequency of oscillations in ring topology-based VCO is varied by changing resistance or capacitance of individual delay cell with external control signal α. The capacitance values can be varied for controlling oscillations using varactors as shown in [41], [42], or with switched capacitor configuration as shown in [43]. The variable resistance in delay cell is implemented by operating transistors in the triode region, whose on resistance is varied by changing the gate potential in [44].

In the case of controlled C ring-VCO, maintaining a high-quality factor for varactor over the entire tuning range is difficult. Also, variation in capacitance with control voltage for varactors is not perfectly linear. Changing the capacitances for frequency tuning using digitally controlled switches demands analog to digital converter and well-controlled switches for converting external control voltage to digital input which is done in [43]. Varying the resistance value in controlled R ring-VCO varies the amplitude of oscillations. Hence, there is a need for a delay cell without varactors while maintaining constant amplitude for oscillations across a given tuning range of VCO.

3.7.1 Proposed Delay Cell

The variable delay cell used in the design of 3-stage ring VCO is the proposed 1st order LPF, in which the effective impedance is varied using the principle of Miller effect. The variable delay LPF is designed with a gain K for sustained oscillations. From small signal representation of the variable delay cell (Figure 3.38(a)) the transfer function $H_\alpha(s)$ and low frequency group delay τ_α of delay cell is given by

$$H(s) = \frac{K(1 + [1+\alpha]sRC)}{(1 + s2RC)}, \tag{3.22}$$

where K is a DC gain of the CML buffer required for sustained oscillations,

$$\tau_g \approx (1-\alpha)RC. \tag{3.23}$$

Thus, varying the control signal α varies delay τ_g and the resulting oscillation frequencies f_{osc} of VCO.

The CML buffer and multiplier introduce various parasitic capacitances at the input and output terminals of each stage which effect the oscillation frequency. To understand its effect, consider small signal representation of the delay cell $\tau_\alpha(n)$ in presence of parasitic capacitances C_{p1},C_{p2} as seen in Figure 3.38(b). The parasitic capacitances C_{p1} and C_{p2} are expressed as

$$C_{p1} = C_{dB(n)} + C_{gB(n+1)} + C_{gM(n+1)}, \tag{3.24}$$

$$C_{p2} = C_{gM(n+1)}. \tag{3.25}$$

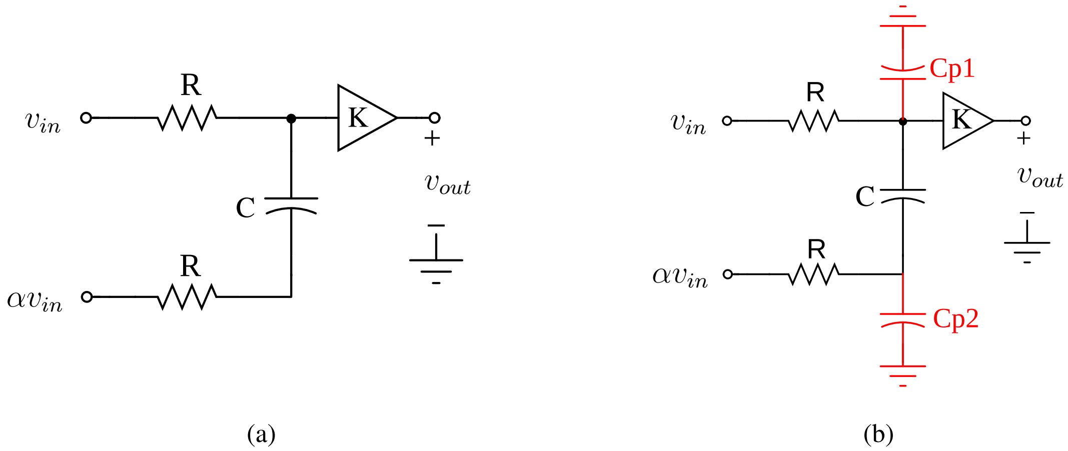


Figure 3.38: Small signal equivalent circuit of the proposed delay cell (a) without and (b) with parasitic capacitances.

The capacitances $C_{dB(n)}$ and $C_{dB(n+1)}$ refers to total drain capacitance of the CML buffer at n^{th} and $(n+1)^{th}$ stage respectively. $C_{dM(n)}$ refers to total drain capacitance of the multiplier of the n^{th} stage. $C_{gB(n+1)}$ refers to total gate capacitance of the multiplier of the $(n+1)^{th}$ stage

The transfer function $H_\alpha(s)$ in presence of C_{p1} and C_{p2} is evaluated to,

$$H(s) = \frac{K(1+s[(1+\alpha)RC + RC_{p2}])}{1+s[2RC+RC_{p1}+RC_{p2}]+s^2[R^2CC_{p1}+R^2CC_{p2}+R^2C_{p1}C_{p2}])}. \tag{3.26}$$

From Eq. 3.26, it is clear that $H_\alpha(s)$ is a 2$^{\text{nd}}$ order system with 2 poles and 1 zero. Hence, the function $H_\alpha(s)$ can be approximated as,

$$H(s) \approx \frac{K(1+s[(1+\alpha)RC + RC_{p2}])}{\Big(1+s[2RC+R(C_{p1}+C_{p2})/2]\Big)\Big(1+s[R(C_{p1}+C_{p2})/2]\Big)} = \frac{K(1+s/z_1)}{(1+s/p_1)(1+s/p_2)}. \tag{3.27}$$

where

$z_1 = 1/[(1+\alpha)RC + RC_{p2}]$ is the variable zero frequency,

$p_1 = 1/[2RC + R(C_{p1}+C_{p2})/2]$ and $p_2 = 1/[R(C_{p1}+C_{p2})/2]$ are the pole frequencies.

If the parasitic capacitances C_{p1} and C_{p2} are related to the capacitor C as $C_{p1} = xC$ and $C_{p2} = yC$, then $H(s)$ can be written as

$$H(s) \approx \frac{K(1+sRC[(1+\alpha+y])}{\Big(1+sRC[2+(x+y)/2]\Big)\Big(1+sRC[(x+y)/2]\Big)} \tag{3.28}$$

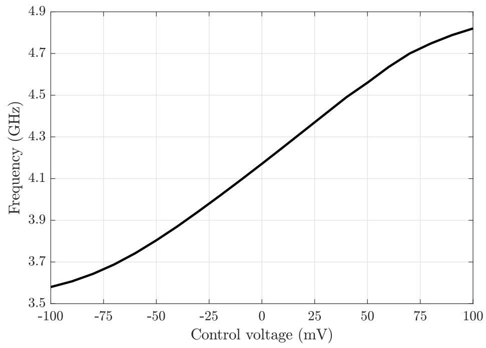
The resulting delay τ_g is approximated to,

$$\tau_\alpha \approx \big[(1+x-\alpha)\big]RC \tag{3.29}$$

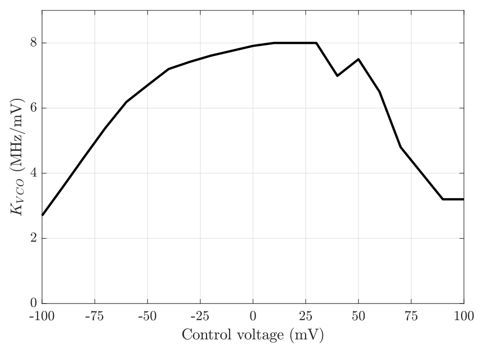
From Eq. 3.29, it can be observed that τ_g increases in presence of parasitic capacitances. Hence, oscillation frequency of VCO reduces.

3.7.2 Results and Discussions

The 3-stage ring VCO with proposed delay element is designed and implemented in TSMC 180-nm technology. The CML delay cell is designed for gain $K = 7$ with R =580 Ω, C=10 fF with bias current I_{BUF}=1.9 mA to maintain required oscillation condition. The differential control voltage $\pm\alpha$ of the Gilbert multiplier is varied from -100 mV to 100 mV over a common mode voltage of 0.85 V. Figure 3.39(a) shows the plot of frequency with the different control voltages. It is observed that VCO frequencies increase from 3.58 GHz to 4.82 GHz when the control signal is varied from -100 mV to 100 mV. The response is linear for the control voltage ranging from -60 mV to 60 mV. This linearity is quantified by the gain of VCO - K_{VCO}, where $K_{VCO} = \partial f_{osc}/\partial\alpha$. The plot of K_{VCO} with differential control voltage is shown in Figure 3.39(b). The gain K_{VCO} varies within 6.2 MHz/mV to 8 MHz/mV for the differential control signal varying from -60 mV to 60 mV. The phase noise plots for different values of α are shown in Figure 3.39(c). For $\alpha = \pm 100\ mV$, the phase noise is approximately -90 dBc/Hz at offset of 1 MHz and with $\alpha = 0$ the simulated phase noise is -86.5 dBc/Hz. The proposed VCO has an average power consumption of 13.32 mW at a supply voltage of 1.8 V. The 3-stage VCO is also designed and implemented in TSMC 65-nm technology. CML delay cell is designed for gain $K = 7$ with R =1.4 $K\Omega$, C=10 fF with bias current I_{BUF}=500 μA to maintain required oscillation condition. The variation of VCO oscillation frequency with the control voltage in 65-nm technology is shown in Figure 3.40(a). The frequencies vary from 4.7 GHz to 8.4 GHz when the external control signal α is varied from -100 mV to 100 mV. But the wide tuning range effects the gain K_{VCO} of the proposed VCO design, which is shown in Figure 3.40(b). The phase noise response across offset frequencies at different control voltages is shown in Figure 3.40(c). The VCO exhibits poor phase noise of -76.76 dBc/Hz at offset of 1 MHz from the oscillation frequency of 8.4 GHz when α=100 mV. The VCO has an the average power consumption of 7.2 mW at the supply voltage of 1.2 V. The proposed VCO design is compared with other existing architectures with CML-based delay cells, and the observations are tabulated in Table 3.8. The performance of VCO is quantified by its figure of merit (FOM) [46], which is



(a)



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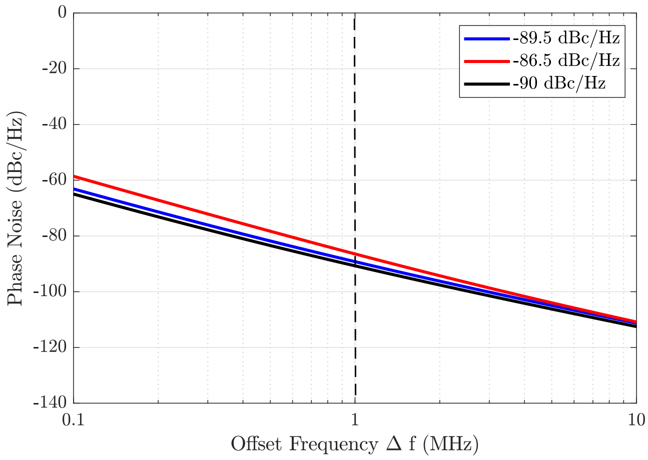


(c)

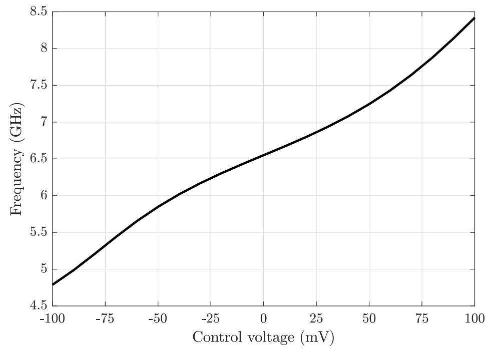
Figure 3.39: Plots for (a) variation of frequency f_{osc} with the control voltage α, (b) variation of gain K_{VCO} with the control voltage α and (c) phase noise at offset of 1 MHz in 180-nm technology.

defined as,

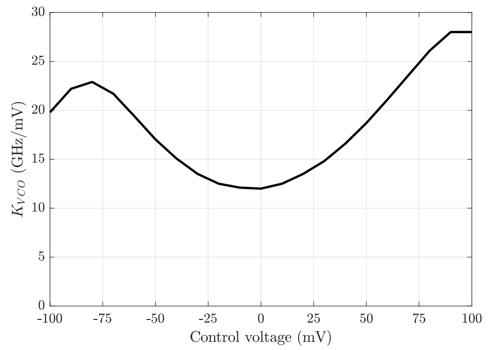
$$FOM = 20\log[f_o/\Delta f] - L(\Delta f) - 10\log(P_{DC}) + 20\log[FTR/10] \tag{3.30}$$

where f_o is the oscillation frequency, Δf is the offset frequency, $L(\Delta f)$ is the phase noise at offset frequency Δf, P_{DC} is the power dissipation in mW and FTR is the frequency tuning range.

The proposed VCO design in 180-nm technology shows increase in tuning range by 53% compared to [12]. The VCO has lower power dissipation compared to that of [45]. The VCO designed in 65-nm technology node has the highest tuning range. From the FOM value calculated according to Eq. 3.30, the proposed VCO shows better performance when compared with other VCO architectures using CML based delay units.



(a)



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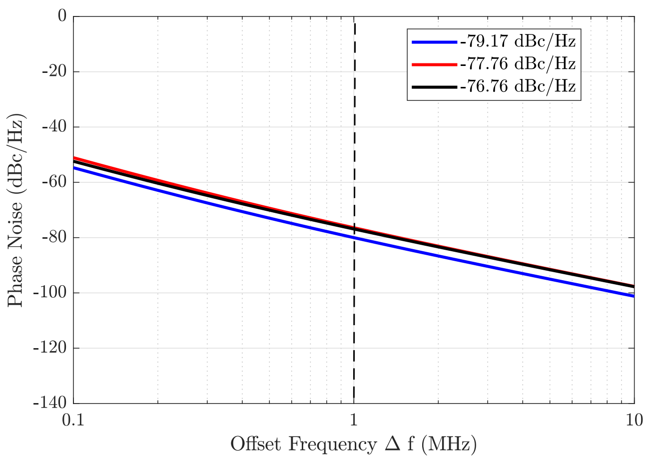


(c)

Figure 3.40: Plots for (a) variation of frequency f_{osc} with the control voltage α, (b) variation of gain K_{VCO} with the control voltage α, (c) phase noise at offset of 1 MHz in 65-nm technology.

Table 3.8: Comparison of proposed ring VCO with existing architectures.

	This work	This work	[12]	[45]
Technology (nm)	180	65	180	180
Frequency range (GHz)	3.58-4.8	4.7-8.4	1.78-2.53	2.2-2.7
Supply voltage (V)	1.8	1.2	1.8	1.8
Phase noise (dBc/Hz @1MHz)	-86.5	-76.76	-92.68	-94.41
Power (mW)	13.3	7.2	26	10.1
FOM (dBc/Hz)	177.41	181.69	159.19	176.04

Chapter 4

Variable-Delay LPF-Pulse Expansion and Compression Circuits

This chapter focuses on the on-chip realization of an expansion circuit for stretching narrow pulses and a compression circuit for compressing wide pulses using a variable delay LPF. The LPF used in these circuits is realized by the proposed 1st order RC circuit with variable C obtained using the Miller effect, which was discussed in Section 3.1 of Chapter 3.

The expansion of high-frequency pulses using LPF is analyzed by applying a narrow Gaussian pulse with half power width of 1 ns to the 1st order RC LPF as shown in Figure 4.1. The

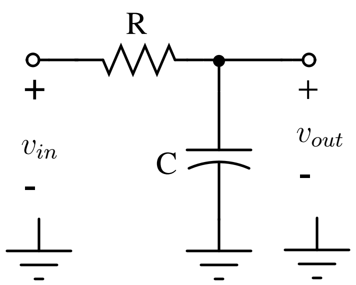


Figure 4.1: 1st order RC LPF.

output of the filter can be observed as the stretched version of the input signal with distortion as shown in Figure 4.2(b). The expansion with distortion is due to frequency-dependent gain and delay of the filter experienced by the input pulse. Hence, for expansion without distortion, the filter should ideally offer increasing frequency-independent delays with constant gain up to required frequency. Similarly, for compression, the filter must exhibit decreasing frequency-independent delays and constant gain.

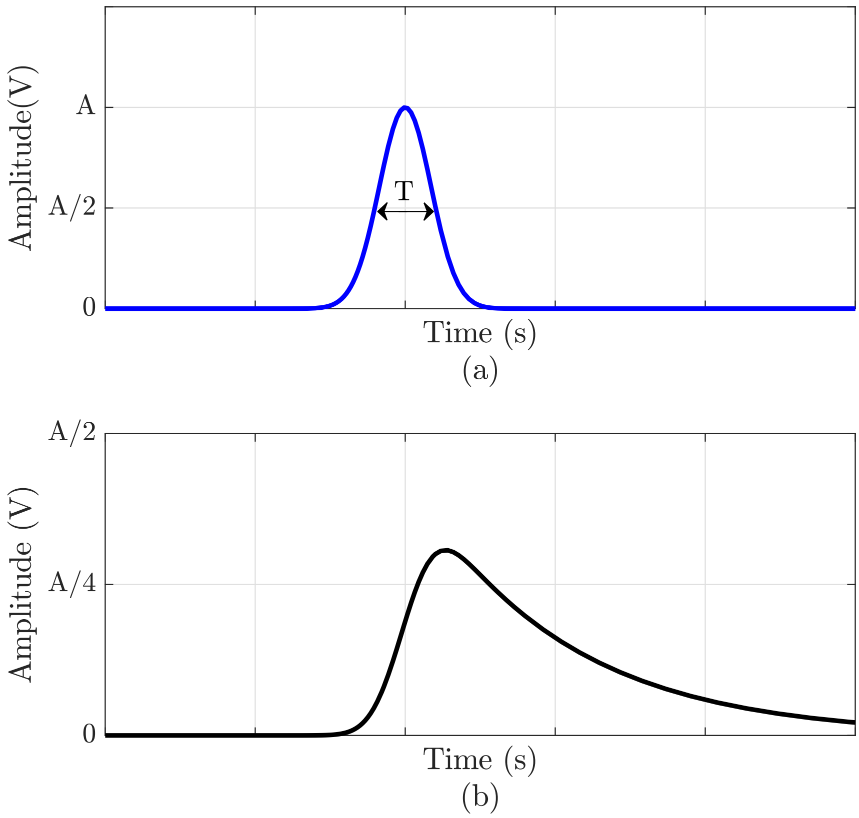


Figure 4.2: (a) Gaussian input with half power width of T s, (b) stretched pulse with distortion on passing through of 1st order RC LPF.

4.1 Proposed Expansion and Compression Circuit

Figure 4.3 shows 1st order RC circuit as variable delay LPF in which the value of C is varied using the principle of Miller effect, which is discussed in Chapter 3. The transfer function of

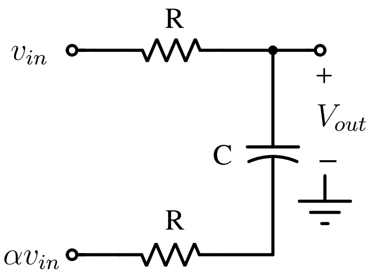


Figure 4.3: 1st order varibale delay LPF.

the circuit $H_\alpha(s)$ and its low frequency group delay τ_α can be shown to be,

$$\begin{aligned} H_\alpha(s) &= \frac{1 + s[1+\alpha]RC}{(1 + s2RC)} \\ \tau_\alpha &\approx (1-\alpha)RC \end{aligned} \tag{4.1}$$

The gain and delay plots for different values of α varying from 1 to -3 is shown in Figure 4.4. It is observed that the group delay of the circuit increases linearly with a minimum magnitude

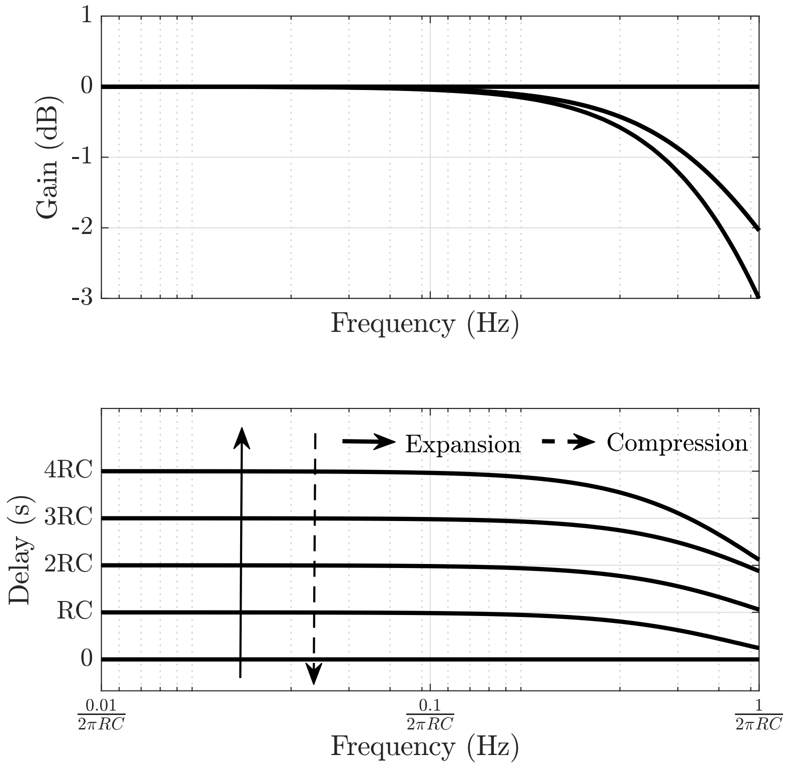


Figure 4.4: Gain and group delay plots of 1-order variable delay RC circuit for different values of α required expansion and compression.

drop in the frequencies of interest as α is decreased from 1 to -3. A continuous time pulse whose bandwidth is within this frequency region will undergo expansion in time if α is a falling ramp. If the control signal α is a rising ramp, then the pulse will undergo compression without distortion.

A single variable delay LPF as an expansion (compression) circuit would stretch (compress) the input pulse by a very small factor. So, to improve the expansion (compression) factor N stages of expansion (compression) circuits are cascaded as shown in Figure 4.5. The required control signal which is a linearly varying ramp signal for the expansion (compression) process

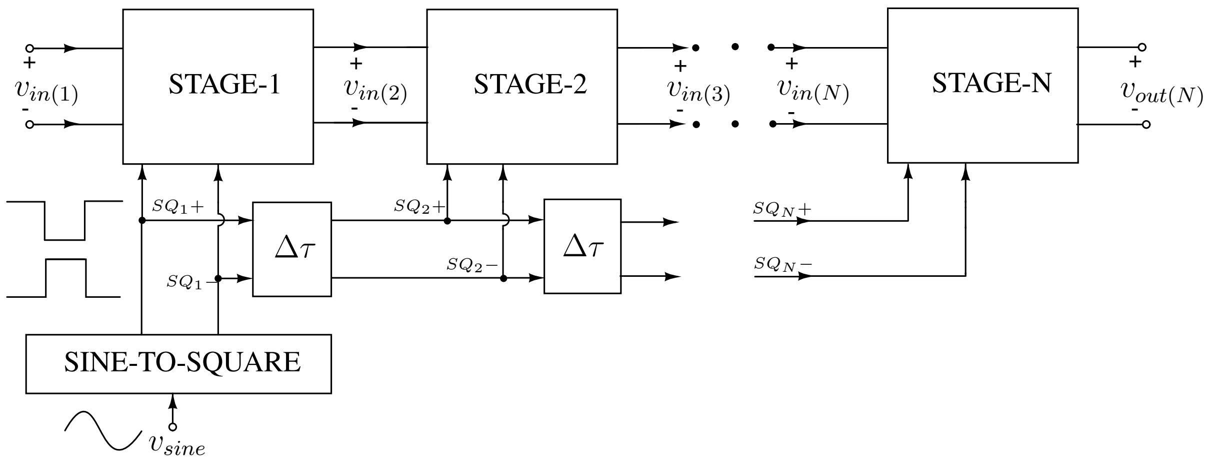


Figure 4.5: Cascaded N stages of expansion (compression) blocks for enhancement of stretch (compression) factor.

is generated by an off-chip sinusoidal signal V_{sine} fed to sinusoidal to differential square wave converter block SINE-TO-SQUARE. The differential square wave is then internally converted to a ramp signal.

A n stage expansion (compression) circuit consists of variable delay LPF and ramp generator unit RAMPGEN as shown in Figure 4.6. The desired differential ramp signal $\pm\alpha$ for

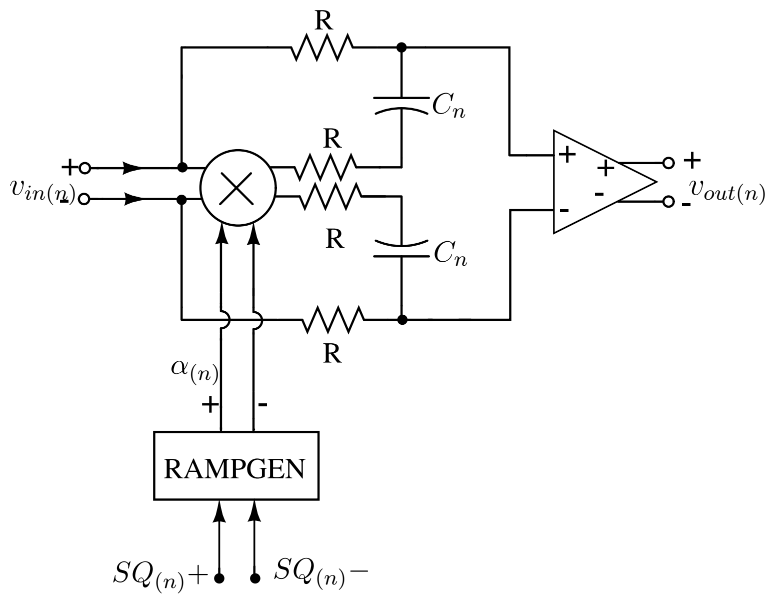


Figure 4.6: n^{th} stage of an expansion (compression) circuit in differential architecture.

expansion (compression) is obtained from differential square signals SQ$\pm$. The ramp signals for succeeding expansion (compression) stages must be delayed by a certain delay $\Delta\tau$ (as seen in Figure 4.7 to ensure the stretched (compressed) pulse at each stage is within the ramp signal range.

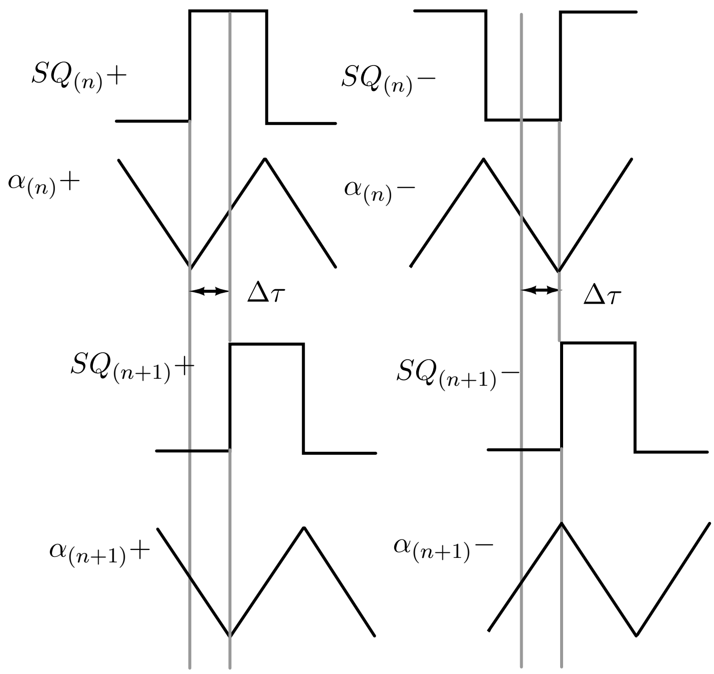


Figure 4.7: Control signals of (n+1)th stage delayed by $\Delta\tau$.

4.1.1 Implementation details

Figure 4.8 shows the block level implementation of variable delay LPF used in an n^{th} stage expansion (compression) circuit using transconductor G_B and multiplier G_{MUL}. The transistor level implementation of these blocks is similar to that discussed in the Section 3.3 of Chapter 3. The implementation details of the ramp generator block RAMPGEN are explained in Appendix A. A sinusoidal input v_{sine} is AC coupled through off-chip capacitor to a self biased amplifier, A_I, in which feedback resistor R across CMOS inverter I_0 ensures A_I is biased at the switching threshold, $V_T = V_{DD}/2$ of the inverter I_0. The output of A_I is passed through the cascaded inverters $I_{1,2,3}$ to obtain a rail to rail square wave. The differential square signals are obtained by adding an additional inverter I_x to the output of I_1. The delay between the signals $SQ_{(n)}+$ and $SQ_{(n)}-$ is controlled by adjusting the size of I_x. The latches connected between the voltage lines $SQ+$ and $SQ-$ ensure rail to rail differential square wave has 50% duty cycle [47]. The differential square signals are delayed by delay unit $\Delta\tau$.

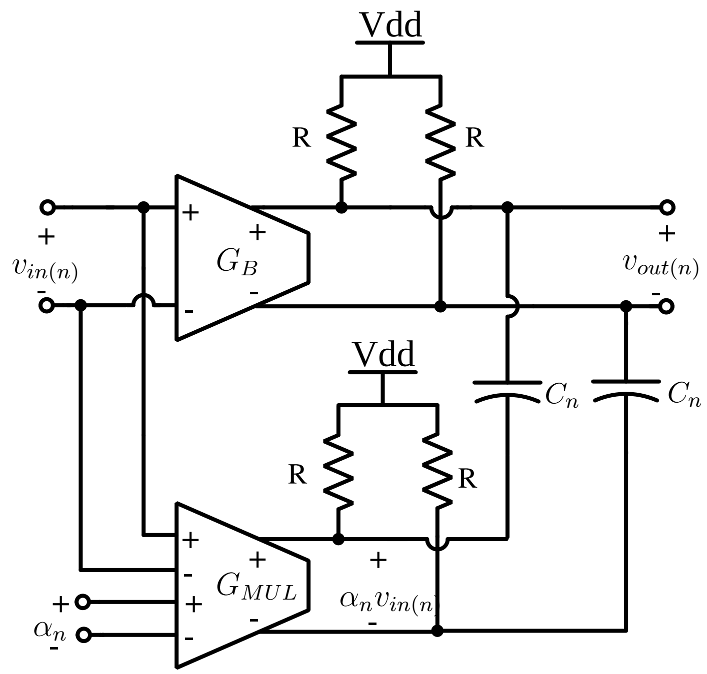


Figure 4.8: Block level representation of variable delay circuit using controlled sources G_B and G_{MUL}.

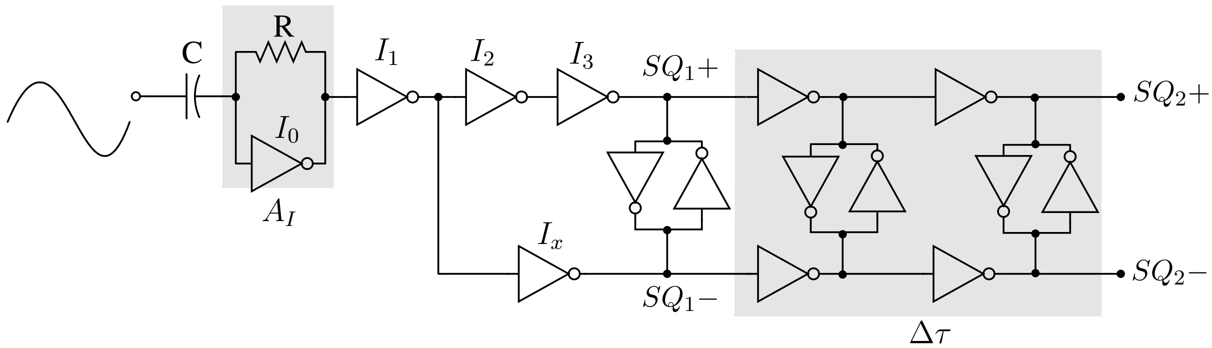
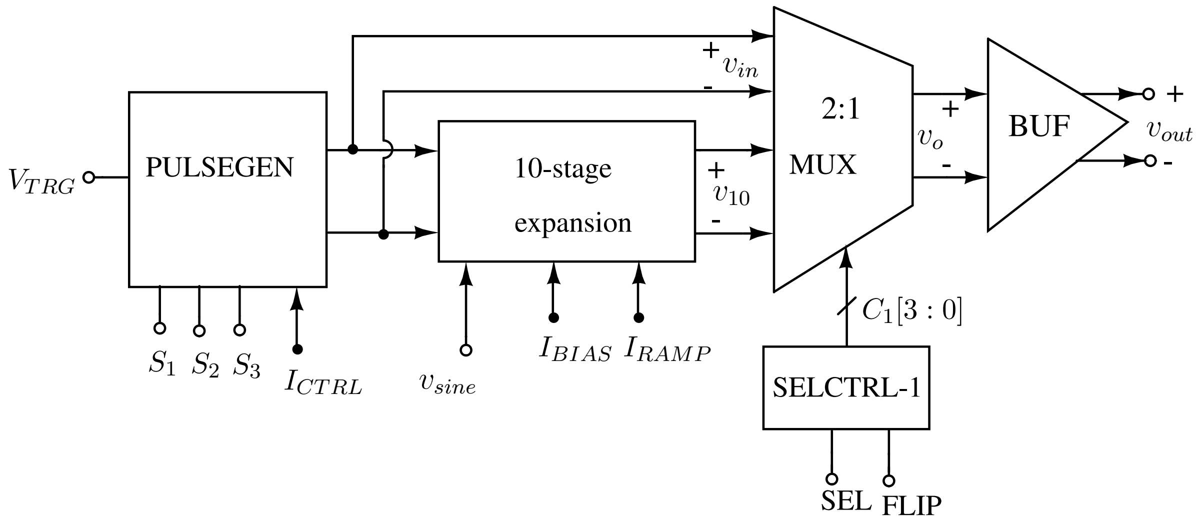


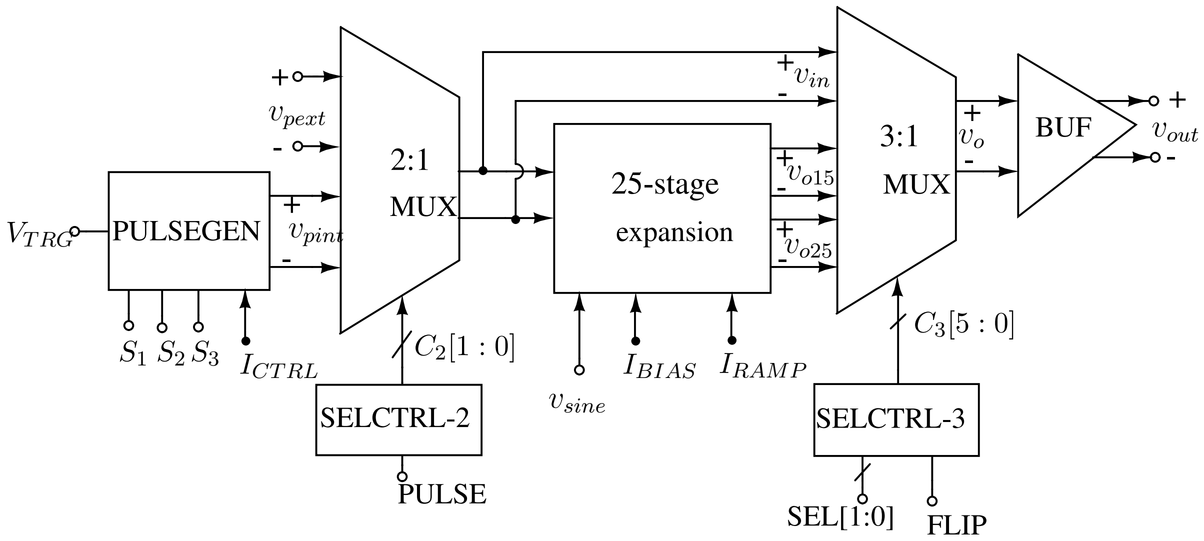
Figure 4.9: Internal circuit diagram of SINE-TO-SQUARE block.

4.2 10 and 25-stage Pulse Expansion Circuits

A 10 and 25-stage pulse expansion circuits are designed and fabricated in UMC 180-nm technology. Figure 4.10a and Figure 4.10b shows simplified architectures of the test setup for verifying the operation of 10 and 25 stage expansion circuits. A Gaussian pulse from the pulse generator block PULSEGEN is applied to the 10 and 25-stage expansion circuits. The PULSEGEN block generates an approximate Gaussian pulses with different pulse widths (controlled by $S_{1,2,3}$) and amplitudes (controlled by current I_{CTRL}). The input pulse from the PULSEGEN and the stretched pulses from the expansion circuit are measured through a buffer BUF with output impedance of 50 Ω. The input and output pulse from 10-stage expansion circuit are se-



(a)



(b)

Figure 4.10: Simplified architecture of test setup for testing (a) 10-stage and (b) 25-stage pulse expansion circuits.

lected for measurement using 2:1 MUX with control bits $C_1[3:0]$ obtained from SEL, FLIP digital inputs and SELCTRL-1 block. In testing of 25-stage expansion circuit, the input and output pulses from 15$^{\text{th}}$, 25$^{\text{th}}$ are selected for measurement using 3:1 MUX with control bits $C_3[5:0]$ obtained from SEL, FLIP digital inputs and SELCTRL-3 block. These signals are measured with flipped polarities with FLIP pin, to cancel the undesired effect of input-output feedthrough [40]. An external pulse v_{pext} generated off-chip can also used as input to the 25-stage expansion circuit using 2:1 MUX with control bits $C_2[1:0]$ obtained from PULSE input and SELCTRL-2 block.

Figure 4.11 shows transistor level schematic of 3:1 MUX. The functional table of SELCTRL-3 with digital inputs SEL[1:0], FLIP and outputs $C_3[5:0]$ is tabulated in Table 4.1.

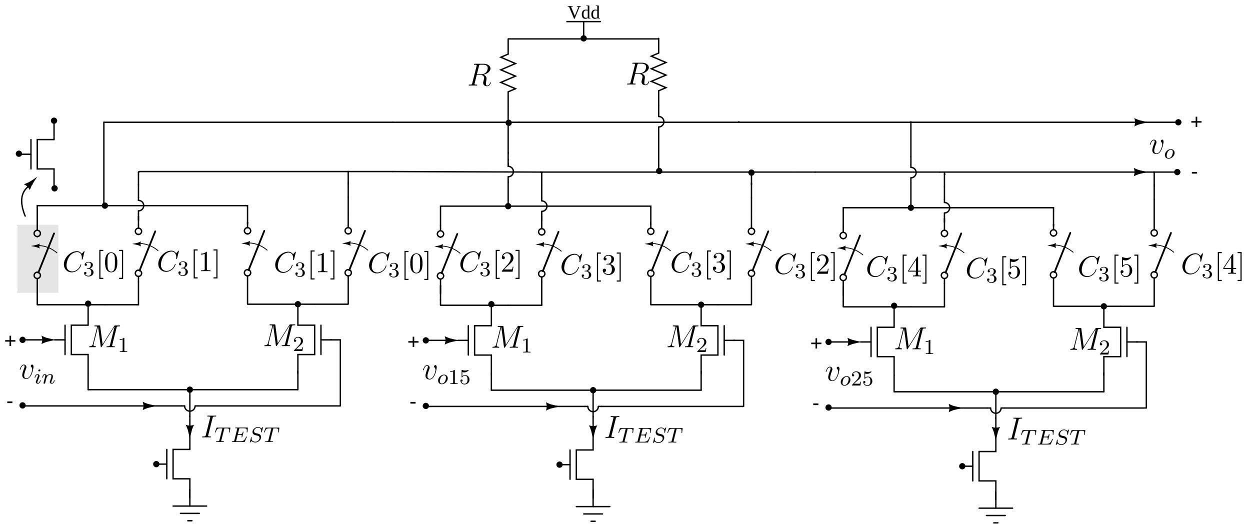


Figure 4.11: Transistor level schematic of 3:1 MUX.

Table 4.1: Functional table of SELCTRL-3 for generating signals $C_3[5:0]$.

SEL[1]	SEL[0]	FLIP	$C_3[5]$	$C_3[4]$	$C_3[3]$	$C_3[2]$	$C_3[1]$	$C_3[0]$
0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	1	0
0	1	0	0	0	0	1	0	0
0	1	1	0	0	0	1	0	0
1	0	0	0	0	0	1	0	0
1	0	1	0	0	0	1	0	0
1	1	0	0	1	0	0	0	0
1	1	1	1	0	0	0	0	0

4.2.1 On-chip Pulse Generator

Pulse with approximate Gaussian shape is generated from 1-order CR differentiator and 1-order RC integrator circuit as shown in Figure 4.12 [7]. The step input v_{step} is passed through differentiator, which passes high frequency signal component and attenuating low frequencies. The differentiated signal V_d is then applied to integrator, a LPF for generating approximated

Gaussian pulse V_p. The output pulse shape is improved by passing V_d through more than one integrator blocks. Typically 4 integrator stages would generate acceptable Gaussian shaped pulse. Figure 4.13 shows internal architecture of PULSEGEN for generating pulses with differ-

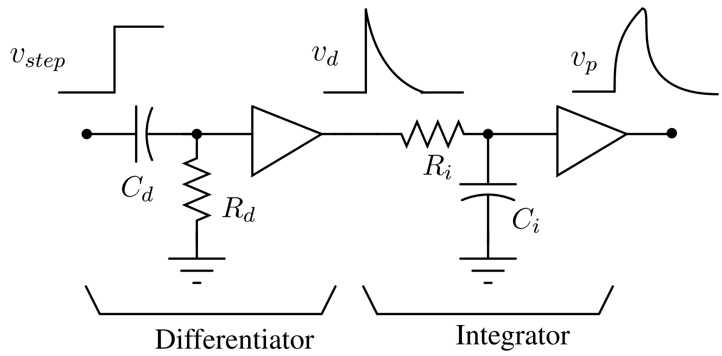


Figure 4.12: Gaussian pulse generation scheme from 1-order differentiation and integrator.

ent widths and amplitudes. A low frequency sqaure wave input V_{TRG} is applied to differential square generator DSQGEN to obtain differential square signals SQ$\pm$. These rail-to-rail signals are converterd to current-mode logic (CML) level signals $\pm$ V_{step} using CMOS-TO-CML converter. The differential step signals $\pm V_{step}$ vary between voltages V_{DD} and $V_{DD} - (I_{CTRL}/2)R$. The amplitude of resulting output pulse is adjusted by varying bias the current I_{CTRL}. The width of Gaussian pulse is controlled by varying time constants of differentiator and integrators. The time constant of differentiator block DIFF is varied by changing R with switches S_1, S_2 and S_3. The differentiator time constant τ_d is high when all switches are closed and is low when all the switches are in open state. These controlled switches are implemented using NMOS transistors as shown in Figure 4.14. Similarly, the time constant of integrator block INT is varied by switching capacitor values as shown in Figure 4.15. The total capacitance value is high when

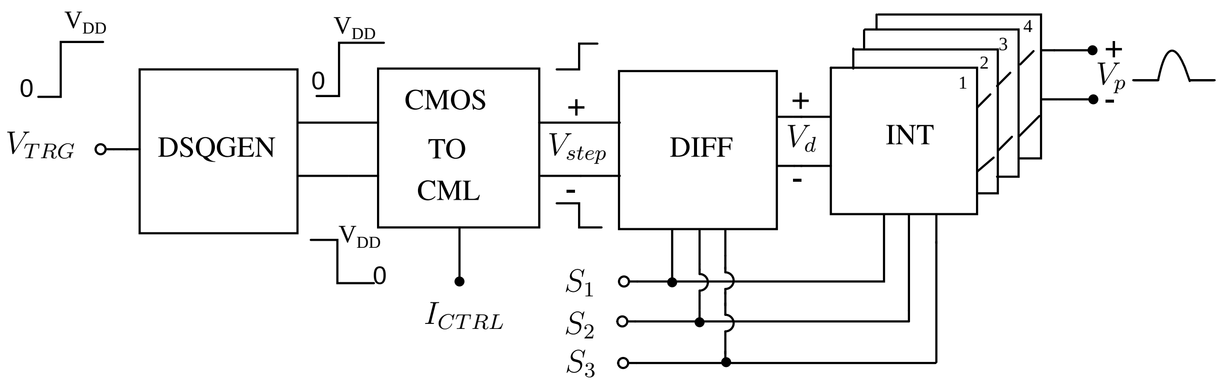


Figure 4.13: Internal architetcure of PULSEGEN for generating Gaussian pulses of different amplitudes and widths.

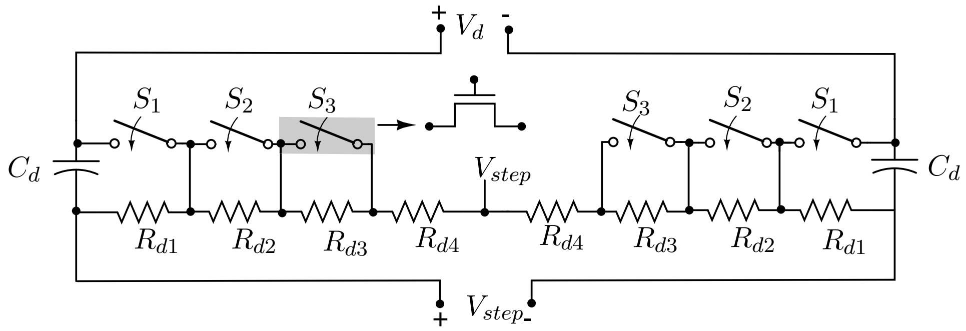


Figure 4.14: Differentiator DIFF with different time constants for changing pulse widths.

all the switches are closed. The integrator has low time constant when all switches are open. These controlled switches in integrator are implemented using PMOS transistors.

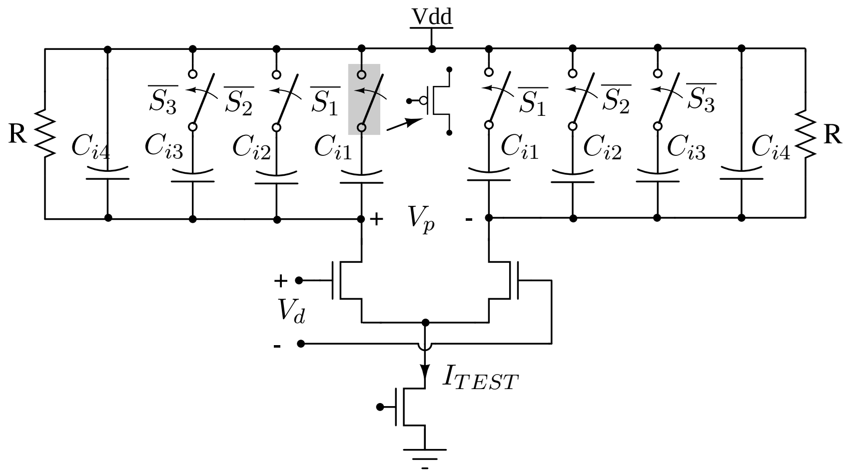
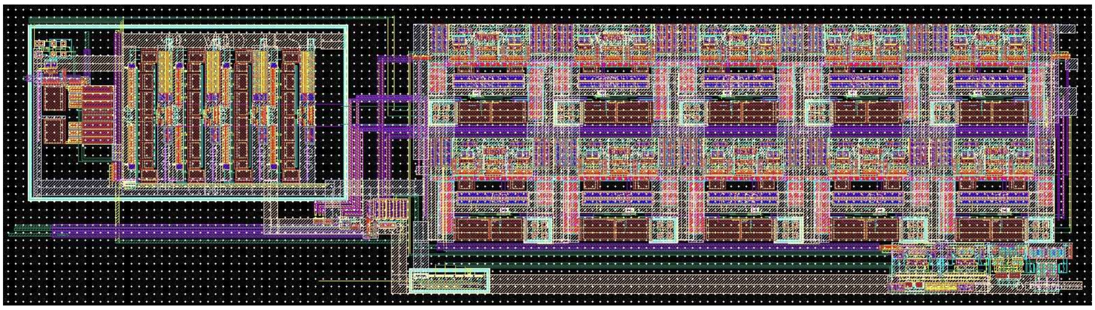


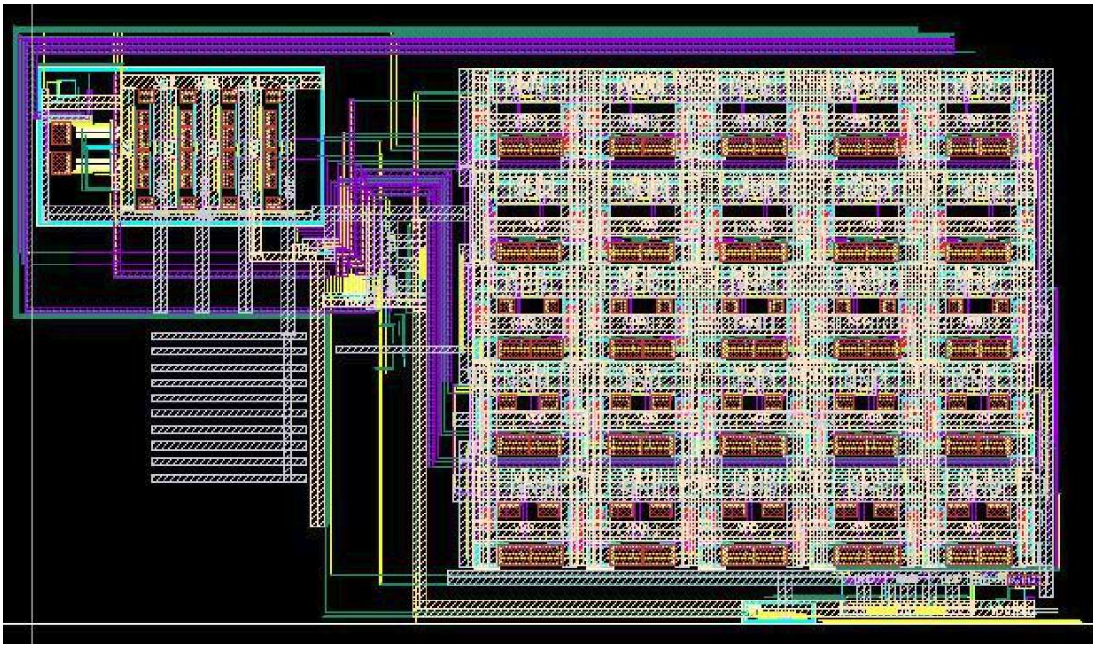
Figure 4.15: Integrator INT with different time constants for changing pulse widths.

4.2.2 Post-Layout Simulation Results

The 10 and 25-stage pulse expansion circuits are designed in UMC 180-nm CMOS technology with supply voltage of 1.8 V. Figure 4.16(a) and Figure 4.16(b) shows layouts of 10 and 25 stage pulse expansion circuits. 10-stage expansion circuit occupies an area of 140 μm $\times$ 426 μm and 25-stage expansion circuit occupies area of 362 μm $\times$ 426 μm. The value of resistors R in G_B and G_{MUL} units of each expansion stage are chosen to 400 Ω. The value of C for every 5 stages is kept constant for design simplicity. In 10-stage expansion circuit, the capacitors $C_{1,2}$ are chosen to 25 fF and 40 fF. Similarly, in 25-stage expansion circuit, the capacitors C_{1-5} are chosen to 25 fF, 40 fF, 60 fF, 85 fF and 110 fF. Each G_B and G_{MUL} blocks are biased with a bias current of I_{BIAS}=1 mA resulting in common mode voltage V_{CM}=1.6 V. With an external



(a)



(b)

Figure 4.16: Layout of (a) 10-stage (b) 25-stage expansion circuits with PULSEGEN and other blocks.

sinusoidal signal v_{sine}=100 MHz and bias current I_{RAMP}=30 μA, the differential control signals $\pm\alpha$ vary linearly from $\pm$ 0.4 with common mode voltage of 1.2 V. A low frequency square signal V_{TRG} of 5 MHz is applied to generate Gaussian pulse from PULSEGEN. The integrators in PULSEGEN block are biased with current I_{TEST}=1 mA and current I_{CTRL} is varied from 0.8 mA to 1.25 mA to obtain pulses with different amplitudes.

Gaussian pulse with half power pulse width of 350 ps (3-dB frequency of 1 GHz) with the input amplitude of 96 mV from pulse generator is applied as test input to expansion block with 10 stages. The output is stretched version of input pulse with a stretch factor of 1.55. The error between ideal stretched and actual samples vary from 7 mV to -5 mV as shown in Figure 4.17. The root mean square error (RMSE) between ideal and actual samples is 1.5 mV.

A similar simulation was done with 25-stage expansion circuit. Here Gaussian pulse with

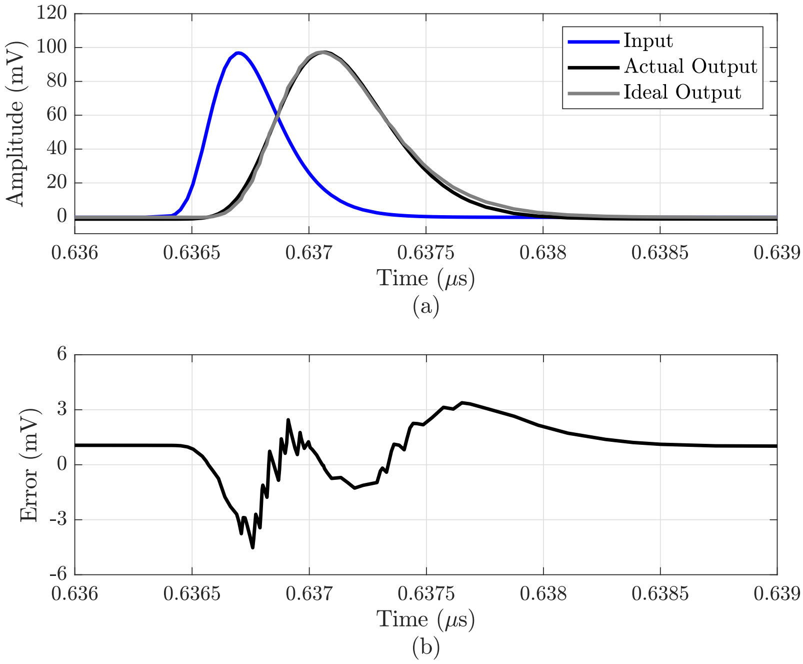


Figure 4.17: (a) Stretched Gaussian pulse of 534.75 ps with a stretch factor of 1.55 for an input pulse width of 350 ps applied to 10 stage expansion circuit designed in UMC 180-nm technology, (b) Plot of error between the actual output samples and ideal output samples.

half power pulse width of 500 ps with the input amplitude of 80 mV from pulse generator is applied as test input to expansion block with 15 stages. The output is stretched version of input pulse with stretch factor of 2.2. The error between ideal stretched and actual samples vary from 4 mV to -2 mV as shown in Figure 4.18. The RMSE between ideal and actual samples is 2.7 mV.

4.3 15-stage Expansion Circuit

A 15-stage pulse expansion circuit is designed in TSMC 65-nm technology. Figure 4.19 shows simplified architecture of the test setup for verifying the operation of 15 stage expansion circuit. A Gaussian pulse from the pulse generator block PULSEGEN as a test input is applied to the 15-stage expansion circuit. The PULSEGEN block generates an approximate Gaussian pulses with different pulse widths (controlled by $S_{1,2,3}$) and amplitudes (controlled by current I_{CTRL}). The input pulse from the PULSEGEN and the stretched pulses from the expansion circuit are measured through a buffer BUF with output impedance of 50 Ω. The input and output pulse

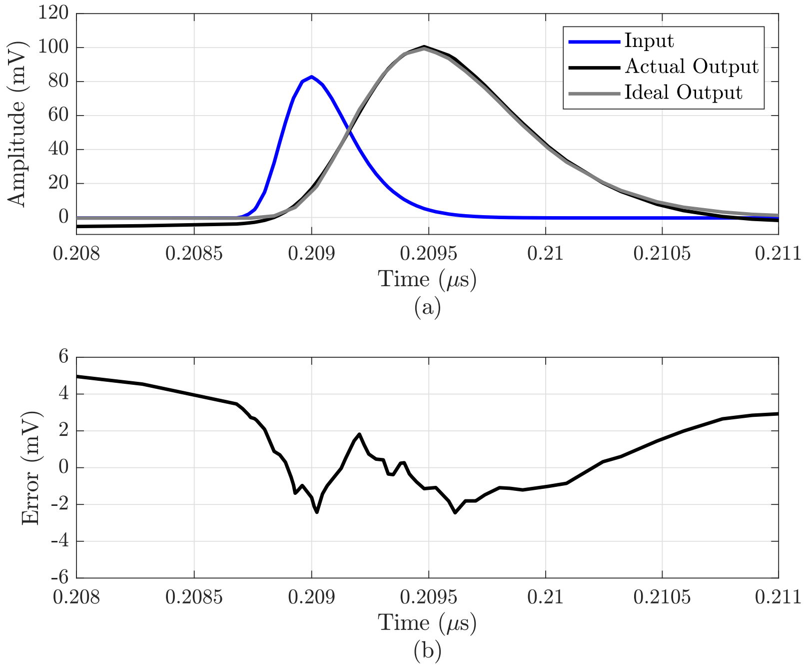


Figure 4.18: (a) Stretched Gaussian pulse of 1.1 ns with a stretch factor of 2.2 for an input pulse width of 500 ps applied to 25 stage expansion circuit designed in UMC 180-nm technology, (b) Plot of error between the actual output samples and ideal output samples.

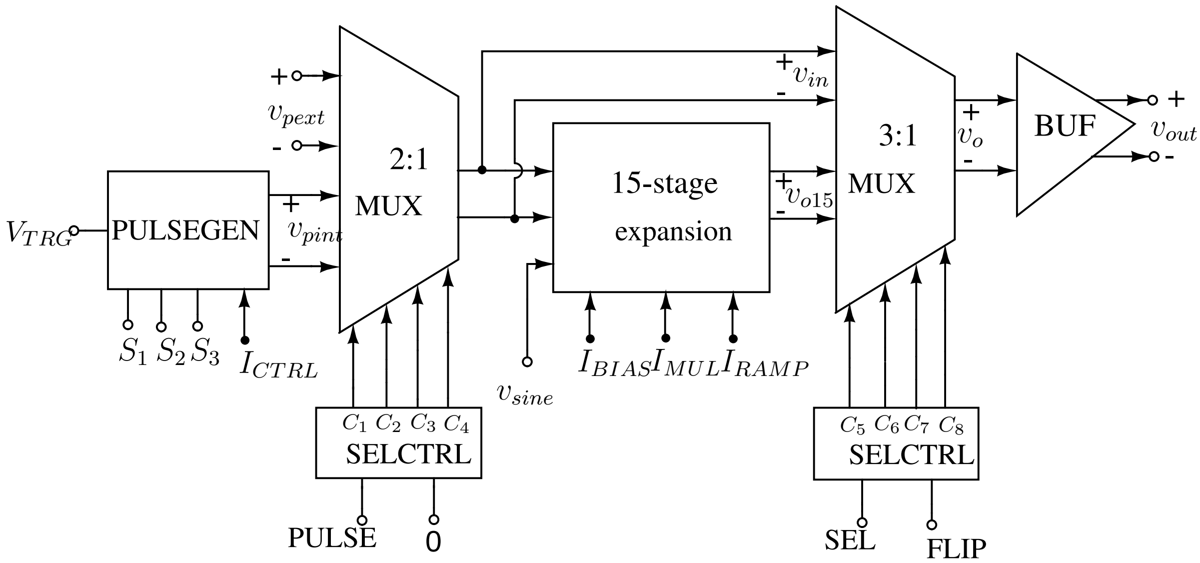


Figure 4.19: Simplified architecture of test setup for testing 15-stage pulse expansion circuit.

from 15-stage expansion circuit are selected for measurement using 2:1 MUX with control signals $C_{5,6,7,8}$ obtained from SEL, FLIP digital inputs and SELCTRL block. These signals are measured with flipped polarities with FLIP pin, to cancel the undesired effect of input-output feedthrough. An external pulse v_{pext} generated off-chip can also used as input to the 15-stage expansion circuit using 2:1 MUX with control signals $C_{1,2,3,4}$ obtained from PULSE input and SELCTRL block. The transistor level implementations details of 2:1MUX, SELCTRL and BUF is similar to that explained in Section 3.5 of Chapter 3.

4.3.1 Post-Layout Simulation Results

The 15-stage pulse expansion circuit is designed in TSMC 65-nm CMOS technology with supply voltage of 1.2 V. Figure 4.20 shows the layout of 15 pulse expansion circuit with PULSEGEN and other circuits, in which 15-stage expansion circuit occupies an area of 315 $\mu m\times$ 313 μm. The value of resistors R in G_B and G_{MUL} units of each expansion stage are chosen to

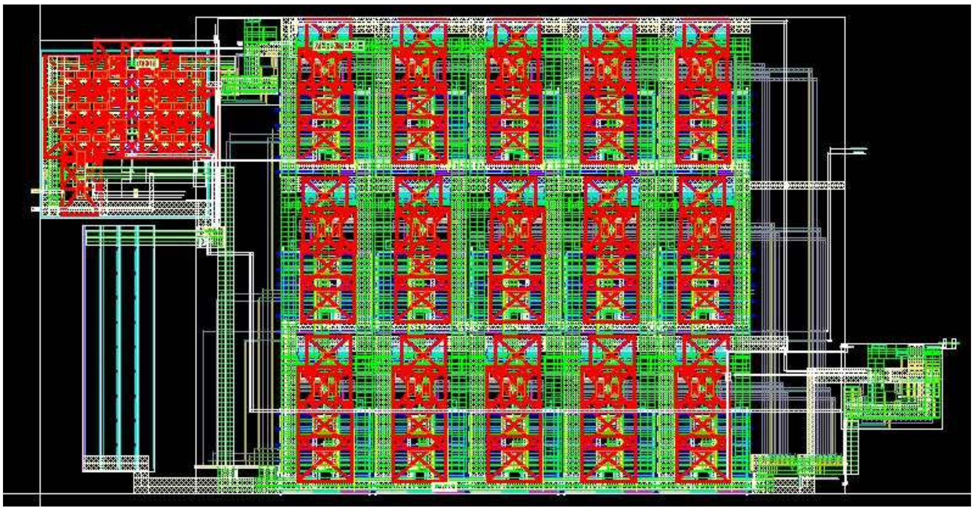


Figure 4.20: Layout of 15-stage expansion circuits with PULSEGEN and other blocks.

400 Ω. The value of C for every 5 stages is kept constant for design simplicity and the value of $C_{1,2,3}$ is chosen to 30 fF, 35 fF and 40 fF. Each G_B and G_{MUL} units in expansion circuit are biased with currents I_{BUF}=500 μA and I_{MUL}= 500 μA, resulting in common mode voltage V_{CM}=1 V. With an external sinusoidal signal v_{sine}=125 MHz and bias current I_{RAMP}=50 μA, the differential control signals $\pm\alpha$ vary linearly from $\pm$ 0.2 V with a common mode voltage of 0.8 V. A low frequency square signal V_{TRG} of 5 MHz is applied to generate Gaussian pulse from PULSEGEN. The integrators in PULSEGEN block are biased with current I_{TEST}=500 μA and the current I_{CTRL} is varied from 100 μA to 500 μA to obtain pulses with different amplitudes.

Gaussian pulse with half power pulse width of 350 ps (3-dB frequency of 1 GHz) with the input amplitude of 50 mV from pulse generator is applied as test input to the expansion block. Post-layout simulation result shows that the output is a stretched version of input pulse with

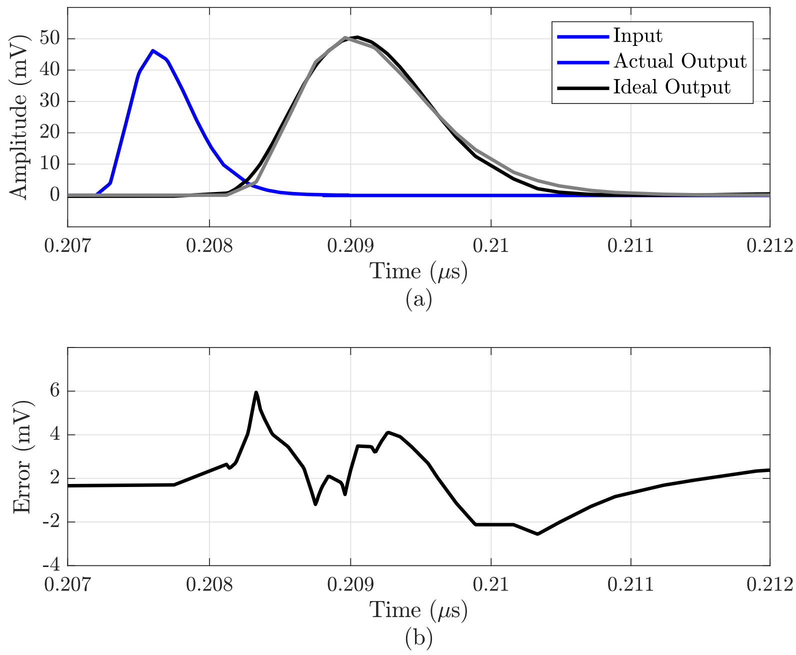


Figure 4.21: (a) Stretched Gaussian pulse of 787.5 ps with a stretch factor of 2.25 for an input pulse width of 350 ps applied to 15 stage expansion circuit designed in TSMC 65-nm technology, (b) Plot of error between the actual output samples and ideal output samples.

stretch factor of 2.25 as seen in Figure 4.21(a). Error between the ideal and actual output samples within the half power pulse width vary from -2 mV to 1.2 mV as shown in Figure 4.21(b). The RMS error between the ideal and actual samples is calculated to 1.5 mV.

4.4 15-stage Compression Circuit

A 15-stage compression circuit is designed and implemented in TSMC 65-nm CMOS technology with supply voltage of 1.2 V. Figure 4.22 shows the layout of 15-stage compression circuit which occupies an area of 318 μm $\times$ 325 μm. The value of resistors R in G_B and G_{MUL} units of each compression stage are chosen to 400 Ω. The value of C for every 5 stages is kept constant for design simplicity and the value of $C_{1,2,3}$ is chosen to 200 fF, 90 fF and 35 fF. Each G_B and G_{MUL} units in compression circuit are biased with currents I_{BUF}=400 μA and I_{MUL}= 300 μA resulting in common mode voltage V_{CM}=1 V. With an external sinusoidal signal v_{sine}=50 MHz and bias current I_{RAMP}=50 μA, the differential control signals $\pm\alpha$ vary linearly from $\pm$ 0.2 V with a common mode voltage of 0.8 V.

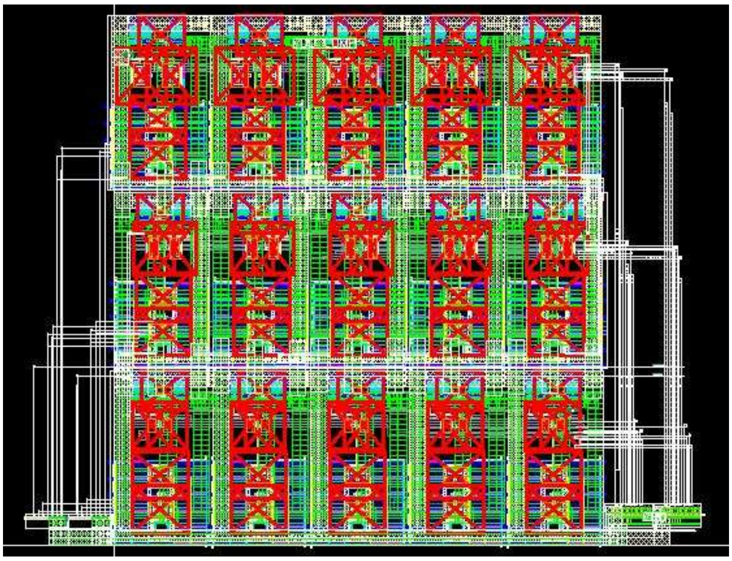


Figure 4.22: Layout of 15-stage compression circuit.

An ideal Gaussian pulse with half power pulse width of 3.32 ns with the input amplitude of 50 mV from pulse generator is applied as test input to compression block. Post-layout simulation result shows that the output is compressed version of input pulse with compression factor of 0.47 as seen in Figure 4.23(a). The error between the ideal and actual compressed pulses within the half power pulse width vary from -2.45 mV to 0.488 mV as shown in Figure 4.23(b). The RMSE between the ideal and actual pulses is calculated to 1.05 mV. The 15-stage compression circuit has a power dissipation of 30 mW.

4.5 Discussions

Table 4.2 summarizes and compares different pulse expansion techniques with the proposed method. The photonic-based method [48], [37] requires lengthy optical fibers in the range of kilometers to obtain a stretch factor of 2-3. In addition, there is great difficulty in obtaining high rate quadrature phase modulation to generate a chirp carrier. Pulse expansion using on-chip dispersive filters [39], requires high linearity in generation of chirp carrier from VCO of 50 GHz. However, the proposed technique does not require any optical fibers or generation of chirp carrier signals. This method does not require an additional clock generation network for the time-interleaved sampling process, as in the case of the sampling-based method [33]. Compared to expansion from higher-order variable bandwidth filter approach [13], the proposed technique need not require the exact information on the arrival of the input pulse. The variable

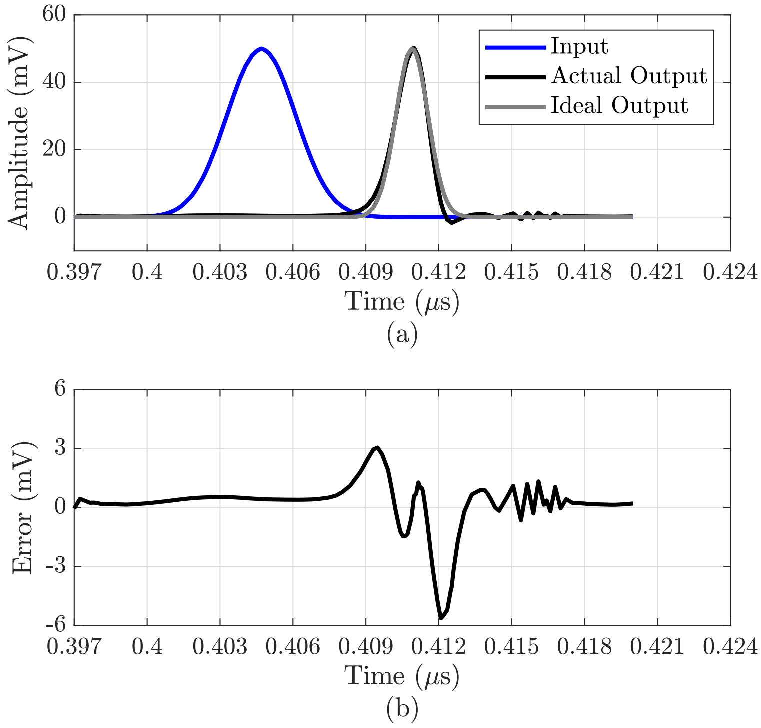


Figure 4.23: (a) Compression of the input Gaussian pulse with compression factor of 0.47, (b) error between actual and ideal output pulses.

Table 4.2: Comparison of proposed pulse expansion technique with existing architectures.

	This work	This work	[13]	[33]	[39]	[48]
Techniques	Variable delay LPF	Variable delay LPF	Variable bandwidth filter	Sampling	DDL	Photonic
Technology (nm)	65	180	130	65	130	Discrete
Carrier/clock	Not required	Not required	Not required	Sampling clock	Chirp carrier from VCO	Optical chirp carrier
Bandwidth (GHz)	1	1	0.87	3.75-4.25	1	5
Stretch factor	2.25	2.2	1.8	400	2	10
Power(mW)	36	110	350.61	170*	-	-

* External clock of 20 GHz is used and clock generation power not included.

delay LPF based expansion circuit has a better stretch factor and lower power consumption compared to [13].

Chapter 5

Variable-delay APF- TTD

This chapter focuses on a G_m-C based variable delay APF as a TTD element. The APF group delay is varied by changing its bias current to modulate its Gm for different delay settings, which avoids the need for high quality varactors. The proposed method emphasizes on designing low-order filters to avoid the usage of large range of capacitance values for design simplicity and reduced power consumption. Identical variable Gm APFs are cascaded to extend the delay range.

5.1 Proposed technique

Figure 5.1 shows the block level architecture of proposed TTD element with 3 stages of variable delay 3-order APFs preceded by an anti-aliasing filter. A 3[rd] order APF is realized from singly

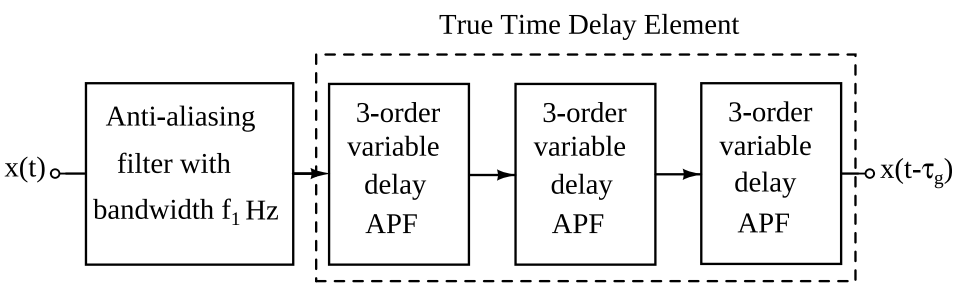
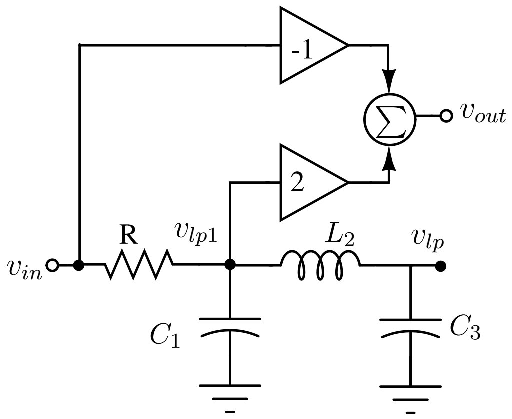
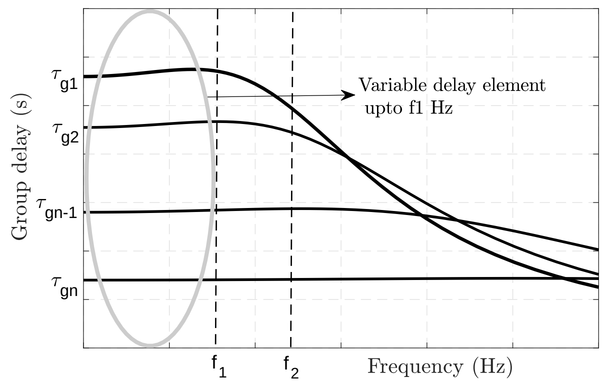


Figure 5.1: Block level diagram of proposed TTD element with anti-aliasing filter.

terminated LC ladder LPF is shown in Figure 5.2(a) [30]. The APF maintains a constant group delay τ_{g1} throughout the frequency f_1 (as seen in Figure 5.2(b)), where f_1 is the 3-dB cut-off frequency of the LPF. This filter architecture serves as a delay element, which introduces a time delay of τ_{g1} for the incoming signal of the filter. Input signal frequencies within the cut-off frequency of the filter undergo a delay of τ_{g1} without any magnitude droop. The filter bandwidth, f_1 can be increased to frequency f_2 by reducing the L and C values. This decreases the τ_g of the filter, maintaining a constant group delay up to frequency f_2. Thus, systematically deceasing the values of L and C, decreases the group delay of the filter from τ_{g1} to $\tau_{gn}\{\ \tau_{g1}>\tau_{g2}>\cdots>\tau_{gn}\}$ as shown in Figure 5.2(b). Thus, the APF can be used as a variable delay element which delays the input signal frequencies up to f_1 Hz from τ_{g1} to τ_{gn}. The delay range of filter, $\Delta\tau_g$ is expressed as $\tau_{g1} - \tau_{gn}$. The frequency f_1 is the maximum bandwidth across which APF behaves as a TTD element. An anti-aliasing filter with a cut-off frequency of f_1 is added before the delay element. This pre-filtering process bandlimits the incoming signal to f_1 Hz. This ensures that the overall bandwidth is constant at f_1 Hz for all delay settings. In general, any number of



(a)



(b)

Figure 5.2: (a) 3rd order APF realized from singly terminated LPF,(b) APF as TTD element up to f_1 Hz.

variable delay APFs with a delay range of $\Delta\tau_g$ can be cascaded to improve the delay range. But the parasitic capacitances at summing circuit decides the limit on number of stages, which will be discussed in the implementation part of the chapter.

5.2 Implementation Details

A lower order Bessel APF of order n= 3, is used as a basic delay element, as it maintains constant group delay within the bandwidth of the filter [49]. The APF is realized from a 3rd order singly terminated Bessel LPF (LPF) [30]. The driving point impedance function $Z_{11}(s)$

and low frequency group delay, τ_{gLP} of the LPF is given as,

$$Z_{11}(s) = \frac{s^2L_2C_3 + 1}{s^3RC_1L_2C_3 + sR[C_1 + C_3]}, \tag{5.1}$$

$$\tau_{gLP} = R[C_1 + C_3]. \tag{5.2}$$

The APF output, v_{out} is expressed as $v_{out} = 2v_{lp1} - v_{in}$ and the resulting group delay of APF, τ_{gAP} in terms of τ_{gLP} is expressed as, $\tau_{gAP} = 2\tau_{gLP}$.

The values C_1, L_2, and C_3 of the LPF are evaluated from the Bessel 3rd order polynomial, $D(s) = 15s^3 + 6s^2 + 15s + 1$ [49]. The driving point impedance function, $Z_{11}(s)$ is expressed as $Z_{11}(s) = D_e(s)/D_o(s)$, where $D_e(s)$ and $D_o(s)$ are the even and odd order polynomials of $D(s)$. Using continuous long division process the values of C_1, L_2 and C_3 are evaluated as 1F, 3H and 5F respectively for the resistance, $R = 1\Omega$ [50]. This prototype filter has a cut-off frequency of f_o Hz. The frequency scaling process shifts the frequency f_o of the prototype filter to the desired frequency [49].

For the complete on-chip implementation of the delay element, the LPF is implemented using its G_m-C equivalent as shown in Figure 5.3. All the operational transconductor amplifiers (OTAs) are designed with the same values of transconductance, Gm. Figure 5.4 shows the

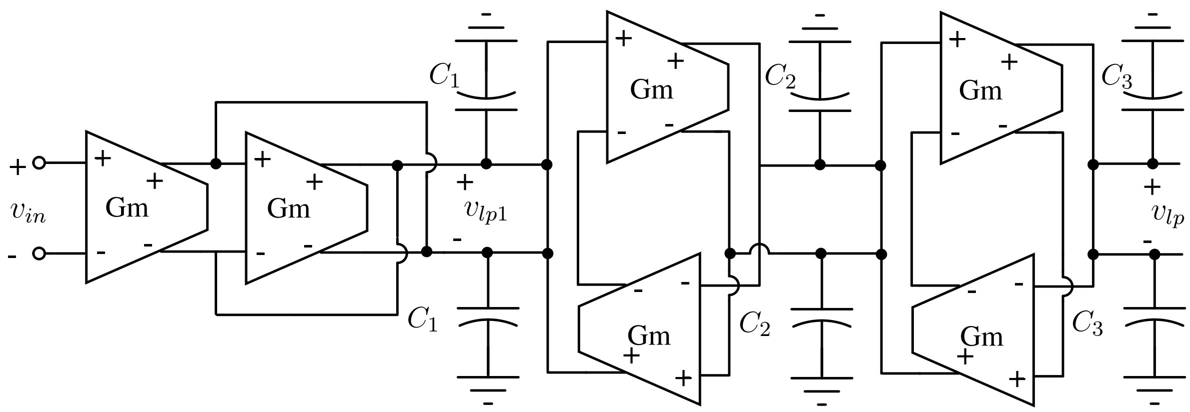


Figure 5.3: Gm-C implementation of 3rd order LPF with differential architecture.

transistor level implementation of the OTA. The transconductor is biased with the current I_o, providing Gm=g_{m1}. The finite output impedance of the OTA corresponds to resistive losses in capacitors and inductors of the LC ladder filter [30]. These nonidealities vary the impedance function $Z_{11}(s)$ and thereby degrading the expected magnitude and delay response. A negative conductance is added to cancel out the finite output conductance of the OTA as done in [51]. The positive feedback circuit adds a negative conductance -g_{m5} to the output conductance of

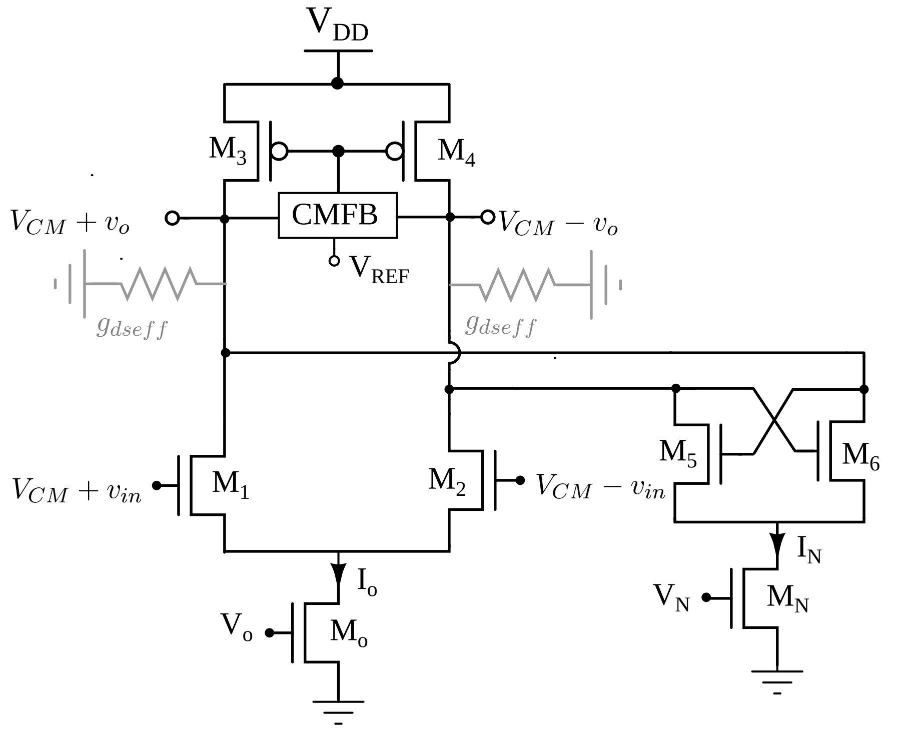


Figure 5.4: OTA with impedance cancellation technique.

OTA. The resulting effective differential output impedance g_{dseff} can be written as

$$g_{dseff} = g_{ds2} + g_{ds4} + g_{ds6} - g_{m5}. \tag{5.3}$$

The bias current, I_N of the positive feedback circuit is generated using the technique reported in [51]. OTA with the output impedance cancellation technique provides a better DC gain of 55 dB. The common mode feedback circuit (CMFB) ensures the output common mode voltage of the transconductor to be same as its input common mode voltage. The summing circuit to generate all-pass signal V_{AP} from V_{LP} and V_I is shown in Figure 5.5. The parasitic capacitance C_{AP} and resistance R_{AP} limits the response of the APF to 3-dB cut-off frequency F_{AP}, where $F_{AP} = 1/(2\pi R_{AP}C_{AP})$. The frequency F_{AP} must be at least 1 decade away from frequency f1, which defines the bandwidth of the delay element.

5.2.1 Delay Variation Technique

As discussed earlier, group delay of the filter can be varied by varying its bandwidth. In its G_m-C equivalent implementation, the bandwidth can be varied either by varying C (constant Gm scaling) or by changing the Gm of transconductors (constant C scaling). In the proposed method, Gm of all the transconductors are varied by changing the bias current, I_o. Increasing

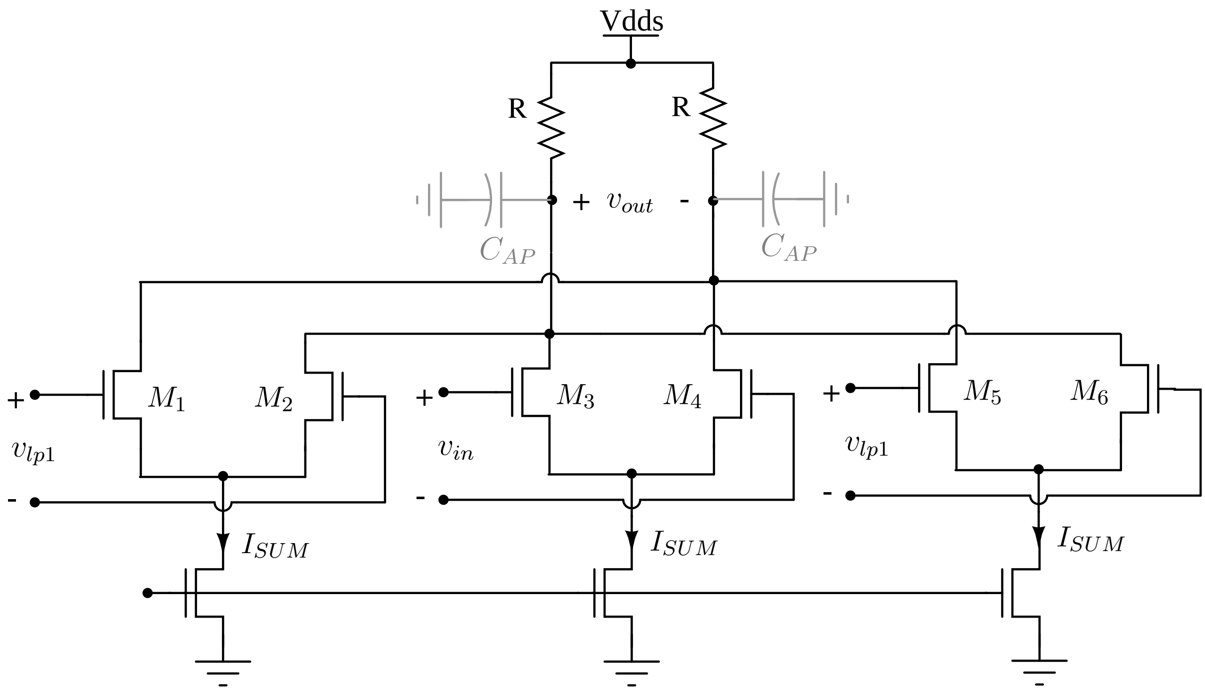


Figure 5.5: Differential current summing circuit for generation of all-passsignal.

the current I_o from I_{o1} to I_{on} i.e. increasing Gm from g_{m1} to g_{mn} decreases the group delay from τ_{g1} to τ_{gn}. The relation between τ_g and Gm for the G_m-C 3rd order APF is expressed as $\tau_g = (2/Gm)(C_1 + C_3)$. By using a constant Gm biasing circuit, in which Gm $\propto 1/R_{Gm}$ [52], τ_g can be controlled linearly as $\tau_g \propto (2R_{Gm})(C_1 + C_3)$. This linear variation in Gm is achieved by changing the resistance R_{Gm} of constant Gm-biasing circuit using N-bit digital input as shown Figure 5.6. Cascading these low-order variable delay APFs increases the delay range.

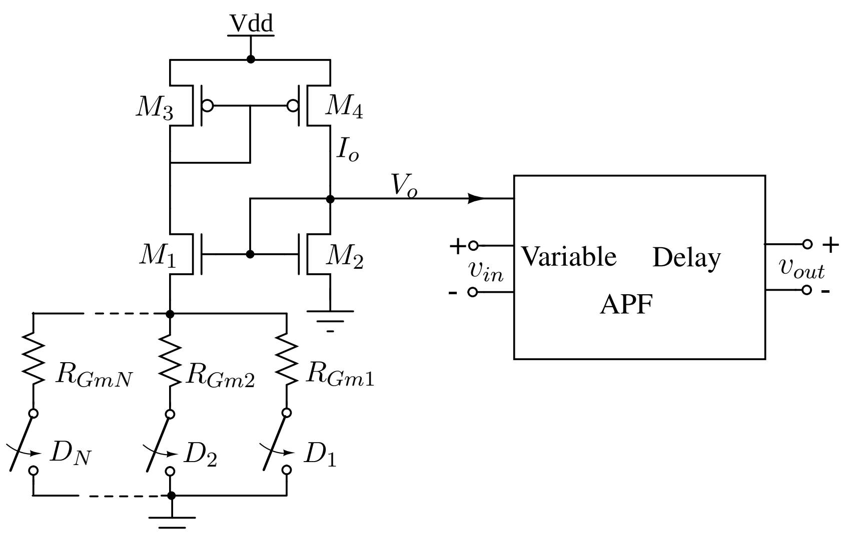


Figure 5.6: Proposed technique of delay tuning by changing the Gm of the APF with N-bit digital input.

If M number of such similar stages are cascaded, then the overall APF response is limited by frequency F_{APM} which can be expressed as,

$$F_{APM} = F_{AP}\sqrt{(2^{1/N} - 1)}. \tag{5.4}$$

Thus, the number of stages, M is to be chosen such that the frequency $F_{APM} > 10f_1$ to avoid the degradation of bandwidth of the delay element.

5.3 Results

The proposed variable time delay element is implemented in TSMC 65-nm technology with a supply voltage of 1.2 V. The APF is designed with 3rd order Bessel LPF whose 3-dB cut-off frequency is 1 GHz. The integrating capacitors are evaluated to be $C_1 = 23$ fF, $C_2 = 50$ fF, $C_3 = 95$ fF for Gm=0.5 mS. The bias current, I_o is varied from 70 μA to 220 μA by switching the value of R_{Gm} from 880 Ω to 378 Ω with 5-bit digital input. Thus, the Gm varies from 0.5 mS to 1 mS which increases the bandwidth of APF from 1 GHz to 2 GHz. The summing circuit is biased with the current I_{SUM} of 225 μA and supply voltage $V_{DDS} = 1.4$ V with R_{AP}=1.35 KΩ to maintain the output voltage at $V_{CM} = 0.95$ V. A 3rd order Bessel LPF with the cut-off frequency of 1 GHz is used as anti-aliasing filter. The variation of τ_g of the APF for different values of R_{Gm} is linear as shown in Figure 5.7. Figure 5.8(a) and Figure 5.8(b) shows the gain and delay plots for a 3-order variable delay Bessel APF.

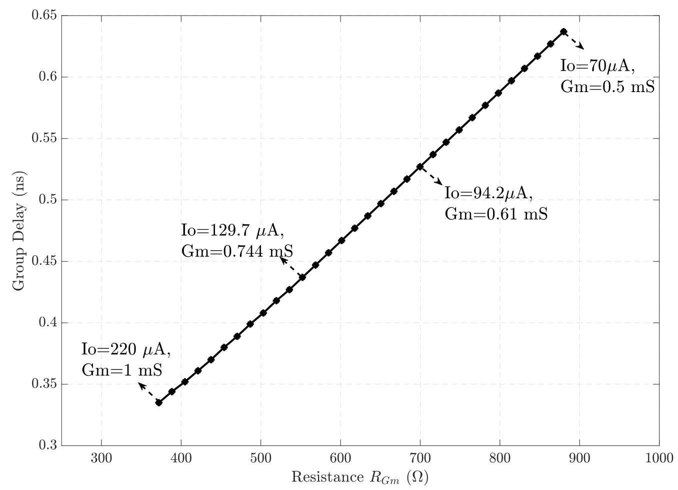
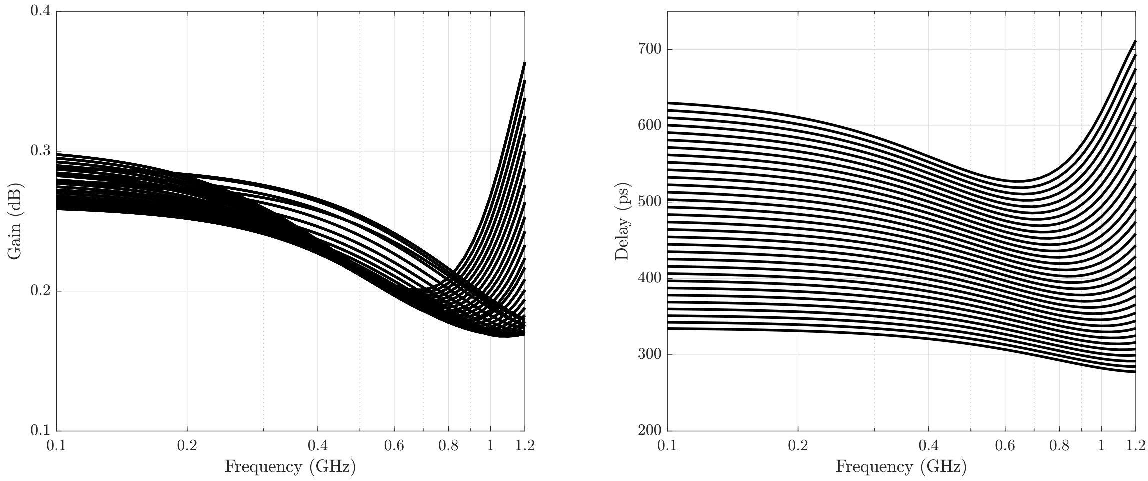


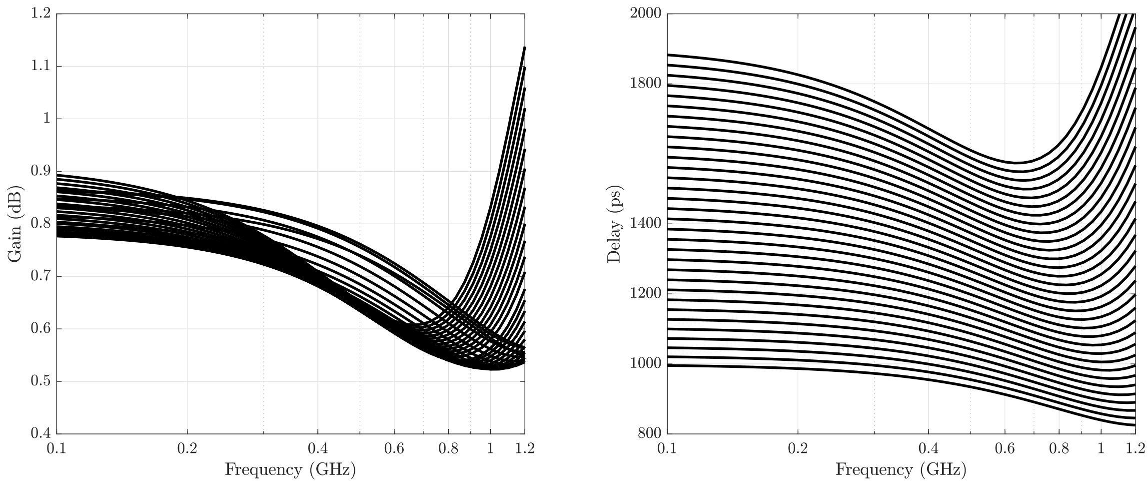
Figure 5.7: Variation of group delay τ_g with the resistance R_{Gm} of one stage variable delay APF .



(a) (b)

Figure 5.8: (a) Gain and (b) group delay plots of a single-stage variable delay APF.

It can be seen that the gain deviation is very small with the average gain of 0.3 dB within 1 GHz. The delay, τ_g varies from 0.35 ns to 0.68 ns with the deviation of ± 40 ps within the signal bandwidth of 1 GHz. The delay range is increased by cascading 3 APFs, with the delay varying from 1.05 ns to 1.85 ns. The gain and group delay plots for the delay element with 3



(a) (b)

Figure 5.9: (a) Gain and (b) group delay plots of cascaded 3 stages of APFs for different delay settings

stages of APFs are shown in Figure 5.9(a), Figure 5.9(b) respectively. The delay variation is within ± 120 ps for maximum delay settings with the average gain of 0.74 dB. The poles introduced by C_{AP} at the node V_{AP} of stage1 and stage3 is around 14 GHz and 8 GHz respectively (as seen in Figure 5.10(a)). This ensures that cascading 3 stages do not effect the bandwidth of the entire delay element as linear phase (seen in Figure 5.10(b)) and uniform magnitude is maintained upto 1 GHz.

Process corner and temperature variation simulations are performed to check the robustness of proposed delay element. Results show that maximum gain and delay variations are within ± 2.12 dB and ± 170 ps respectively, across all the process corners when temperature is varied from 0°C to 70°C.

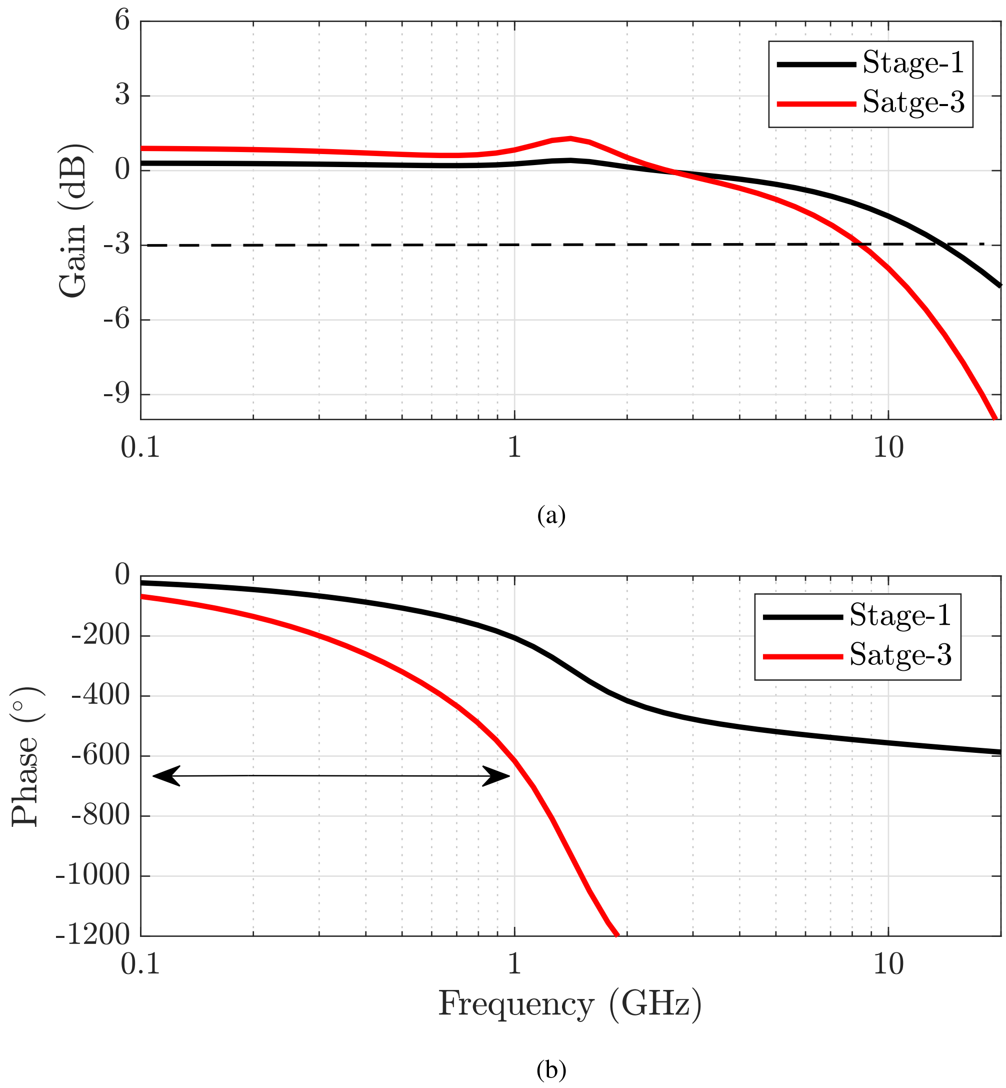
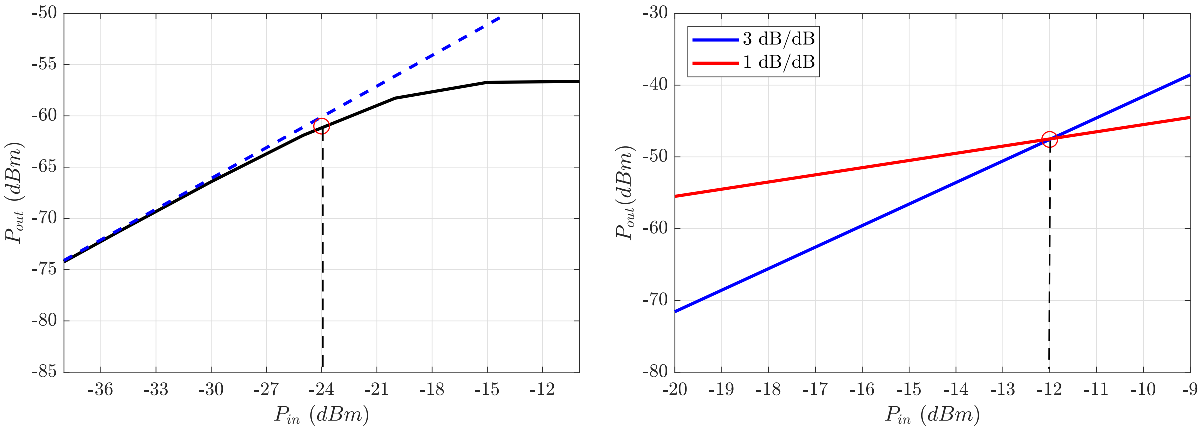


Figure 5.10: (a) Gain and (b) phase plots of 1st and 3rd stage APF with linear phase response up to bandwidth of 1 GHz)

5.3.1 Noise and Distortion

To quantify the amount of distortion introduced by the proposed variable-delay APF, the 1-dB compression point (P1-dB) and input third-order intercept point (IIP3) have to be determined. The APF shows a P1-dB $=-24$ dBm as shown in Figure 5.11(a). IIP3 is determined by exciting the filter with 2 tones separated by 1 MHz (990 MHz, 1000 MHz). Figure 5.11(b) shows the APF has an IIP3 value of -12 dBm. Figure 5.12 shows the input referred noise plot of the APF.



(a) (b)

Figure 5.11: (a) 1 dB compression point of -24 dBm and (b) IIP3 of -12 dBm obtained for variable-delay APF.

The APF has a noise voltage density of 23 nV/$\sqrt{Hz}$ at 1 GHz, with a total integrated noise of 876.2 μV.

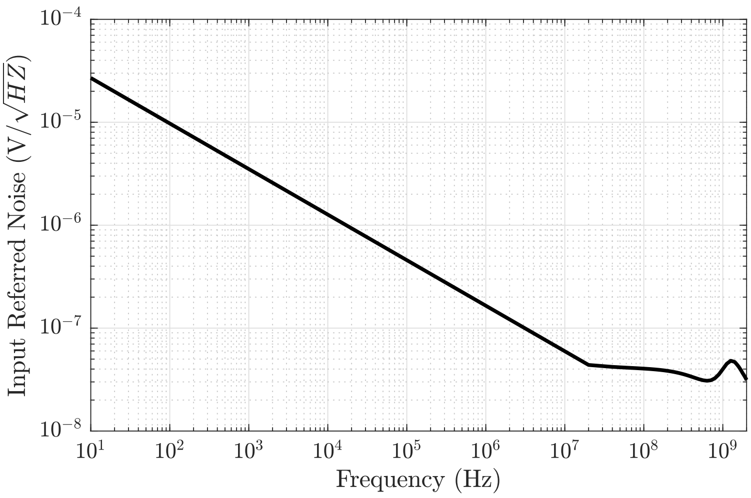


Figure 5.12: Input referred noise plot of the proposed variable-delay APF.

The dynamic range of the APF is determined by the maximum and minimum signal levels that the filter can handle. The maximum value is determined from the P1-dB value, which corresponds to a RMS value of 14.1 mV. The minimum signal value is determined from the noise floor of the input-referred noise plot. Considering the signal bandwidth from 10 KHz to 1 GHz, the noise corresponding to 10 KHz is the heighest. This value corresponds to 806 nV/$\sqrt{Hz}$ and hence, the filter's dynamic range is found to be 84.88 dB. Table 5.1 summarizes the distortion and noise values of the proposed variable-delay APF, LPF, and other reported continuous-time filters.

Table 5.1: Comparision of distortion and noise presented by variable-delay APF, LPF with various reported continuous-time active filters.

Parameters	This work	This work	[28]	[30]	[29]
Technology (nm)	65	65	140	130	65
Architecture	LPF with corr.	G_m-C APF	G_m-C APF	G_m-C APF	G_m-RC APF
$P1dB_{50\Omega}$ (dBm)	-28	-24	-	-14 to -4	-
$IIP3_{50\Omega}$ (dBm)	-17	-12	3	-4.6 to 4.9	2
Input referred noise (nV/$\sqrt{Hz}$)	$13.1_{@1GHz}$	$23.08_{@1GHz}$	$6.2_{@2.63GHz}$	8 to $18_{@2GGHz}$	
Total integrated noise (uV)	422.8	876.2	-	550	-
Power (mW)	36	32	450	364	2.47

5.4 Discussions

Table 5.2 compares the proposed delay element with the existing architectures. The delay element with variable order APF as reported in [30] has ratio of maximum to minimum capacitance, C_{max}:C_{min} $\approx$ 17:1. Stabilizing the CMFB circuit in OTA for these wide ranges of capacitors is challenging. Also, with the large C values, the corresponding Gm value required also increases. This increases the overall power consumption of the circuit. The proposed design with low-order APF maintains smaller C_{max}:C_{min} $\approx$ 5:1, which essentially requires low G_m values and has reduced power consumption of 32 mW. In addition, the proposed method of delay tuning does not require high-quality factor varactors for delay tuning as done in [30], [28]. The delay element has a maximum group delay deviation of 6% from its average. The variable

Table 5.2: Comparison of proposed delay element with existing architectures.

Parameters	This work	This work	[22]	[28]	[30]	[29]
Technology (nm)	65	65	130	140	130	65
Architecture	G_m-C APF	LPF with corr.	Passive APF	G_m-C APF	G_m-C APF	G_m-RC APF
Maximum delay(ps)	1800	1150	400	550	1700	55
Bandwidth (GHz)	DC-1	DC-1	1-20	1-2.5	DC-2	0.4-2
Delay Range (ps)	800	400	350	550	1450	27
Delay Variation (ps)	±120	±5	±8	±10	±140	±2
Gain Variation (dB)	±0.16	±2.5	±15	±1.4	±0.7	±0.375
Power (mW)	32	36	6	450	364	2.47
FOM-1^{+}	-27.95	-20.92	-60.89	-5.26	-18.02	-24.85
FOM-2*	-38.09	-47.95	-57.69	-31.25	-30.91	-40.69

FOM-1^{+}=20log(Power/(delay range $\times$ bandwidth))

FOM-2*=FOM1+ 20log ($\frac{2\times\text{delay variation}}{\text{delay range}}$)+2$\times$gain variation

delay element has the lowest gain deviation compared to other reported architectures. The same figures of merit (FOM-1 and FOM-2) that were discussed for LPF based TTD in Capter 3 are used here. The proposed active APF based TTD shows competing FOM-2 when compared with other reported APF based TTD circuits. However, due to its high delay variations and high power consumption, the LPF based TTD has superior performance.

Chapter 6

Variable-delay APF- Pulse Expansion and Compression

This chapter discusses on-chip circuit implementations of expansion and compression circuits for stretching (compressing) continuous-time pulses using variable delay APFs. The principle of changing the delay of a APF is similar to the technique explained in section 5.1 of Chapter 5. The circuit realization of APF focuses on designing low order filter to avoid the use of large range of capacitance for design simplicity and low power consumption. Identical APFs are cascaded to improve the expansion (compression) factor. A detailed comparison between LPF and APF based expansion techniques is also reported in this chapter.

6.1 Principle of Expansion and Compression

Figure 6.1 shows the block diagram of proposed expansion (compression) circuit using variable delay APF. It consists of 3 cascaded stages of 3^{rd}-order variable delay APFs. This 3^{rd} APF is

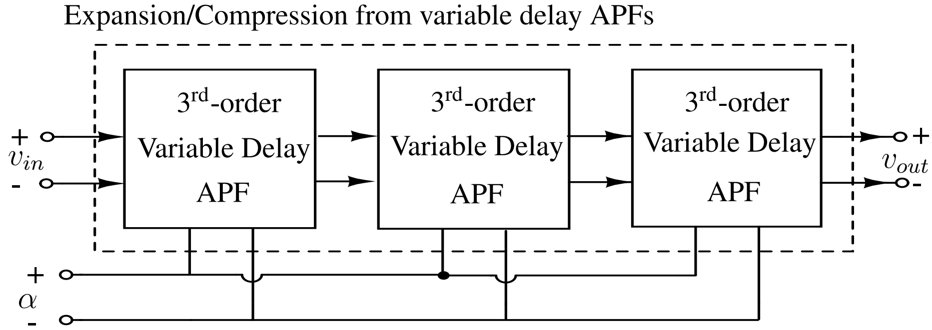
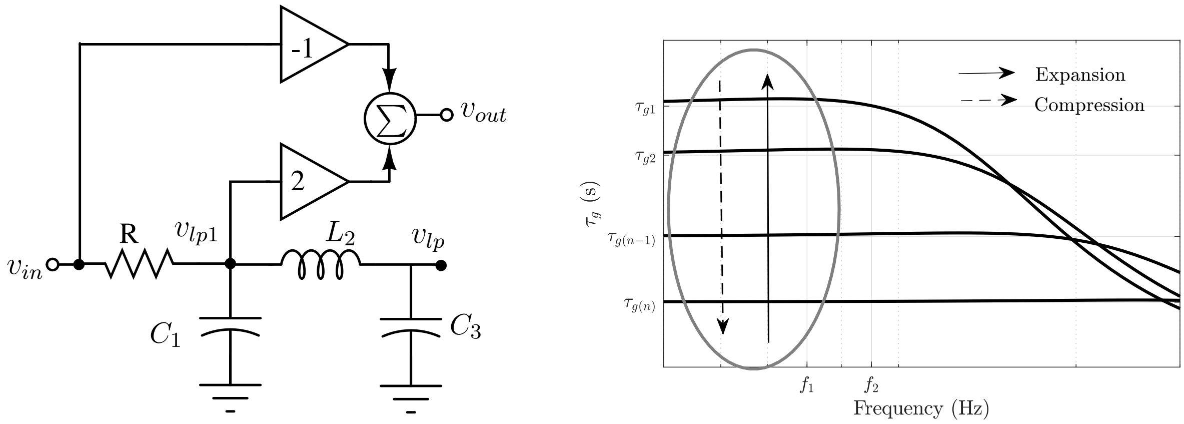


Figure 6.1: Block level diagram of proposed expansion (compression) circuit using variable delay APF.

realized from singly terminated LC ladder architecture as shown in Figure 6.2(a). The principle of changing the delay of the APF is already explained in section 5.1 of Chapter 5. Figure 6.2(b) shows the group delay plots of 3rd-order filter for different values of L and C. A continuous



(a) (b)

Figure 6.2: (a) 3rd-order APF realized from singly terminated LPF, (b) APF as expansion (compression) circuit up to f_1 Hz.

time high frequency sporadic input pulse to the tunable delay APF will undergo expansion when group delay of the filter is increased from τ_{gn} to τ_{g1}. Similarly, compression of wide pulse occur when the delay is decreased from τ_{g1} to τ_{gn}. The expansion (compression) operation is controlled with a linearly increasing (decreasing) ramp signal α.

6.2 Implementation Details

A lower order Bessel APF of order n= 3, is used as a core expansion (compresssion), as it maintains constant group delay within the bandwidth of the filter [49]. The filter realization procedure for evaluating the values of L and C of Bessel LPF is explained in Section 5.2 of Chapter 5. For the complete on-chip implementation of the delay element, the LPF is implemented using its G_m-C equivalent as shown in Figure 6.3.

In the above G_m-C implementation, the delay of LPF is varied with constant-C scaling, in which the transconductance Gm of all the OTAs are varied. For expansion and compression techniques, the Gm of OTAs are controlled with an external control signal α. A OTA with controlled transconductance Gm(α) and high DC gain with impedance cancellation technique is shown in Figure 6.4.

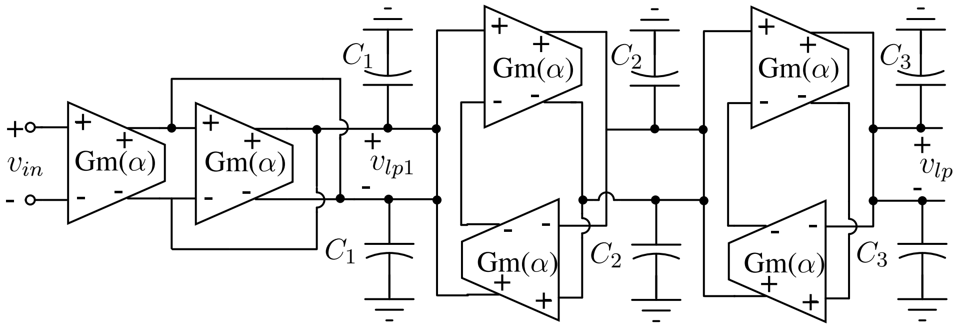


Figure 6.3: Gm-C implementation of 3rd-order LPF with differential architecture.

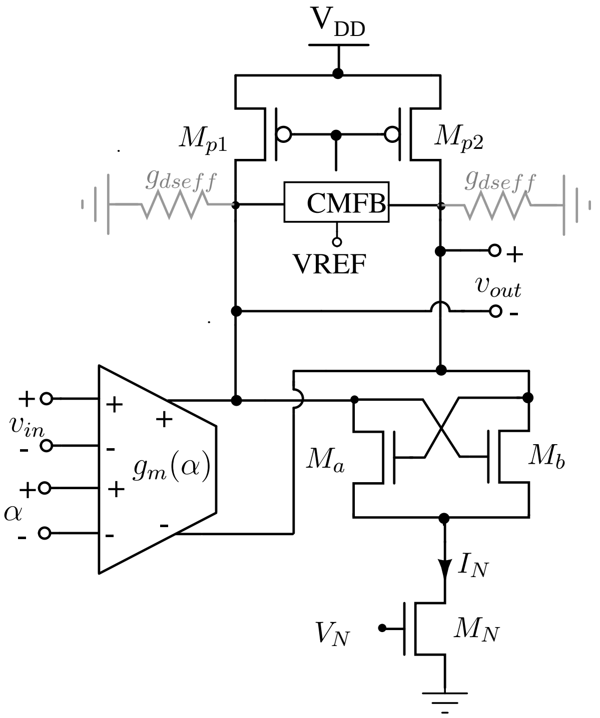


Figure 6.4: OTA with controlled transconductance Gm(α) and high DC gain with impedance cancellation technique.

Figure 6.5 shows the transistor level schematic of Gm(α) block used in G_m-C LPF. The circuit is a conventional Gilbert multiplier with small signal output current $i_o\pm$ is proportional to αvin. The detailed analysis of the circuit is as follows: If g_{m3} and g_{m5} represents transconductances of $M_{3,4}$ and $M_{5,6}$ respectively, then the current i_o is expressed as,

$$i_o = (g_{m3} - g_{m5})v_{in} \tag{6.1}$$

The differential pairs $M_{3,4}$ and $M_{5,6}$ are biased with currents I_o+$g_{m1}\alpha$ and I_o-$g_{m1}\alpha$ respectively. Hence, the transconductances g_{m3} and g_{m5} is expressed as,

$$g_{m3} = \sqrt{(I_o + g_{m1}\alpha)K_{vin}} \tag{6.2}$$

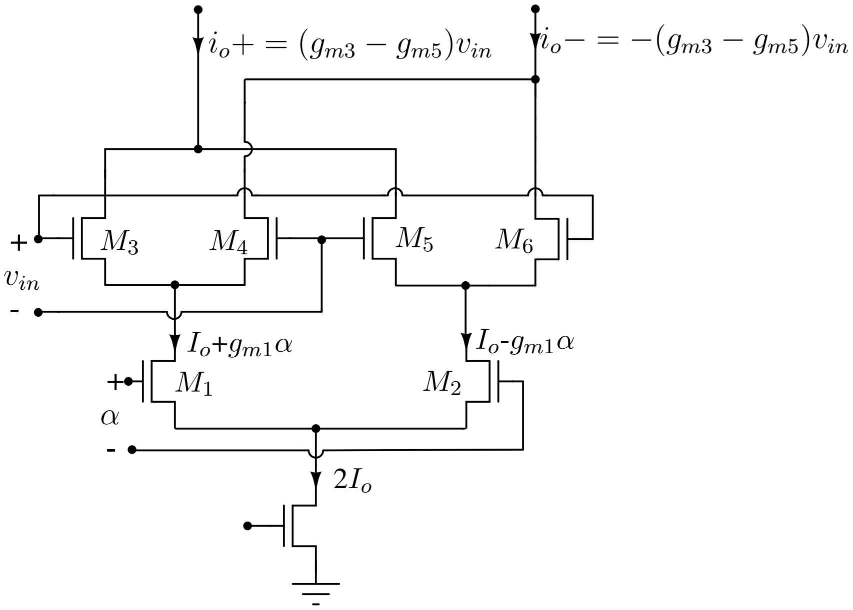


Figure 6.5: Transistor level schematic of Gilbert multiplier for realizing Gm(α).

$$g_{m5} = \sqrt{(I_o - g_{m1}\alpha)K_{vin}} \tag{6.3}$$

where K_{vin} represents current factor of $M_{3,4,5,6}$. Using, $(1+x)^{1/2} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16}..$ and $(1-x)^{1/2} = 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16}$ and neglecting higher order terms, g_{m3} and g_{m5} can be approximated to,

$$g_{m3} \approx \sqrt{I_o K_{vin}}(1 + g_{m1}\alpha/2I_o) \tag{6.4}$$

$$g_{m5} \approx \sqrt{I_o K_{vin}}(1 - g_{m1}\alpha/2I_o) \tag{6.5}$$

With these approximations, Eq. 6.1 can be written as,

$$i_o = \left(2\sqrt{K_{vin}/I_o}\right)g_{m1}\alpha v_{in} \tag{6.6}$$

If K_α represents current factor of $M_{1,2}$, then transconductance of $M_{1,2}$ is expressed as,

$$g_{m1} = \sqrt{2I_o K_\alpha} \tag{6.7}$$

Using Eq. 6.7 in Eq. 6.6, the current i_o is expressed as,

$$i_o = 2\sqrt{2K_{vin}K_\alpha}\alpha v_{in} \tag{6.8}$$

From Eq. 6.8, it can be seen that the variable transconductance $G_m(\alpha)$=$2\sqrt{2K_{vin}K_\alpha}\alpha$ of the OTA can be controlled with an external signal α. The effective conductance g_{dseff} is reduced to 0 to increase the DC gain of OTA by setting the value of I_N [51]. The circuit architecture of the summing circuit required for generating the all-pass output v_{out} is similar to that shown in section5.2 of chapter 5.

6.3 Results and Discussions

The proposed variable time delay element is implemented in TSMC 65-nm technology with a supply voltage of 1.2 V. The programmable OTA is designed with bias current $2I_o$=200μA with common mode voltage V_{CM} set to 0.95 V. The control signal α of Gm(α) is varied from 50 mV to 150 mV across the common mode voltage of 0.8 V. This results in variation of Gm from 532.6 uS to 1015 uS as shown in Figure 6.6. The integrating capacitors of the LPF are

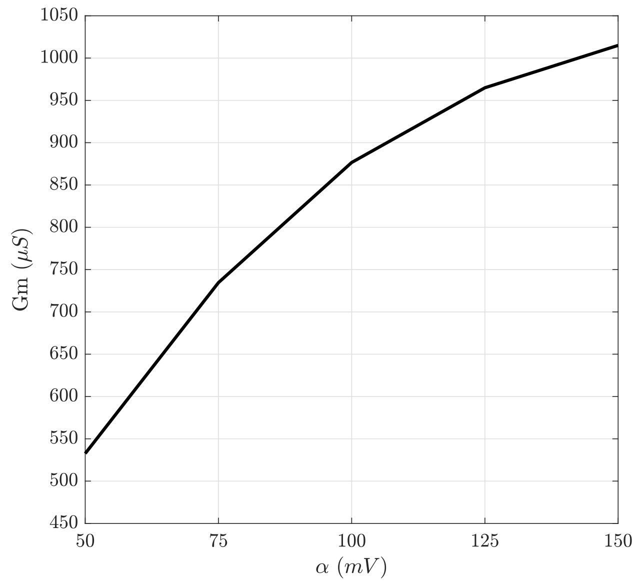
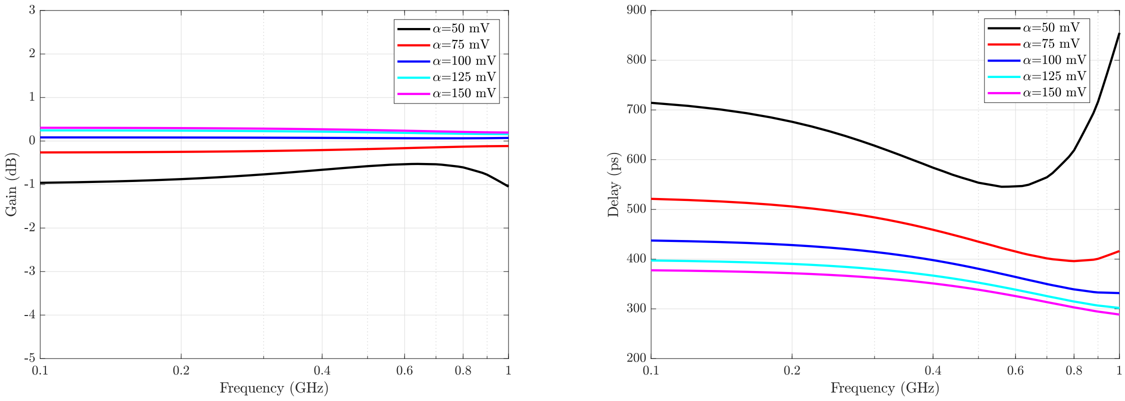


Figure 6.6: Different Gms from programmable OTA with the control voltage varied from 50 mV to 150 mV.

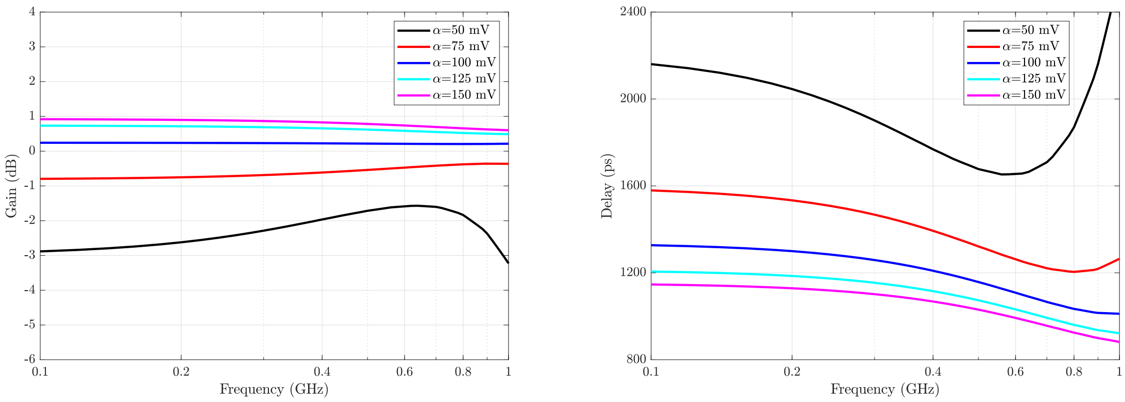
chosen to $C_1 = 23$ fF, $C_2 = 50$ fF, $C_3 = 95$ fF. The summing circuit is biased with the current I_{SUM} of 225 μA and supply voltage $V_{DDS} = 1.4$ V with R_{AP}=1.35 KΩ to maintain the output voltage at $V_{CM} = 0.95$ V.

Figure 6.7(a) and Figure 6.7(b) show the gain and delay plots of a 3rd-order variable delay Bessel APF for different values of α. It can be seen that the gain of the APF varies between -1 dB to 0.3 dB and the delay varies between 380 ps to 650 ps with a maximum delay deviation of $\pm$ 50 ps within the bandwidth of 0.5 GHz. Figure 6.8(a) and Figure 6.8(b) show the gain and delay plots of a cascaded 3 stages of APFs for different values of α. It can be seen that the gain of the APF varies between -3 dB to 1 dB and the delay varies between 1100 ps to 1820 ps with a maximum delay deviation of $\pm$ 200 ps within the bandwidth of 0.5 GHz. With different input values of α, the minimum gain from 1-stage APF and 3-stage APFs is obtained for α=50 mV.



(a) (b)

Figure 6.7: (a) Gain and (b) group delay plots of a 3[rd]-order variable delay APF.

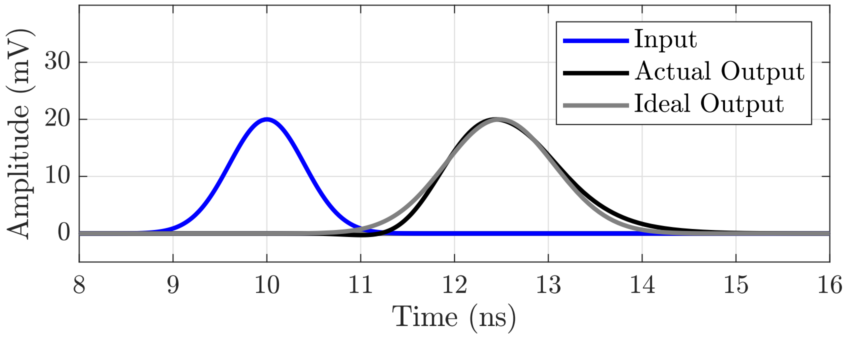


(a) (b)

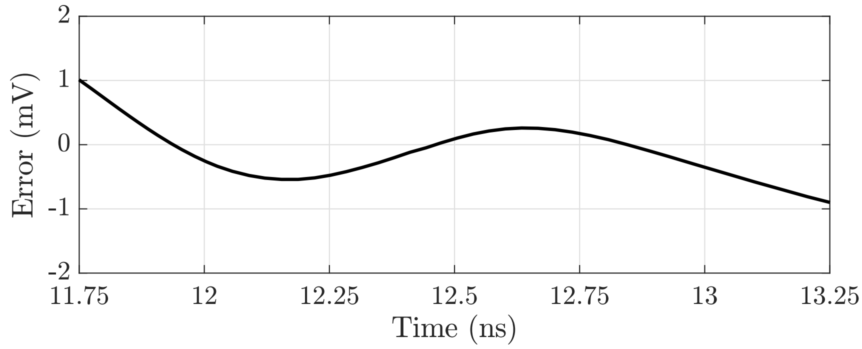
Figure 6.8: (a) Gain and (b) group delay plots of 3 cascaded stages of variable-delay APFs.

This is due to decreased DC gain ($\approx$35 dB) of the programmable OTA blocks.

An ideal Gaussian pulse with a half power pulse width of 0.944 ns (3 dB frequency of 450 MHz) is applied as test input to verify the expansion operation. The control signal α to Gm(α) is decreased linearly from 150 mV to 50 mV within the pulse duration. The output pulse is a stretched version of the input pulse with half power pulse width of 1.4 ns (stretch factor=1.48) as seen in Figure 6.9(a). The RMSE between the actual and ideal stretched pulse is around 0.353mV. The total power dissipation of the expansion circuit is 22.31 mW. In order to verify compression operation, the control signal α to Gm(α) is increased from 50 mV to 150 mV. With the same input pulse of 0.944 ns, the output is a compressed pulse with a compression factor of 0.87 as seen in Figure 6.10. The RMSE between the actual and ideal stretched pulse is around 0.416 mV.



(a)



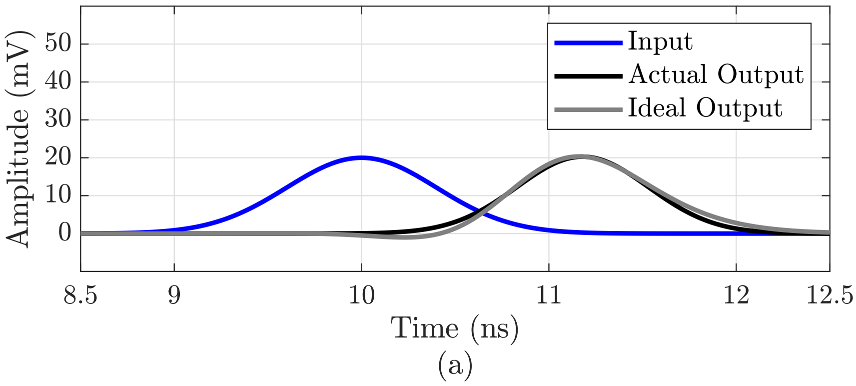
(b)

Figure 6.9: (a) Expansion of the input Gaussian pulse with a stretch factor of 1.48 (b) plot of error between actual and ideal output samples.

6.4 Comparison with LPF based expansion technique

The performance of APF based expansion technique is compared with LPF based expansion operation which was discussed in Chapter 4. To have a fair comparison between these two circuits, a variable delay LPF based expansion circuit is designed to have the same stretch factor of 1.04 as that obtained with 1-stage variable delay G_m-C APF when an input Gaussian pulse of 0.944 ns is applied. It is observed that for the same technology node, The 3-stage variable delay LPF based expansion circuit designed with R=500 Ω, C_1=30 fF, C_1=43 fF and C_3=58 fF produces same stretch factor. Simulation results of these two expansion techniques are tabulated in Table 6.1.

Results show that RMSE value for APF based expansion technique is lower than LPF based technique. But the power dissipation minimum for LPF based expansion with better FOM. Hence, there is a trade-off between accuracy and power parameters for LPF and APF based expansion techniques.



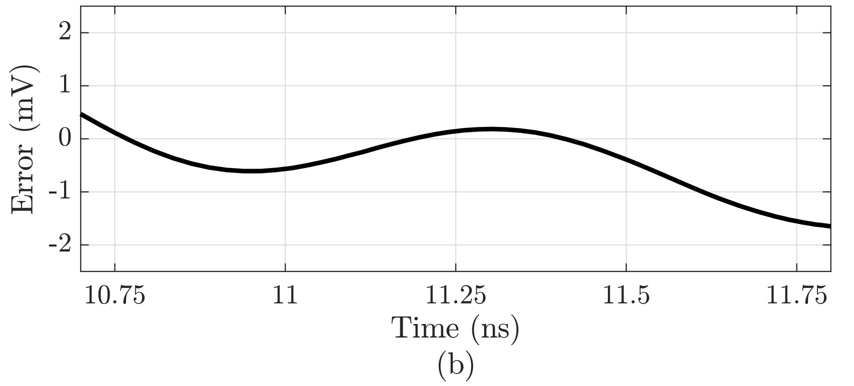


Figure 6.10: (a) Output pulse with half power pulse width of 0.825 ns with compression factor of 0.87, (b) plot of error between actual and ideal output pulses.

Table 6.1: Performance comparison of APF and LPF based expansion circuits.

Technique	APF	LPF
Pulse width (ns)	0.944	0.944
Stretch factor	1.04	1.04
RMSE (mV)	0.318	0.533
Power dissipation (mW)	7.38	6.78
FOM(mW/mV)	23.21	12.72

Chapter 7

Analysis of Practical Implications of TTD based Beamforming

With ever-increasing demand for large bandwidth and high data rates, wireless technology operating at mm-wave frequencies is a promising solution for next-generation communication systems. State-of-the-art transceivers with MIMO technology use phased (or timed) antenna array architectures to enable multi-user access by establishing directional links between the base station and mobile devices. In these receivers, the signals from each antenna in the array are delayed and combined constructively to form a directional radiation beam; i.e. beamforming. In addition to having high gain for the intended user, beamforming also reduces interference in the spatial domain. The beamforming process can be broadly classified into three types, namely: (i) analog beamforming, (ii) digital beamforming, and (iii) hybrid beamforming.

7.1 Survey on Various Beamforming Techniques

In analog beamforming, the complete beamforming is carried out in the analog domain before digitization. The delay elements used in this beamforming can be realized either using phase shifters (PS) or TTD elements. Depending on where in the signal path the required phase delays or time delays are applied, analog beamforming can be further classified into three different categories, namely: (a) radio frequency (RF) path beamforming [53–62], (b) local oscillator (LO) path beamforming [63–70], and (c) analog beamforming at baseband (BB) [71, 72]. In some recent reported works, the beamforming is done partially using PS and partially with TTD's [73]. This allows the authors to perform a coarse beamforming with PS and residual correction is done with a TTD. Figure 7.1 shows the conceptual block diagrams of the three different types of analog beamforming architectures, which are discussed as follows.

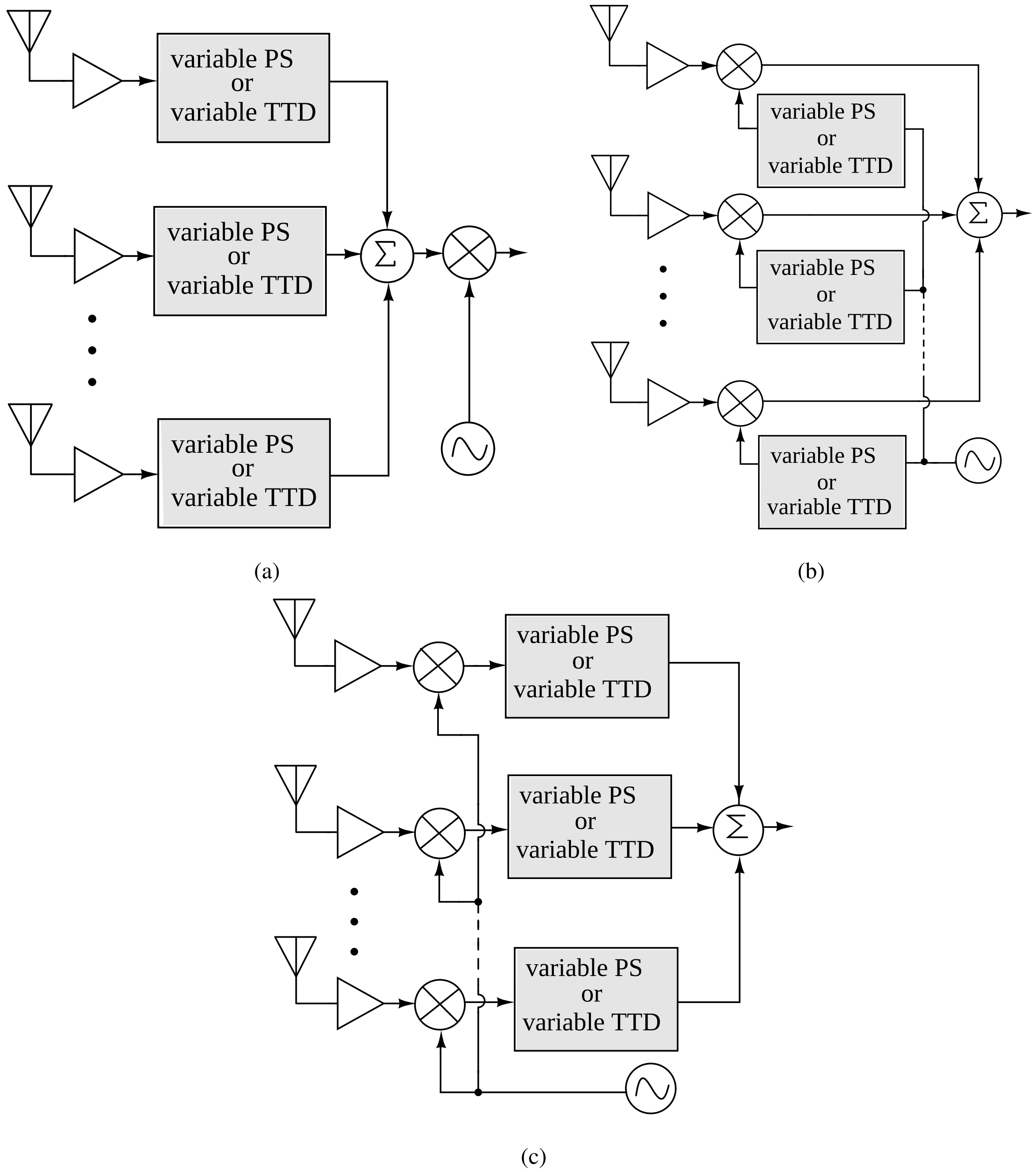


Figure 7.1: Different analog beamforming configurations used in phased (or timed) array receivers: (a) RF path beamforming, (b) LO path beamforming, (c) Beamforming at BB frequency.

In RF path beamforming (Figure 7.1(a)), the incoming modulated signal at each antenna is delayed and added constructively before downconversion. In mm-wave applications, which are around 24 GHz and above, on-chip phase shifting at RF becomes easy, allowing low-cost integration [74–76]. As beamforming is performed at RF, there is a reduced linearity constraint on the mixer and subsequent blocks in the receiver as blockers are rejected before the mixer in the signal path. This technique also uses fewer components compared to other configurations, as a single mixer is shared among multiple antenna paths with one LO for downconversion. However, PS at these frequencies incurs huge losses and requires other compensation circuits that also operate at RF leading to high power consumption [72].

In LO path beamforming, the signal from the LO is provided to the mixers with different phase shifts. The delayed versions of signals after downconversion are added constructively to achieve beamforming. With LO path architecture, the linearity requirement on the PS is reduced as the PS works with a single tone LO signal. However, the linearity requirement on the mixer is increased in this configuration. The major drawback of this architecture is that it requires a power-hungry and complex LO routing network [71].

In BB beamforming, the incoming signal at all the antennas is first downconverted, and suitably delayed and combined for beamforming. BB beamforming is chosen in applications with wide bandwidth [30, 71]. Beamforming at baseband frequencies features fine delay resolution and this configuration is more robust than RF and LO beamforming as it is less sensitive to high-frequency parasitic effects. But unlike RF, the interference is not filtered spatially before downconversion in BB beamforming. Hence, this increases the linearity requirement on the mixer in each antenna path [74]. BB beamforming can also be performed on discrete signal obtained after sampling the downconverted signal using resampling technique [77]. The discrete delays are controlled by varying the relative delay between LO and resampler clock.

It may be noted that the architectures of beamforming shown in Figure 7.1 are for beamforming at the reciever. However, by simply reversing the sequence of operations, similar architectures for beamforming at the transmitter can be realised. For example, consider Figure 7.1(a), wherein beamforming is performed using the signals from multiple antennas that are received, amplified with low noise amplifiers, suitably delayed and added before downconversion. For realising a similar beamforming transmitter, an upconverted signal can be equally split into multiple channels, suitably delayed, amplified with power amplifiers and fed to antennas.

For the purpose of discussion of different types of beamforming in this work, we will consider beamforming at the receiver only.

In the case of digital beamforming configuration (as shown in Fig 7.2(a), the incoming RF signal from each antenna is downconverted and digitized using analog-to-digital converters (ADCs), and beamforming is performed in the digital domain using digital signal processing

(DSP) [78, 79]. Digital beamforming configurations allows easy implementation as the entire process is carried out in digital domain. However, this architecture demands highly linear ADCs with a large input dynamic range at each antenna path. The sampling rate of the ADC depends not only on the signal bandwidth, but also on the delay resolution required for beamforming. The ADC should be operated at a sampling rate that is higher of the above two requirements. Hence, meeting this constraint increases total power power consumption of the receiver.

Figure 7.2(b) shows the block diagram of a fully connected hybrid beamforming architecture, which combines the techniques of both analog and digital beamforming [80, 81]. In this configuration, beamforming is done in two stages. In the first stage, multiple channels of analog beamforming are realised using all antennas (for fully connected architecture). These multiple channels are downconverted and digitized. These multiple digitized signals are then used in the second stage to perform digital beamforming. When a subset of antennas are used for the first stage of beamforming it is called sub-array hybrid beamforming [81]. These hybrid techniques harness the advantages of both analog and digital beamforming to have precise, flexible, and power efficient architecture.

7.2 Beamforming with Phase Shifters

In phased or timed antenna array receivers, the signals arriving from a distinct source at an angle θ_0 at each antenna experience a time delay of $\Delta\tau$ relative to adjacent antennas in the array as shown in Figure 7.3(a). If d represents the spacing between the antennas, $\Delta\tau$ can be expressed as

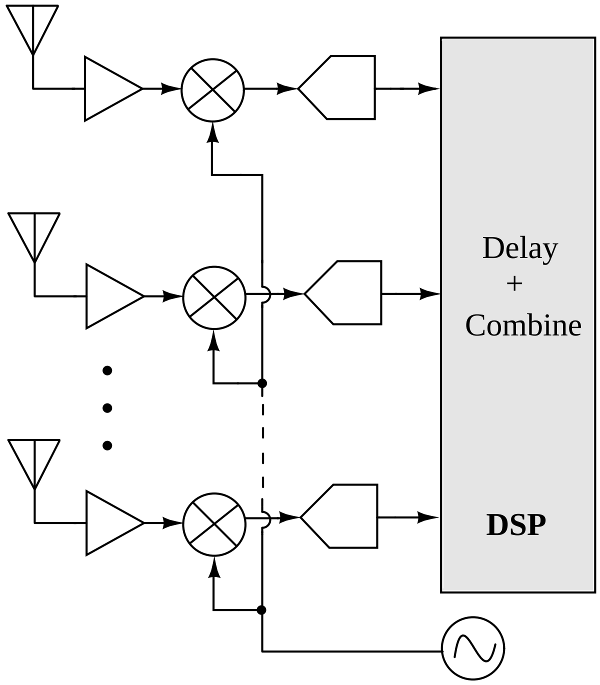
$$\Delta\tau = (d/c)\cos\theta_0. \tag{7.1}$$

If the maximum spacing between the antennas is $\lambda/2$, where λ is the wavelength of the incoming RF signal with frequency f_c, then the time delay $\Delta\tau$ can be rewritten as

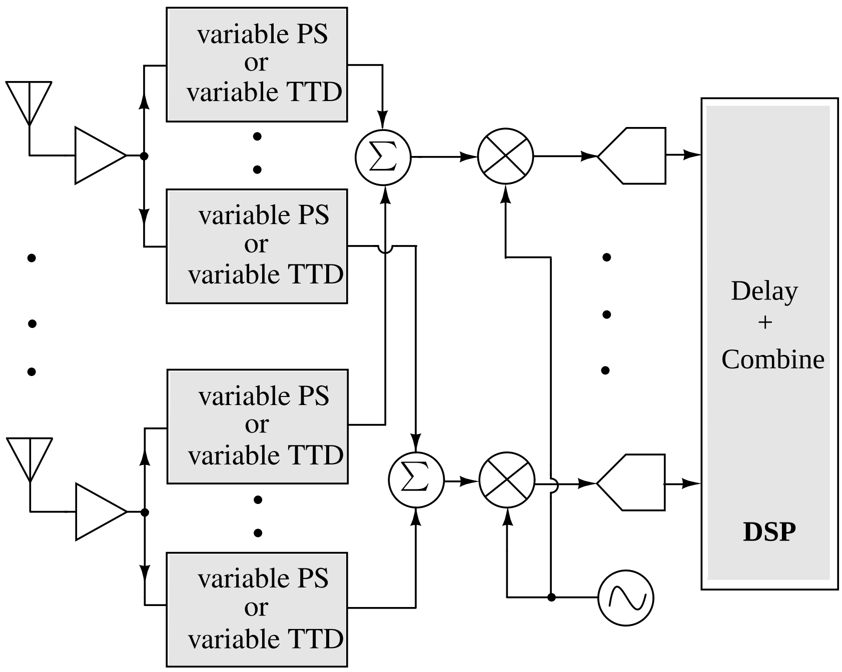
$$\Delta\tau = (1/2f_c)\cos\theta_0. \tag{7.2}$$

The above time delay $\Delta\tau$ is expressed in the frequency domain as a frequency-dependent phase shift $\Delta\phi(f)$, where $\Delta\phi(f) = \pi(f/f_c)\cos\theta_0$. If f_{BW} represents the bandwidth of the modulating or baseband signal, then $\Delta\phi(f)$ can be expressed as

$$\Delta\phi(f) = \pi\left(\frac{f_c \pm (f_{BW}/2)}{fc}\right)\cos\theta_0. \tag{7.3}$$



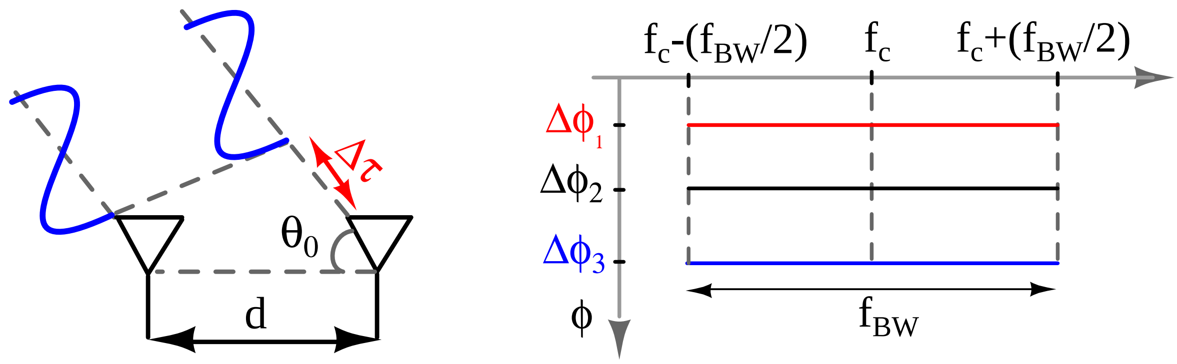
(a)



(b)

Figure 7.2: (a) Digital beamforming, and (b) Hybrid beamforming.

For narrowband beamforming applications, the bandwidth $f_{BW} << f_c$. With the assumption of f $\approx f_c$, the frequency independent phase shift $\Delta\phi$ can be expressed as $\Delta\phi = \pi \cos\theta_0$. Hence, desired time delays can be realized using phase shifters, in which the phase shifts are



(a) (b)

Figure 7.3: (a) Incoming signal arriving with different delays on antennas in phased and timed antenna array receivers, (b) Phase shifters with frequency independent phase shifts across the narrow frequency range.

assumed to remain constant across the narrow bandwidth f_{BW} as shown in Figure 7.3(b). The following subsections discuss different PS architectures reported in the literature and non-ideal effects associated with practical PS.

7.2.1 Various reported PS Architectures

PS can be realized using active or passive devices and with distributed or lumped elements. Considering these, the PS are classified into reflection type [82–85], loaded line [86–88], switched type [89], [90], and vector sum [91, 92] phase shifters. The reflection type PS consists of a 3-dB hybrid coupler with the complex impedances Z_a and Z_b connected to port 2 and port 3 respectively as shown in Figure 7.4(a). Varying the values of Z_a and Z_b will alter the level of reflections from these ports. This changes the phase of the output signal for different impedance values.

In case of the loaded line phase PS, the transmission line with length ℓ and characteristic impedance Z_0 is terminated with different complex impedance values as shown in Figure 7.4(b), to achieve different phase shifts. The phase shift range can be improved by cascading these identical stages.

The switched type PS consists of two transmission lines with different lengths ℓ_1 and ℓ_2 as shown in Figure 7.4(c). The input signal is passed to either of the transmission lines using switches to provide different phase shifts. The transmission lines could be modeled with shunt/series combinations of L and C for the required phase shifts. Identical or non-identical switched line sections are cascaded to improve the phase shift range. Another topology of switched type PS is a cascade of digitally controlled variable phase shifters realized with T or Π low-pass, high-pass, and switched filters [90].

Figure 7.4(d) shows a block level architecture of a vector sum PS. In the vector sum PS, the weighted sum of I-phase and Q-phase signals is taken which results in an output whose phase can be controlled over the entire 360° range. IQ signals can be generated using hybrid couplers [93] or polyphase filters [94]. For implementing the weighted sum of the IQ signals Gilbert multipliers [95] can be used. Another way of realizing the weighted sum is by using digital attenuators followed by combining amplifiers [96].

A fairly detailed review along with implementation details of other existing PS architectures is reported in [74]. The selection of PS depends on various parameters like phase shift range with resolution, insertion loss across different phase shifts, power consumption, bandwidth, and die area.

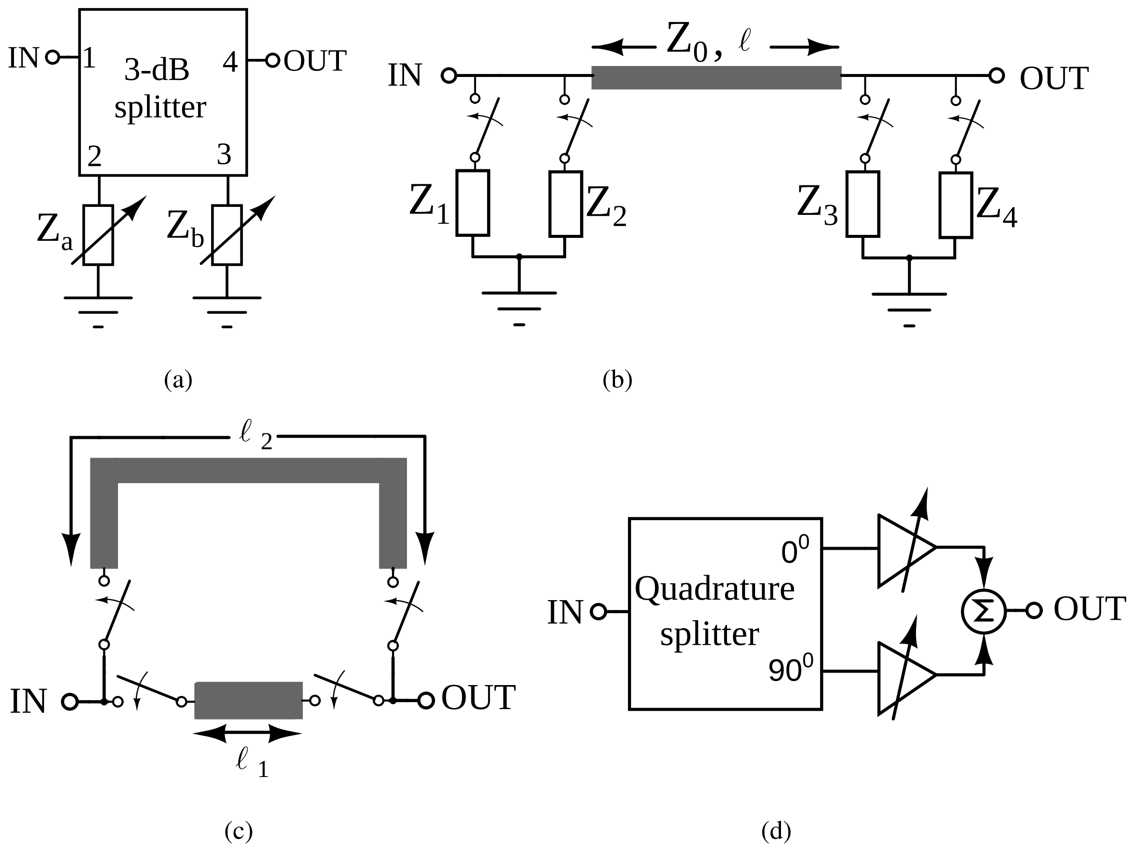


Figure 7.4: Different types of PS architectures: (a) reflection type PS, (b) loaded line PS, (c) switched type and (d) vector sum PS.

7.2.2 Beam-squinting with PS

As discussed earlier, the assumption of $f \approx f_c$ is no longer valid in baseband signals with large bandwidth, especially in wideband beamforming applications [97, 98]. PS used in these beamforming configurations will shift the antenna array's radiation pattern to another angle θ instead of θ_0. This changes the direction of the steered beam which results in gain deviation of

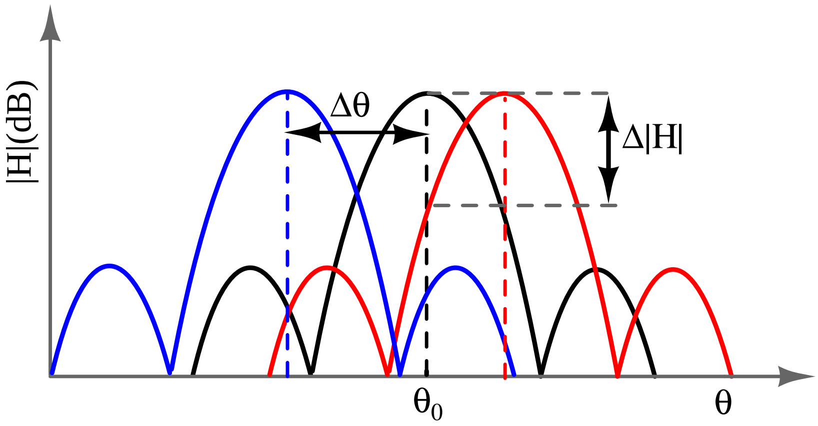


Figure 7.5: beam-squint in wideband beamforming due to PS.

$\Delta|\mathrm{H}|$ at θ_0. This is referred as beam-squinting and the amount of squint $\Delta\theta$ is expressed as

$$\Delta\theta = \cos^{-1}\left[\frac{f_c}{f}\frac{\Delta\Phi}{\pi}\right] - \theta_0 \tag{7.4}$$

The error is a maximum at the edges of the frequency band where the frequency deviates the most from f_c.

The performance of some of the previously reported PS including phase shift range, phase deviation, and gain deviation parameters are summarized in Table 7.1. It is observed that the practical PS have maximum phase deviation of about $\pm 25^{\circ}$ around the center frequency with $\pm$2dB of gain deviation across the given phase shift range. The resulting beam-squint in addition to the gain variation will degrade the receiving system performance. Hence, the PS must be replaced with variable time delay blocks or TTDs which delay the signals in time domain.

7.3 Effect of TTD nonidealities

An ideal TTD element must maintain constant gain and flat delay across all the frequencies. Hence, the timed array receivers are expected to have maximum gain at a given steering angle

Table 7.1: Performance summary of some existing PS in literature.

	Process-Technology	Topology	Bandwidth (GHz)	Phase Shift Range	Phase deviation	Gain variation(dB)
[84]	CMOS 180-nm	Reflection type	22-26	360°	$\pm 25°$	± 2
[75]	CMOS 65-nm	Reflection type	27-30	360°	0.3° (see 1)	± 1.25
[87]	CMOS 180-nm	Loaded line	10-13	360°	$\pm 10°$	± 0.5
[89]	CMOS 180-nm	Switched type	11.6-12.6	360°	$\pm 2.6°$	± 0.5
[99]	CMOS 55-nm	Switched+Reflection	30-36	360°	2.19°	2
[95]	SOI 130-nm	Vector sum	8-12	360°	0.29°	0.24
[96]	CMOS 65-nm	Vector sum	70-90	360°	6.74°	± 10.6

(1) The reported phase deviation in [75] is at the center frequency viz. 28 GHz.

θ_0 without beam-squinting. But practical TTD circuits exhibit gain and delay variations across a given frequency range as seen in Table 2.1 and Table 2.2 of chapter 2. Among the various reported passive and active TTD circuits, the maximum delay variation of about 24% is observed in [30] and maximum gain variation of ± 15 dB is reported in [22]. Some reported TTD circuits with excellent gain and delay profiles (like [20,29]) have very low delay range, delay resolution and bandwidth. The frequency dependent delay and gain characteristics of TTD circuits are due to parastics, undesired mutual inductances, non-ideal switches and finite Q of inductors and capacitors in on-chip implementations.

In this work, the effect of TTD non-idealities on the beamforming process is studied. This detailed study motivates in designing of good variable delay lines by quantifying tolerable delay and gain deviations in practical TTD circuits. In particular, the non-ideal effects of TTDs on RF and BB beamforming architectures are analyzed here. The detailed study includes the individual effects of delay and gain variations and the combined effect of delay-gain deviations on the gain and directivity of the receiver antenna array.

7.3.1 Beamsquint and Gain deviation with practical TTD circuits

Following analysis provides the expression for beam squint $\Delta\theta$ and gain deviation $|H(j\omega)|_{\theta_0}$ with frequency dependent time delays.

Consider a K $\times$ 1 linear timed array receiver system in which signal from a distinct source arrives at an angle θ_0 as shown in Figure 7.6(a). With τ being the delay experienced between the adjacent antennas, The programmable delay element in each antenna path is configured with different delay values:τ_1=(K-1)τ, τ_2=(K-2)τ, $\tau_{\text{(K-1)}}$=τ, τ_{K}=0. With these frequency independent time delays, the resulting antenna array pattern steers to the desired steering angle θ_0 without beam-squint.

Practical TTD circuits have frequency dependent delays, which cause deviation in the ideal antenna radiation pattern. Hence, the receiver beam-pattern shifts to other angles θ from θ_0. This can also be interpreted as the incoming signal S(t) arriving at different angles θ, while the delay elements are programmed with different delays at these incident angles for beamforming. With this understanding, the amount of beam-squint $\Delta\theta$ is evaluated with non-ideal TTD elements as follows.

If the frequency dependent delays are expressed as,

$$\begin{aligned}\tau_1(\omega) &= (K-1)[\tau \pm \Delta\tau(\omega)] \\ \tau_2(\omega) &= (K-2)[\tau \pm \Delta\tau(\omega)] \\ \tau_{(K-1)}(\omega) &= [\tau \pm \Delta\tau(\omega)]\end{aligned} \tag{7.5}$$

and the delay characteristics of $\tau_{(\text{K}-1)}(\omega)$ is as shown Figure 7.6(b), then following observations can be made:

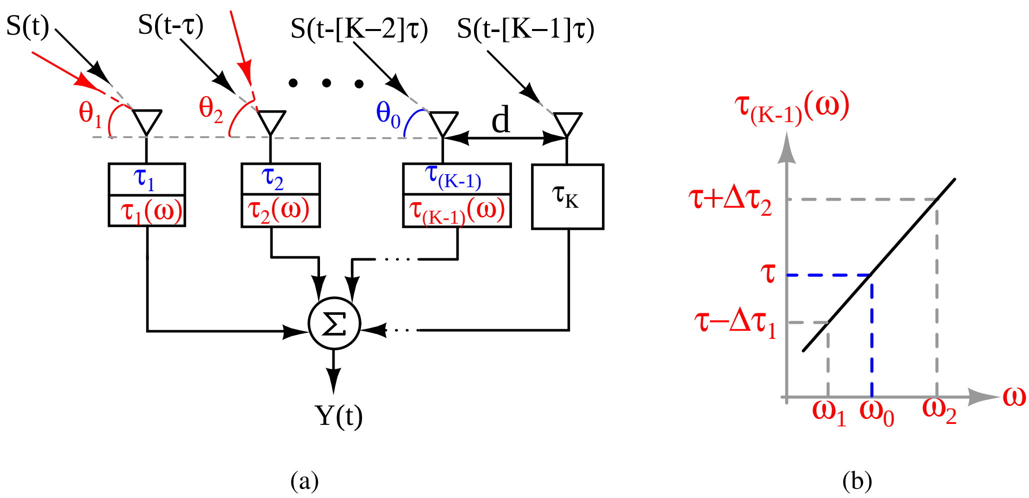


Figure 7.6: (a)Frequency dependent TTD circuits in K $\times$ 1 linear timed antenna array receiver, (b) Delay characteristics of

At frequency $\omega = \omega_0$, the delay element $\tau_{(\text{K}-1)}$ has a ideal value of τ. If the incident angle is θ_0 and d is the spacing between each antennas in the array, then θ_0 is expressed as,

$$\theta_0 = \cos^{-1}\left(\tau\frac{c}{d}\right) \tag{7.6}$$

At frequencies ω_1 and ω_2, the delay values are $[\tau - \Delta\tau_1]$ and $[\tau + \Delta\tau_2]$ respectively. These frequency dependent delays would change the value of θ_0. So in general, relation between the frequency dependent delay $\tau(\omega)$ and the incident angle $\theta(\omega)$ can be expressed as

$$\theta(\omega) = \cos^{-1}\left([\tau \pm \Delta\tau(\omega)]\frac{c}{d}\right) \tag{7.7}$$

Then the beam-squint $\Delta\theta$ resulting due to non-ideal TTDs used in the timed array receiver for the beam-steering angle of θ_0 is expressed as,

$$\begin{aligned}\Delta\theta &= \theta(\omega) - \theta_0 \\ &= \cos^{-1}\left([\tau \pm \Delta\tau(\omega)]\frac{c}{d}\right) - \theta_0\end{aligned} \tag{7.8}$$

Using this expression $\Delta\theta$ can be estimated from the delay variation $\Delta\tau$.

Considering TTDs with ideal gain characteristics with gain A and frequency dependent delay characteristics as seen in Figure 7.6(b), the expression for transfer function of the timed array receiver H(jω) with K $\times$ 1 linear antenna array is written as,

$$H(j\omega) = Ae^{-j\omega(K-1)\tau}\left[1 + \sum_{i=1}^{K-1} e^{\pm j\omega i\Delta\tau(\omega)}\right] \tag{7.9}$$

Using Taylor series of $e^{\pm j\omega i\Delta\tau(\omega)}$ and neglecting higher order terms, gain of the receiver $|H(j\omega)|$ is written as,

$$\begin{aligned}|H(j\omega)| &\approx A\sqrt{K^2 \pm \omega^2\Delta\tau^2(\omega)[1 + 2 + \cdots + (K-1)]^2} \\ &\approx A\sqrt{K^2 \pm (\omega^2\Delta\tau^2(\omega)K^2(K-1)^2/4)}\end{aligned} \tag{7.10}$$

From Eq. 7.10, it is observed that the gain at the beam-steering angle deviates from the ideal value $|H(j\omega)|$=AK and the deviation can be measured from $\Delta\tau$.

7.3.2 Case 1: RF beamforming with non-ideal TTDs

The RF path beamforming architecture with 4x1 linear timed antenna array is shown in Figure 7.7. The modulated signal $S(t)$ incident on each antenna with an angle θ_0 in the array is provided with different delays of τ_{0-3} by TTDs with the gains of A_{0-3}. The final beamformed signal Y(t) is expressed as $\sum_{k=0}^{3} A_k S(t - k\tau - \tau_k)$. The gain of the receiver is maximum at θ_0 if the TTDs are programmed with equal gains and delay values that yield constructive interference for incoming signal at angle θ_0, that is $\tau_0 = 3\tau$, $\tau_1 = 2\tau$, $\tau_2 = \tau$, and τ_3=0.

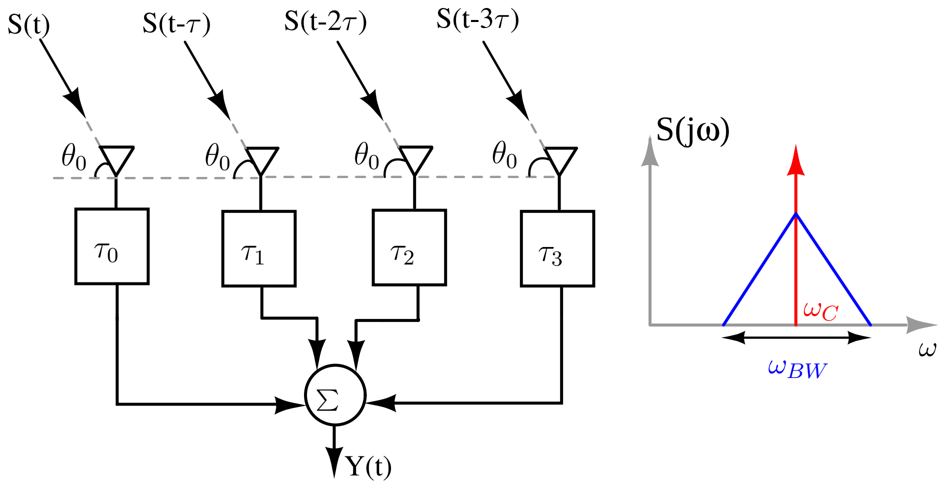


Figure 7.7: 4x1 linear timed antenna array receiver with RF path beamforming architecture.

We consider a typical 5G mm-wave receiver system with a carrier frequency of 28 GHz which is modulated with a baseband signal of 1 GHz [100]. The modulated signal S($j\omega$) has upper and lower cut-off frequencies at 29 GHz and 27 GHz respectively with center frequency f_c at 28 GHz. To maximize the gain of the receiver for the beam steering angle of $60°$, the TTDs are programmed with these delay values: τ_0=26.784 ps, τ_1=17.856 ps, τ_2=8.928 ps and τ_3=0. An ideal TTD must maintain flat group delay and constant gain across frequencies from 27 GHz to 29 GHz. For this ideal scenario, gain of the receiver is maximum at $60°$ at all the frequencies without any beam-squint ($\Delta\theta = 0$) as shown in Figure 7.8. However, using practical TTDs with

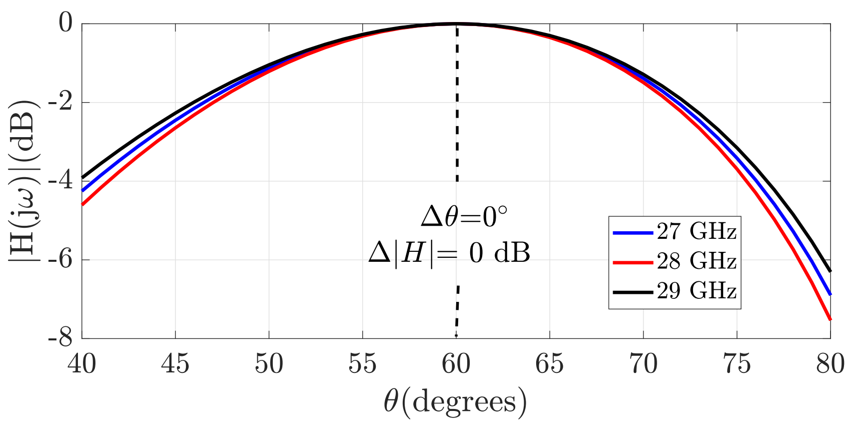


Figure 7.8: Radiation pattern of 4x1 antenna array with ideal TTDs in RF beamforming structure.

frequency-dependent delay and gain characteristics would introduce beam-squint and reduce the gain of the receiver gain at θ_0. To analyze the effect of TTD non-idealities, the delay and gain responses are assumed to be linearly decreasing from 27 GHz to 29 GHz with different levels of variations. Figure 7.9(a) and Figure 7.9(b) shows the 10%, 20% decrease from the ideal delay and gain values.

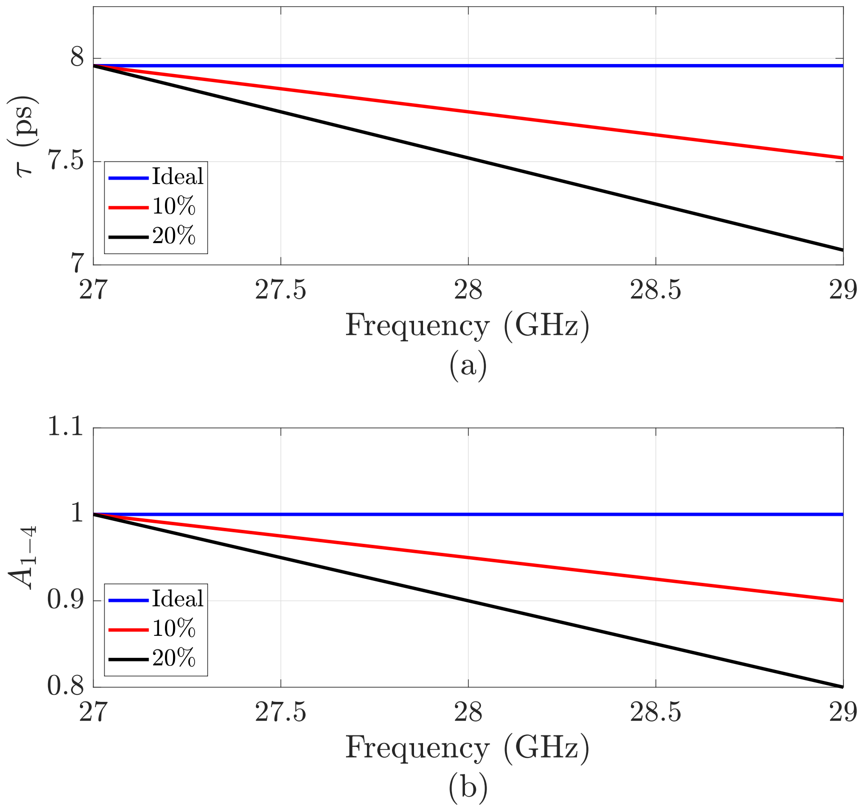


Figure 7.9: TTD with frequency dependent delay and gain characteristics showing 10% (a) and 20% (b) variations from ideal value.

Table 7.2 summarizes individual and combined effects of delay and gain variations in TTDs with the RF path beamforming architecture. Considering 10% and 20% gain variations with frequency-independent delays in the TTDs, the observed gain deviations at 60° are -0.915 dB and -1.93 dB respectively without any beam-squint. Using TTDs with ideal unity gain and 10% delay variation, a beam-squint of 3° with $\Delta|\mathrm{H}|$ of -0.144 dB at 60° is observed. A maximum beam-squint $\Delta\theta_0$ of 6° with a gain deviation $\Delta|\mathrm{H}|$ of -0.583 dB is observed in the radiation pattern considering 20% delay variation with ideal unity gains across the frequency range as seen in Figure 7.10. The combined effects of 20% delay and gain variations in practical TTDs on the receiver radiation pattern are shown in Figure 7.11. The results show a maximum gain drop of -2.521 dB at 60° with the beam-squint of 6°.

Table 7.2: Δ|H| and $\Delta\theta$ observed for different levels of delay and gain variations in the RF path beamforming architecture.

Delay variation(%)	Gain variation (%)	Δ|H|(dB)	$\Delta\theta°$
10	–	-0.144	3
20	–	-0.583	6
–	10	-0.915	0
–	20	-1.93	0
10	10	-1.059	3
20	20	-2.521	6

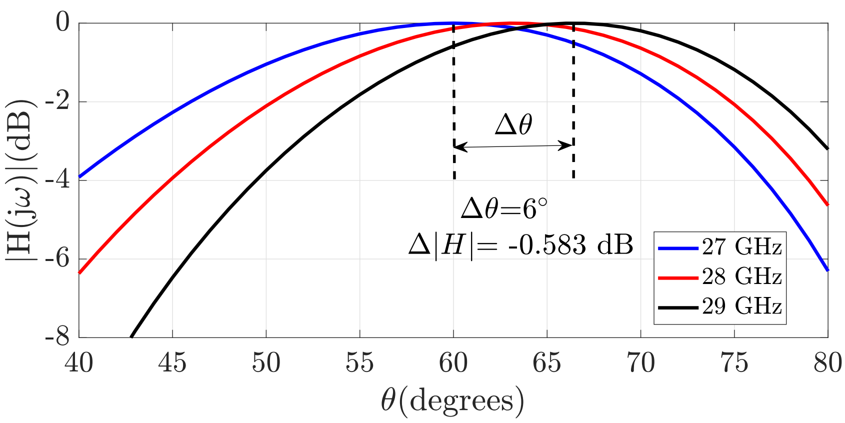


Figure 7.10: Effect of delay variation in TTDs on gain and directivity of the antenna array with 20% decrease from ideal values.

7.3.3 Case 2: BB beamforming with non-ideal TTDs

Figure 7.12 shows timed array receiver with BB beamforming technique in which the modulated signal S(t)=X(t)$e^{j\omega_c t}$ and its delayed versions on each antenna are downconverted with LO of frequency ω_c. The downcoverted signals $X_k(t) = X(t - k\tau)e^{-j\omega_c k\tau}$, (k=0,1,2,3) are then provided with different delays of τ_{0-3} by TTDs with the gains of A_{0-3} to enable beamforming at BB frequencies. The beamformed signal Y(t) can be expressed as $\sum_{k=0}^{3} A_k X(t - k\tau - \tau_k)e^{-j\omega_c k\tau}$.

The RF signal delays are much smaller than the baseband signal time periods. Hence $X(t) \approx X(t - k\tau)$ for small values of τ. To maximize the gain of the system during the beam-

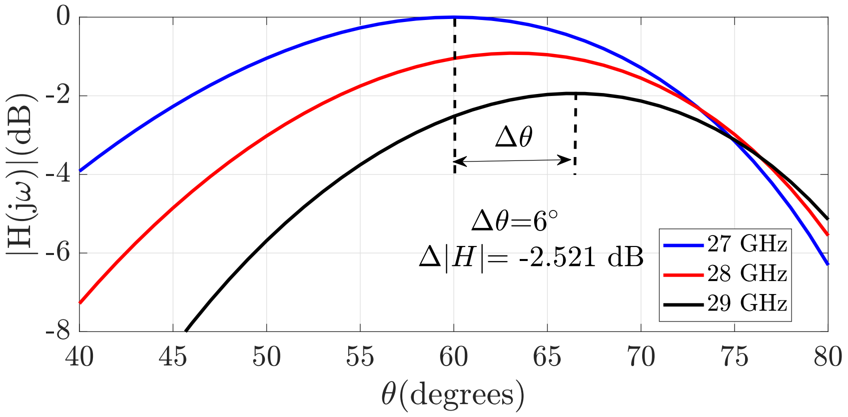


Figure 7.11: Combined effect of delay and gain variations in TTDs on gain and directivity of the antenna array with 20% decrease from ideal values.

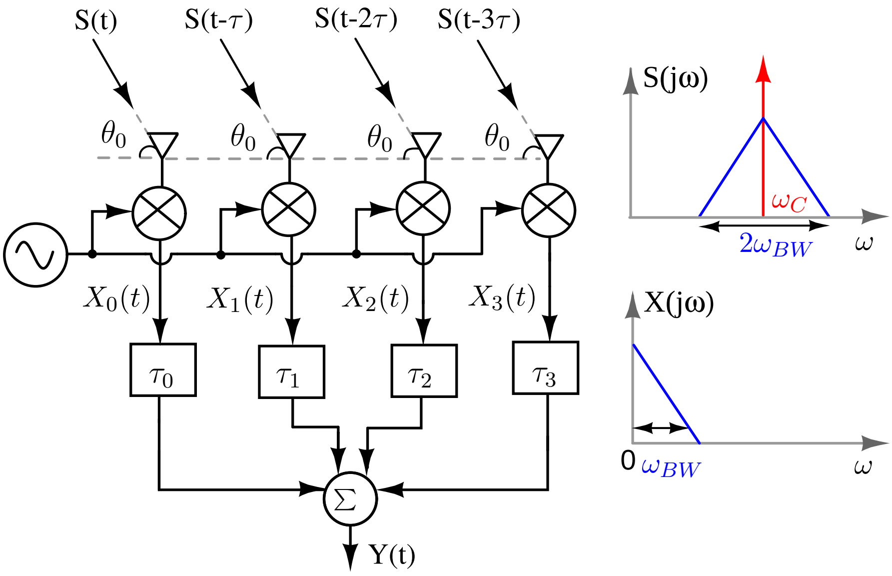


Figure 7.12: 4x1 linear timed antenna array receiver with BB beamforming architecture.

forming process at baseband frequencies, the TTDs with equal gains are configured with the following delay values: τ_0=3Nτ, τ_1=2Nτ, τ_2=Nτ and τ_3=0 where N=$\omega_c/\omega_{BW} \approx \omega_c/\omega$. With BB beamforming used for 5G mm-wave receivers as discussed earlier, the TTDs must be configured with these delay values: τ_0=750 ps, τ_1=500 ps, τ_2=250 ps and τ_3=0 to steer the radiation pattern at 60°. With the frequency-independent delay and gain characteristics of TTDs, gain of the receiver is maximum across all the frequencies of baseband signal as shown in Figure 7.13. To study the effect of TTD non-idealities in BB beamforming architecture, the delay and gain characteristics are assumed to be frequency dependent, in which the delay and gain values are

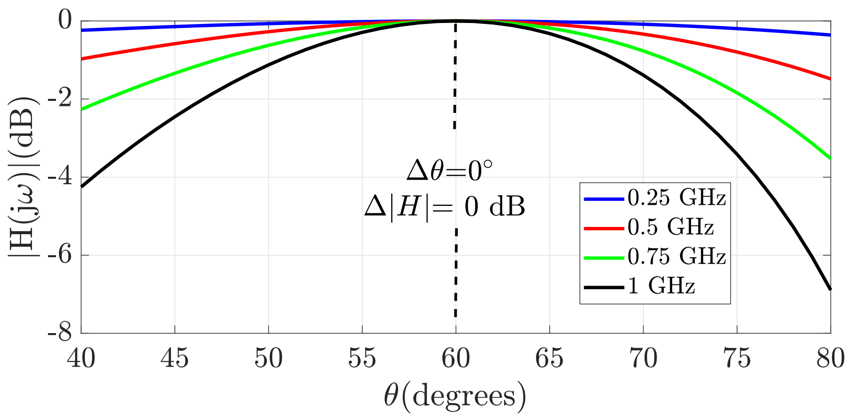


Figure 7.13: Radiation pattern of 4x1 antenna array with ideal TTDs in BB beamforming structure.

decreasing linearly from 0.25 GHz to 1 GHz with 10% and 20% variations as shown in Figure 7.14(a) and Figure 7.14(b).

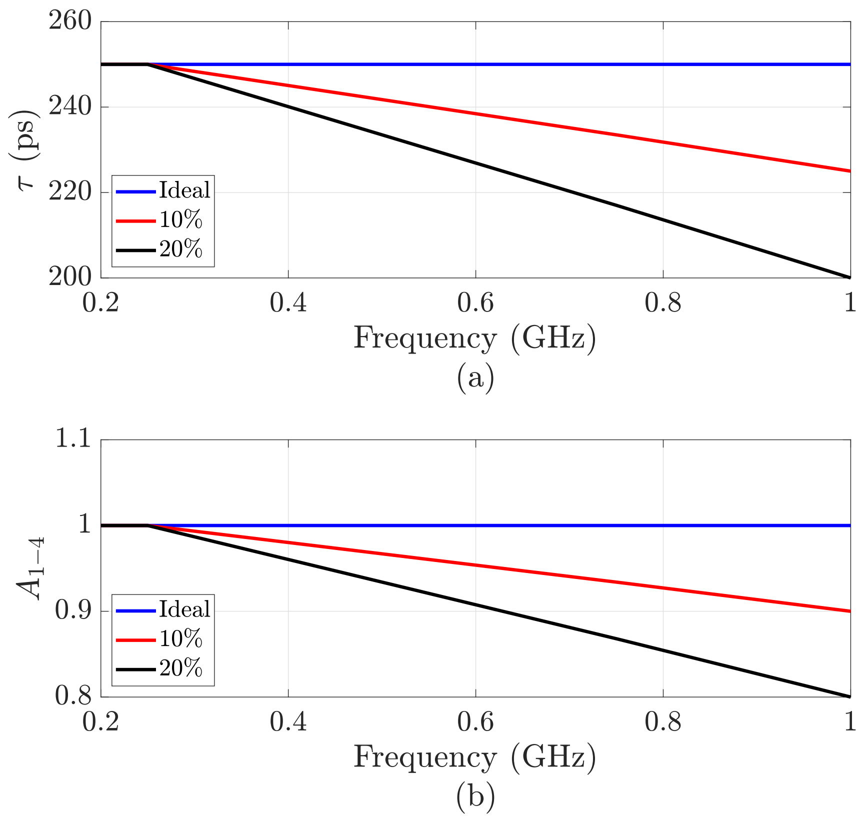


Figure 7.14: TTD with frequency dependent delay and gain characteristics showing 10% and 20% variations from 0.25 GHz to 1 GHz.

Table 7.3 summarizes individual and combined effects of delay and gain variations in

TTDs with the BB beamforming architecture. Considering 10% and 20% gain variations with frequency-independent delays in the TTDs, the observed gain deviations at 60° are -0.915 dB and -1.93 dB respectively without any beam-squint. Using TTDs with ideal unity gain and 10% delay variation, a beam-squint of $3°$ with $\Delta|\mathrm{H}|$ of -0.134 dB at $60°$ is observed. A maximum beam-squint $\Delta\theta_0$ of $6°$ with a gain deviation $\Delta|\mathrm{H}|$ of -0.543 dB is observed in the radiation pattern considering 20% delay variation with ideal unity gains across the frequency range as seen in Figure 7.15. The combined effects of 20% delay and gain variations in practical TTDs on the receiver radiation pattern are shown in Figure 7.16. The results show a maximum gain drop of -2.481 dB at $60°$ with the beam-squint of $6°$. The assumption of $X(t)$=$X(t-k\tau)$, where

Table 7.3: $\Delta|\mathrm{H}|$ and $\Delta\theta$ observed for different levels of delay and gain variations in the BB beamforming architecture.

Delay variation(%)	Gain variation (%)	$\Delta\|\mathrm{H}\|$(dB)	$\Delta\theta°$
10	–	-0.134	3
20	–	-0.543	6
–	10	-0.915	0
–	20	-1.938	0
10	10	-1.059	3
20	20	-2.481	6

$\tau = (d/c)\cos\theta_0$ is not valid for largely spaced antenna arrays. With $X(t) \neq X(t-k\tau)$ and considering 20% variations in delay and gain characteristics, the effect of TTD non-idealities is analyzed. Results show a maximum $\Delta|\mathrm{H}|$ of -2.697 dB with a beam-squint of $7°$ is observed in the radiation pattern. Hence, compared to scenario which is illustrated with Figure 7.16, there is an increase in $\Delta\theta$ by $1°$ with further degradation in $\Delta|\mathrm{H}|$ by -0.216 dB as seen in Figure 7.17

7.4 Quantifying delay and gain variations in TTDs

As discussed earlier, practical TTD circuits with delay and gain variations degrade the gain of radiation pattern of the receiver. Hence, it is important to quantify the maximum delay and

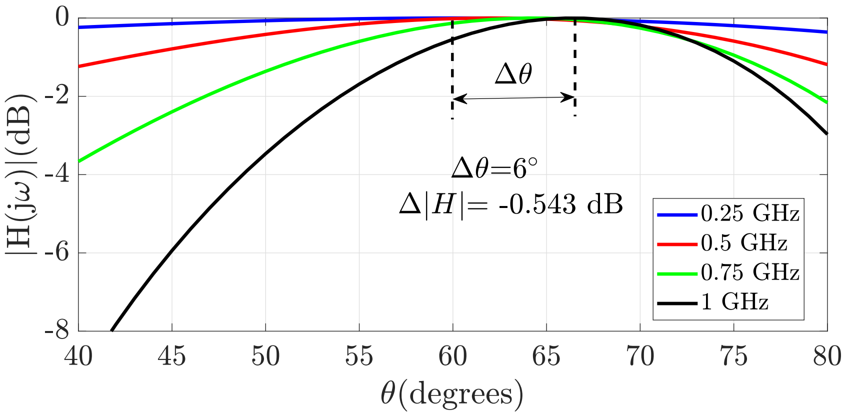


Figure 7.15: Effect of delay variation in TTDs on gain and directivity of the antenna array with 20% decrease from ideal values.

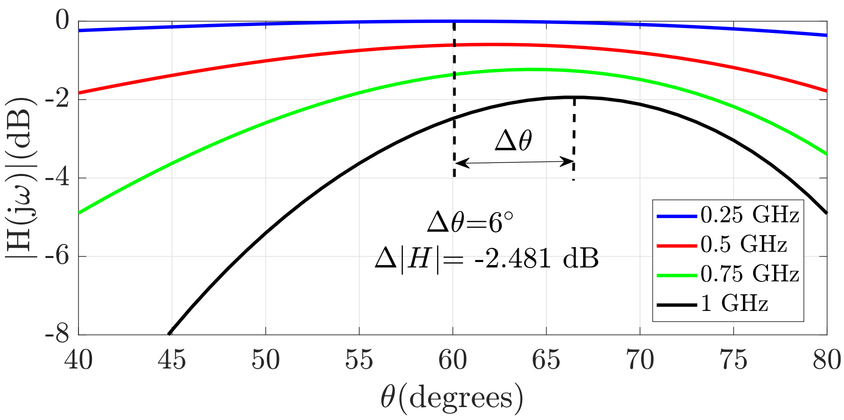


Figure 7.16: Combined effect of delay and gain variations in TTDs on gain and directivity of the antenna array with 20% decrease from ideal values.

gain variations in non-ideal TTDs. One way could be by comparing the value of the gain deviation Δ|H| in the beam-pattern resulting from non-ideal PS with the same level of variation as in practical TTDs. Table 7.4 tabulates maximum gain drop Δ|H| observed in linear 4x1 phased and timed antenna array receiver with RF beamforming configuration. Different values in Δ|H| are due to non-ideal PS and TTD elements with different levels of phase shift, delay, and gain variations. It is observed that TTDs with 20% variation in both delay and gain values show slightly lower Δ|H| compared to that of PS with 20% variations in phase shift and gain characteristics. Hence, an additional 1% variation in TTD delay and gain characteristics is tolerable to obtain an approximate value of Δ|H| as seen with PS. PS with 10% and 20% phase and gain variations show beam-squints of 5° and 10° respectively. But TTDs with the same

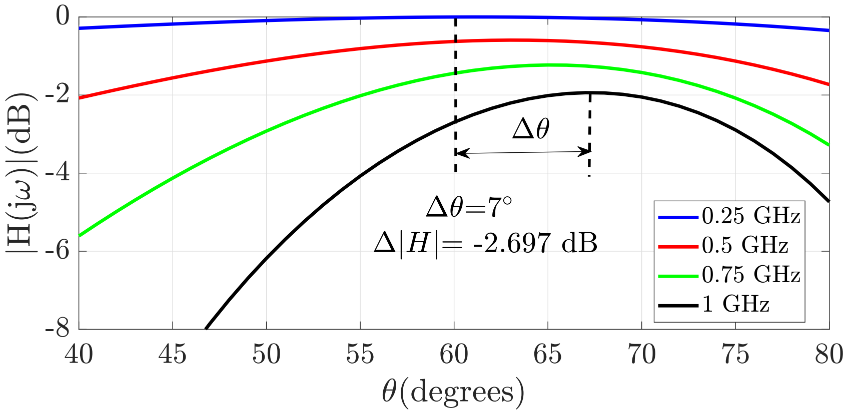


Figure 7.17: Radiation pattern of antenna array with 20% variations in delay and gain characteristics considering $X(t) \neq X(t - n\tau)$.

Table 7.4: Different Δ|H| and $\Delta\theta^{\circ}$ observed in beam-pattern for linear 4x1 phased and timed array receiver with RF beamforming architecture.

PS	Phase shift variation(%)	Gain variation(%)	Δ|H|(dB)	$\Delta\theta^{\circ}$
	10	10	-1.256	9
	20	20	-2.825	15
TTD	Delay variation(%)	Gain variation (%)	Δ|H|(dB)	$\Delta\theta^{\circ}$
	10	10	-1.059	3
	20	20	-2.521	6
	22	22	-2.866	7
	23	23	-3.045	8

level of non-idealities in delay and gain characteristics have 3° and 6° of beam-squints.

Another way of quantifying TTD non-idealities is by limiting the value of gain drop at θ_0 to some fixed value for a given bandwidth. If the maximum gain drop of -3 dB is allowed at the steering angle of 60° with frequency ranging from 27 GHz to 29 GHz, then the TTD elements can have 23% variations in both delay and gain characteristics.

With the detailed study of understanding the effect of TTD non-idealities in RF and BB beamforming configurations, it is observed that the problem of beam-squint also exists with practical TTDs. After studying the resulting beam-patterns of timed array receivers with different delay and gain variations in TTDs, it can be concluded that both delay and gain variations contribute to gain reduction of the antenna array at beam steering angle θ_0 while the beam-squint $\Delta\theta$ occurs only due to delay variation. Hence, designing TTDs with minimum gain and delay variations across the desired frequency range is important so that the combined effect of these would result in minimum beam-squint and less gain deviation in the beam pattern of the receiver.

Chapter 8

Conclusions and Future work

8.1 Conclusions

In this thesis, different types of delay-tuning techniques were discussed, which are used in changing the delay of continuous-time LPF and APF. A novel delay tuning technique for 1^{st}-order RC LPF is presented in which the delay of the circuit is varied with an external control signal using the principle of the Miller effect. The control signal with different time profiles is chosen to use the variable delay LPF for different applications. A certain range of fixed values of control signals are selected to use the variable delay LPFs in TTD and VCO circuit implementations. A time-varying control signal which is of decreasing or increasing ramp signals is used for expansion or compression operation respectively.

A technique of tuning the delay of 3^{rd}-order G_m-C Bessel APF using programmable Gm is also discussed in this work. Variable delay APFs realized with this delay technique are used in the design of TTD circuit for beamforming applications, expansion and compression of continuous time pulses. For TTD circuit implementation, Gms are tuned using programmable resistors in a constant-Gm biasing scheme. In the case of expansion and compression circuits, programmable Gms are realized with Gilbert multipliers with linearly decreasing and increasing control signals. In addition, a technique of calibrating negative impedance load in high-gain OTA with an impedance cancellation technique is also presented in this work. Unlike other techniques that use replicas for calibration, the proposed correction method utilizes the same OTA for correction hence avoiding the problem of mismatch from replicas.

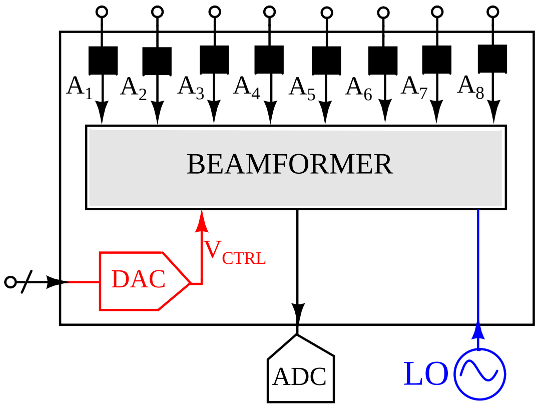
Finally, this work provides an overview of different beamforming architectures used in phased and timed array receivers. Different types of phase shifters and true time delay elements used in beamforming are discussed. A detailed analysis of true time delay-based beamforming circuits is presented. In many reported works on TTD's, immunity to beam squints is used as a motivating reason for the choice. In this work, it is shown how non-idealities in terms of delay and gain variations of true time delay elements do introduce beam squints in receivers. This work reports analysis and simulation results for these non-idealities and quantifies the same. This quantification helps in benchmarking true time delay based beamforming against phase shifter based beamforming, giving useful guidelines for improving the overall performance of timed antenna array receivers.

8.2 Future work

8.2.1 Exploration of TTD circuits in Timed-Array Recievers

A wideband TTD circuit for beamforming application with variable delay LPF was designed in this work. A prototype receiver with baseband beamforming configuration using this TTD circuit needs to be fabricated and tested. The proposed receiver is an 8-channel timed array receiver, in which the BB beamforming is performed on 8×1 linear antenna array as shown in Figure 8.1(a). Beamformer used in the receiver has tunable TTDs for the steering beam at the desired source. The TTDs are programmed with required delays using a control signal V_{CTRL} from a 4-bit DAC. Figure 8.1(b) shows the internal architecture of the beamformer used in the receiver. The input signal incident on each antenna in the array is a 28 GHz RF signal modulated with a baseband signal of 1 GHz arriving with different time delays. This incoming modulated signal from each antenna element is passed through a variable gain low noise amplifier (LNA). The amplified signal is then demodulated using an external local oscillator of 28 GHz to obtain the baseband signal. The BB signals after demodulation are delayed with the variable delay TTDs and then added constructively to obtain a beamformed signal at a given beam-steering angle. It is important to note that the gain of individual signal paths from each antenna will also affect the receiver array pattern. Hence, gain tuning of the signal path is also important as delay control. The programmable gain feature in LNA accomplishes the required gain control. The beamformed signal at BB frequency is then digitized using ADC for further digital signal processing.

TTD circuits were realized with fixed N-stage variable delay LPFs which showed a limited



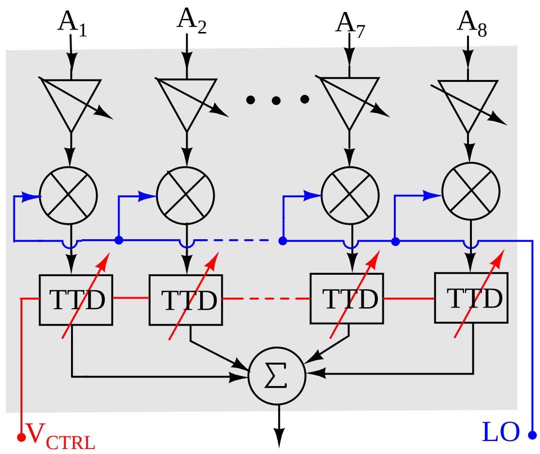


Figure 8.1: (a) Proposed 8-channel receiver module with beamforming on 8 × 1 linear antenna array (b) Internal architecture of beamformer used in the receiver.

delay range. Programmable order variable delay LPFs with controlled N-stages used in the realization of TTD circuit to have coarse and fine delay tuning features which improve the delay-range as well delay resolution of the TTD circuit.

8.2.2 Testing of Proposed Designs with Prototype Chip

Post-layout simulations of the TTD, expansion, and compression circuits in TSMC 65-nm technology were done. Testing of all these designs with a prototype chip needs to be done for final verification after fabrication.

The inline calibration technique for changing the negative conductance in high-gain OTA with impedance cancellation technique was discussed in this work. The use of these OTA architectures in the design of variable- G_m, G_m-C APF for implementing APF based TTD circuits and testing of these designs is left for future work.

8.2.3 Delay Tuning with Complex Impedances

The principle of the Miller effect was used in the delay tuning of 1st-order RC LPF. These delay-tuning techniques can be used to realize equalizers and tunable band-pass filters used in GNS/GPS receivers. The method can also be generalized to complex filters with complex impedance Z and these variable delay complex filters can be explored for different applications requiring tunable delay filters.

Appendix A

Differential Ramp Generator Circuit

From the discussion of the application of variable delay low-pass filter (LPF) in the realization of the pulse expansion (compression) circuit (Chapter 4), it is noted that the control signal α to LPF must decrease (increase) for expansion (compression) operation. Thus, the signal α can be a falling or rising ramp signal. Two different circuit-level architectures for generating differential ramp signals from differential square inputs are discussed here.

The control signal α must vary linearly from +1 to -1 for pulse expansion. This corresponds to α varying linearly from a fixed higher voltage level, V_H to a lower voltage level, V_L in the final circuit implementation. The basic idea of ramp generation is illustrated in the Fig. A.1. The charging and discharging currents of capacitor C is controlled by switch S_1. When a switch

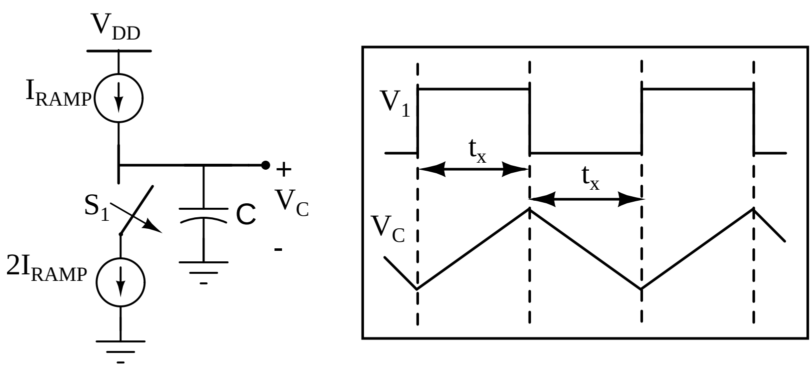


Figure A.1: Ramp generation using current sources and capacitor.

is open for a duration of time t_x, the constant current I_{RAMP} flows into the capacitor C resulting in a linear increase in voltage V_C. If the switch is closed a current of I_{RAMP} flows out of the capacitor which reduces the voltage V_C linearly. The charging and discharging voltages $V_c(t)$

can be expressed as,

$$V_c(t) = \pm \frac{I_O t_x}{C} \tag{A.1}$$

The circuit level implementation of the ramp generator with the differential architecture is shown in Fig. A.2(a). The transistors M_1, M_2 act as switches that control the charging and discharging of the capacitor C_L at each arm. The signals V_1 and V_2 are applied at the gates of M_1, M_2 respectively. The operation of ramp generation is discussed as follows:

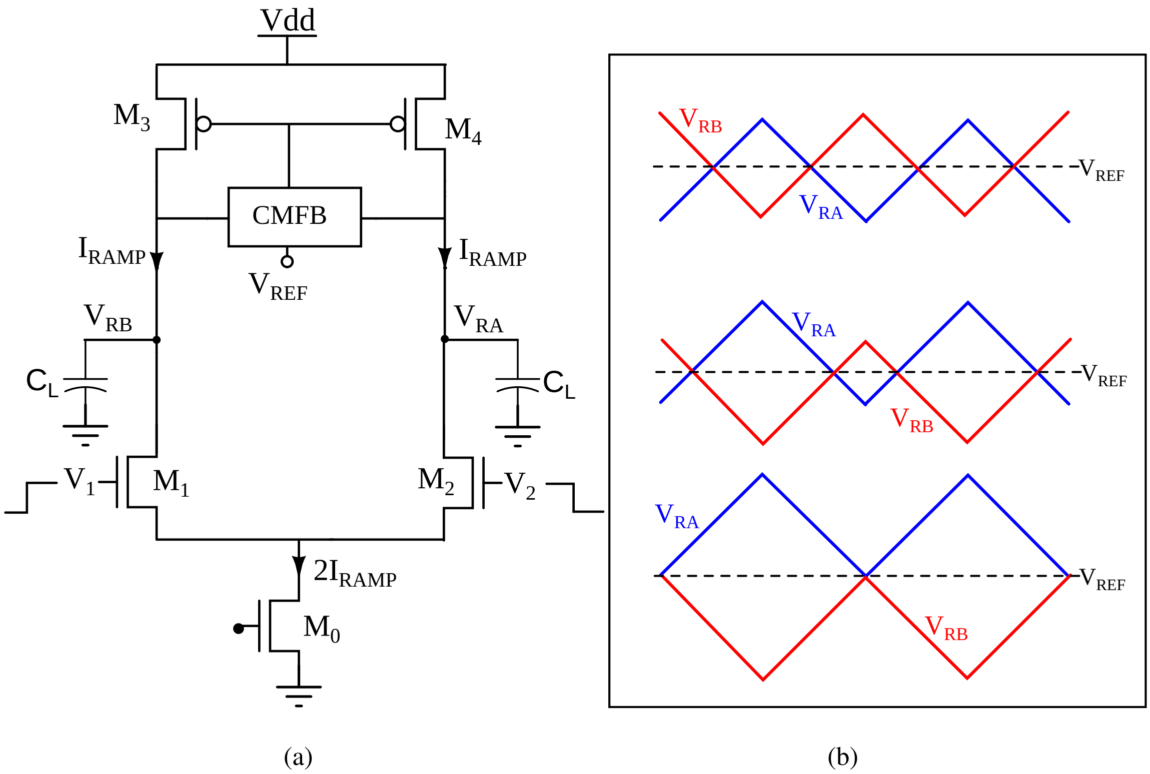


Figure A.2: (a) Ramp Generator with differential architecture, (b) output ramp signals with different steady states.

With $V_1 = 1$, $V_2 = 0$ the transistors M_1=ON and M_2=OFF states (these transistors are in linear/cutt-off region). As the current through $M_1 = 2I_{RAMP}$, this forces the capacitor C_L to discharge with the current I_{RAMP}. This causes the voltage V_{RB} to decrease linearly. On the other arm the capacitor C_L charges with the current I_{RAMP}, which results in a linear increase in voltage V_{RA}. The transistors M_3 and M_4 in the saturation region provide a constant current of I_{RAMP}. The differential ramp signals must swing about fixed reference voltage V_{REF}, which is set by the CMFB circuit.

The ramp generator circuit is a switched capacitor circuit with charging and discharging currents. Thus, it is necessary to initialize the capacitor nodes to fixed reference voltages to avoid different steady states as seen in Figure A.2(b).

The common-mode feedback only ensures the average between the differential signals to be maintained at the reference voltage V_{REF}. But it is also necessary for the average of the difference to be zero.

A.1 Implementation 1

Fig. A.3 shows the first implementation technique for generating symmetric ramp signals varying between fixed voltage values. Each negative feedback loop forces the average value of the

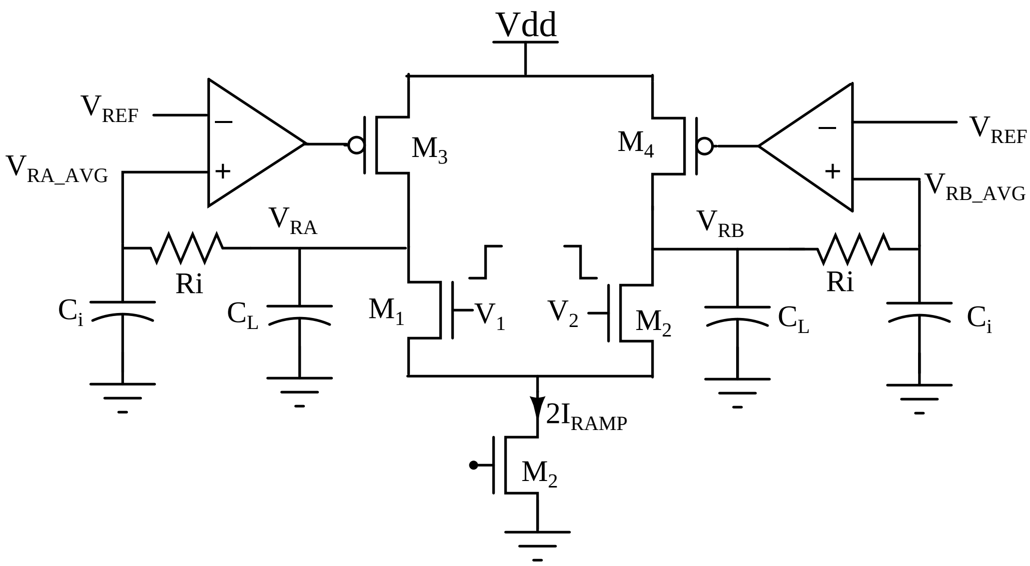


Figure A.3: Ramp generation with 2 negative feedback loops.

individual output signals V_{RA}, V_{RB} equal to the reference voltage V_{REF}. This also ensure the average between these output signals is held at a voltage V_{REF}. A simple RC integrator is used to find the average of the output signal. A passive integrator has to be used for stability reasons. Fig A.4 shows final output signals with symmetric swings across fixed reference voltage. The large integrating capacitor values and parasitic capacitances from resistor R_i limit the slope of the output ramp signal. Hence, an alternate circuit architecture is required to overcome this problem.

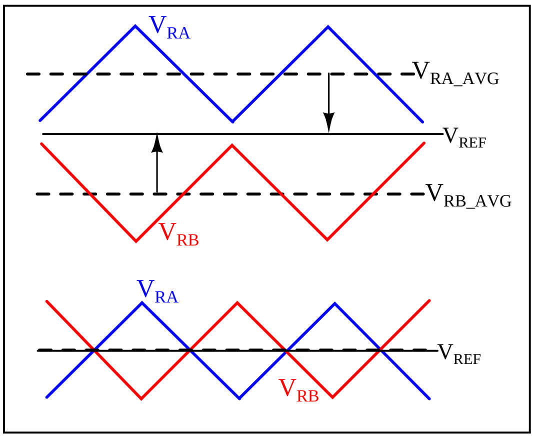


Figure A.4: Symmetric ramp signals from above implementation technique.

A.2 Implementation 2

Fig. A.5 shows the second implementation technique for generating symmetric ramp signals varying between fixed voltage values without affecting the slope of ramp signal. The archi-

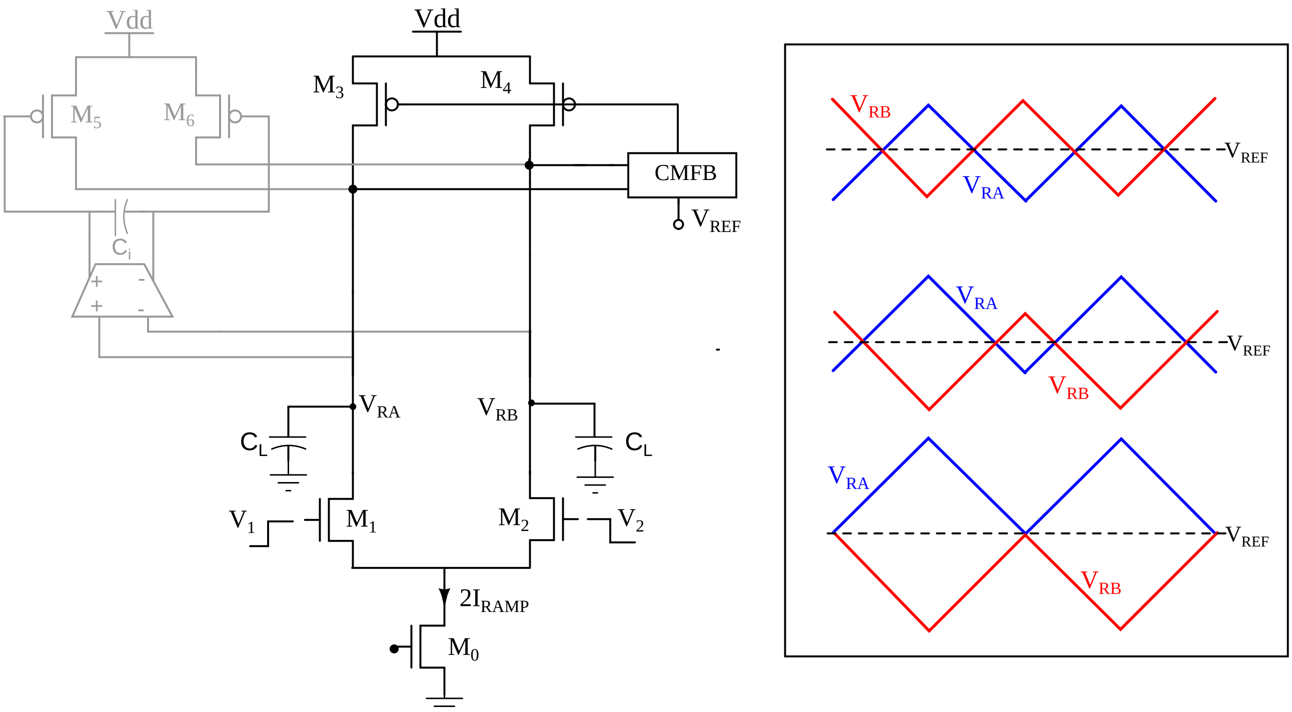


Figure A.5: Feeback loop with CMFB circuit maintaining $(V_{RA} + V_{RB})/2 = V_{REF}$ with different possible steady states.

tecture consists of 2 negative feedback loops which work in tandom. As discussed earlier, the circuit may settle to different steady states while maintaining an average value of V_{RA} and V_{RB} at fixed reference voltage V_{REF}.

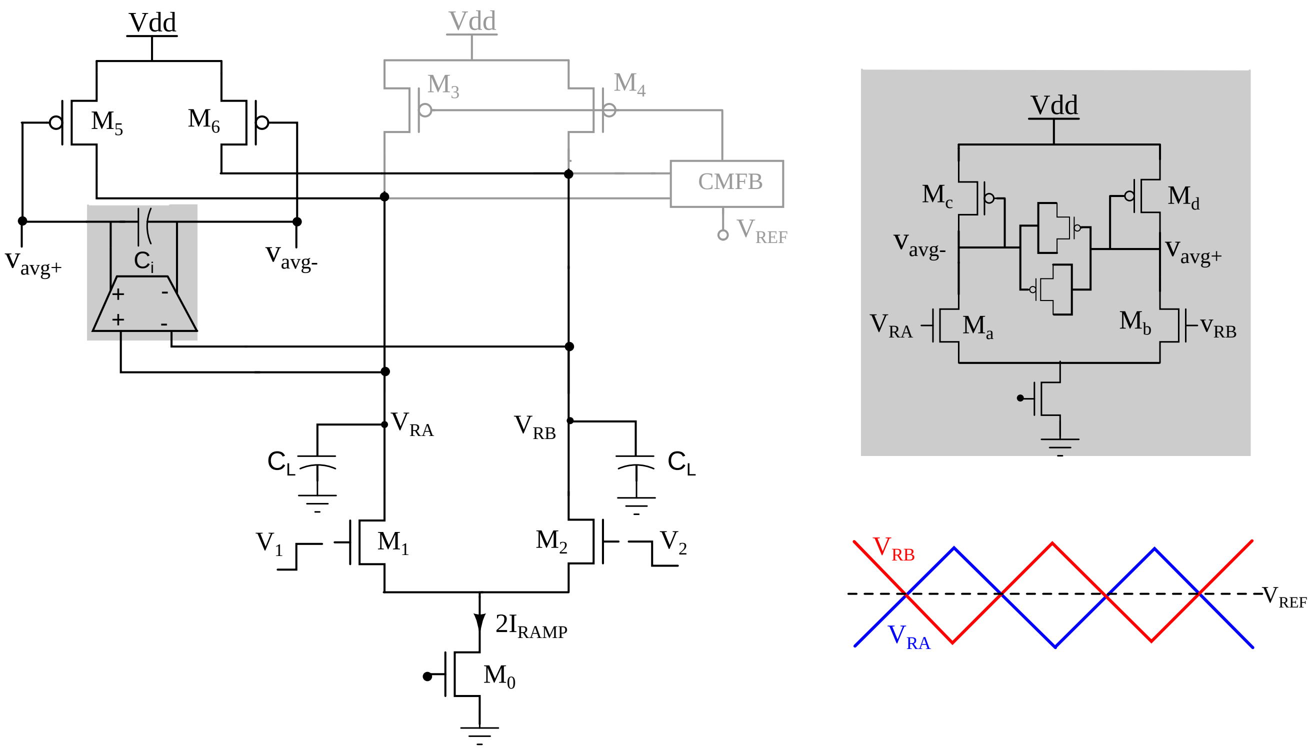


Figure A.6: Feedback loop with differential integrator active for differential signals.The transistor level schematic of integrator is highlighted.

The differential feedback loop consists of a differential integrator as highlighted in Fig. A.6 and is active for differential inputs of V_{RA}, V_{RB}. The transistor level schematic of the differential integrator is shown separately. The integrator block forces the average of differential inputs to 0 which eventually results in symmetric ramp signals over fixed reference voltage V_{REF}.

The RAMPGEN block used in the expansion (compression) circuit for converting differential square signals to differential ramp signals uses the implementation 2 circuit architecture.

Appendix B

Mismatch Tolerant Negative Conductance Load Tuning for High-Gain OTAs

OTA forms a key component of analog and mixed signal designs. Ideal OTAs have an infinite output impedance and most practical circuit designs using OTAs are designed assuming a very high output impedance. Practical OTA's have finite output impedance and hence a finite voltage gain. In scaled technologies, the output impedance of OTAs (and hence the DC gain) is quite low. The DC gain of OTA can be improved by cascading multiple stages or by using cascode configuration. Cascading limits the bandwidth and deteriorates the phase response of the OTA while cascode configuration implementation presents its own challenges due to the limited headroom in scaled technologies [101]. The DC gain of the OTA can also be increased by output impedance cancellation using a negative transconductance (i.e. a weak positive feedback) load. In this technique, an incremental negative conductance is added at the output node to cancel the output conductance of OTA. If done precisely, infinite DC gain is, in principle, possible. Different circuits for gain enhancement with impedance cancellation have been reported in the literature. In [102] the output impedance of a differential pair of inverter based transconductors is cancelled using matched cross coupled inverters at the output (see Figure B.1(a)). The negative conductance is tuned by adjusting the supply voltage (V_{DDtune}) of two self biased inverters as shown. Due to the use of inverter based OTA and self biased inverters, this technique has poor supply noise rejection. The required tuning voltage V_{DDtune} is evaluated via a replica that is used in a VCO circuit, that is tuned for highest Q. In [103], [104] output impedance of differential to single ended OTA is cancelled by adding effective negative conductance generated by Miller effect of an amplifier with a PMOS as a resistor in feedback (Figure B.1(b)). The amplifier 'A' used in the Miller circuit needs to have a tunable gain that is greater than 1 for generating the exact negative impedance required for the cancellation. In [103], the amplifier gain is adjusted manually in a test chip to demonstrate the high gain. Negative conductance generated using another transconductor g_N in positive feedback at the output node is reported in [105] (Figure B.1(c)). The value of $-g_N$ is controlled by voltage V_N to cancel the output conductance, g_o of G_m. In this technique, V_N is detected on chip by using a replica transconductor [105].

In all of the above impedance cancellation techniques, the tuning of negative conductance is done on-chip using an additional replica circuit. Any variations in this reference due to mismatch, would alter the gain of the OTA. Typically the standard deviation of the gain can be high as 9 dB for a OTA with a gain of around 50 dB [105]. In this work, a new technique of

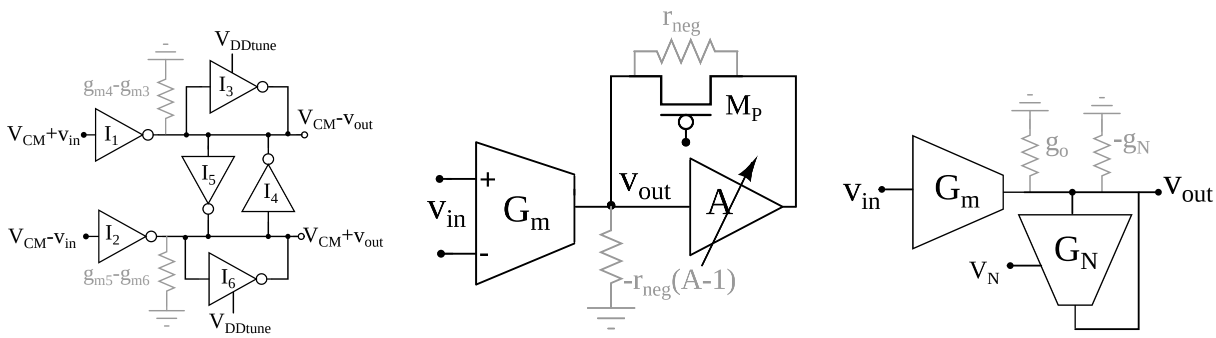


Figure B.1: Various techniques for generating negative conductance reported in literature. Negative impedance from (a) matched cross coupled inverters (b) Miller effect (c) other OTA in positive feedback

tuning negative load impedance for OTA output impedance cancellation is proposed. Here, V_N is detected without using any replica. In addition, the method uses digital feedback system for tracking this tuning voltage. Hence, it will be an added advantage for implementing in scaled technologies.

B.1 Proposed technique

In a high-gain OTA with impedance cancellation scheme (Figure B.1(c)) the effective output conductance, g_{oN} and DC gain, A_{oN} are expressed as,

$$g_{oN} = g_o - g_N \tag{B.1}$$

$$A_{oN} = G_m/g_{oN} \tag{B.2}$$

The negative transconductance, $-g_N$ of G_N is controlled by voltage V_N, which effectively reduces g_{oN} for increasing the DC gain of the OTA.

The basic principle of the proposed technique for determining V_N, as shown in Figure B.2, is by reducing the steady state error of the OTA configured in unity gain feedback to 0. This ideally increases the DC gain of OTA to infinity. Consider an incremental voltage Δv_i as input

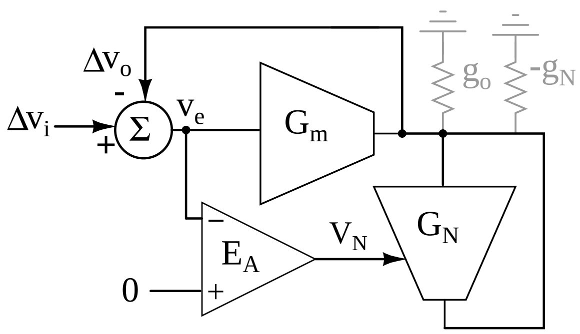


Figure B.2: Proposed V_N detection scheme by reducing steady state error of OTA.

as shown in Figure B.2. The resulting output voltage (Δv_o) and the steady state error (v_e) can be expressed as,

$$\Delta v_o = A_{oN} v_e \tag{B.3}$$

$$v_e = \Delta v_i - \Delta v_o \tag{B.4}$$

From Equations B.2, B.3 and B.4 voltage v_e can be rewritten as

$$v_e = \Delta v_i \; g_{oN}/(G_m + g_{oN}) \tag{B.5}$$

There is a negative feedback acting in parallel which drives this error, v_e to 0 and this happens only when $g_{oN} = 0$, i.e $g_o = g_N$. Thus, it satisfies the condition for output conductance cancellation for DC gain enhancement of OTA. As discussed in previous section, the proposed method does not use any replica circuit for tracking negative conductance when compared to other existing techniques. As the gain in negative feedback is finite. There is always a residual steady state error, implying that an over-correction is not possible.

For OTAs with DC gain of 18 dB the steady state error, v_e is about 100 mV. So to enhance its gain to a value of 60 dB one must reduce this error to around 0.9 mV. This requires a very high-gain error amplifiers E_A which is challenging at low technology nodes [106]. The proposed method addresses this problem by tracking V_N using digital circuits in a negative feedback loop.

B.2 Implementation Details

Figure B.3 shows the proposed high-gain OTA in differential architecture with internal V_N tracking scheme for improving the gain. The proposed OTA operates in 2 modes, namely (a) Calibration mode, (b) Normal mode.

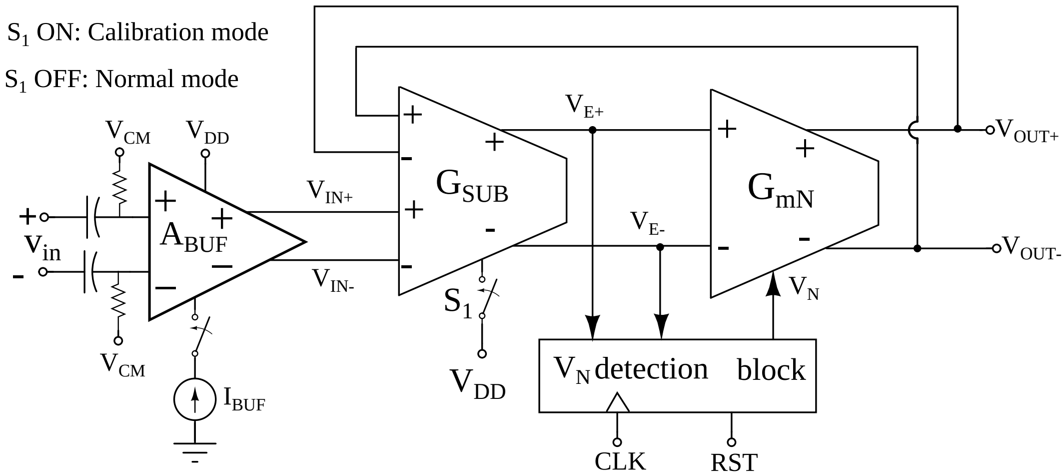


Figure B.3: Block level representation of proposed high-gain OTA operating in different modes of operation.

Referring to Figure B.3, in calibration mode, the differential steady state error signals V_E+, V_E- are obtained from G_{SUB}, by configuring it as subtractor when S_1 is ON. In this condition, one pair of differential inputs of G_{SUB} are held at V_{DD} by deactivating bias current of previous buffer stage A_{BUF} and the other input pair is connected to differential outputs of OTA. It is important to note that the incremental input voltage Δv_i for calibration is generated internally without using any switches in the signal path. The differential error signals, V_E+, V_E- are fed to V_N detection block for comparing and identifying the desired value of V_N for tuned negative conductance.

Figure B.4 shows the internal block diagram of V_N detection block. The comparator must be able to switch its output state for differential inputs of around 1 mV for minimizing the steady state error of OTA. Comparators with minimum resolution of 1 mV and internal offset cancellation techniques can be used for comparison. Comparators with such specifications are commonly used in ADC circuits, like the one reported in [107]. Figure B.5 shows the timing diagram of different signals in V_N detection block with respect to clock transitions. For an initial CMP OUT of logic 0, the value of N-bit up/down counter increments to increase the output voltage of N-bit DAC. This happens every clock cycle until comparator switches the signal

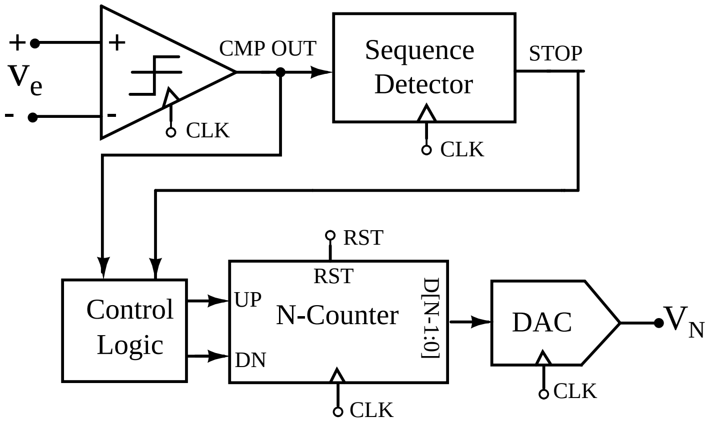


Figure B.4: Internal architecture of V_N detection block.

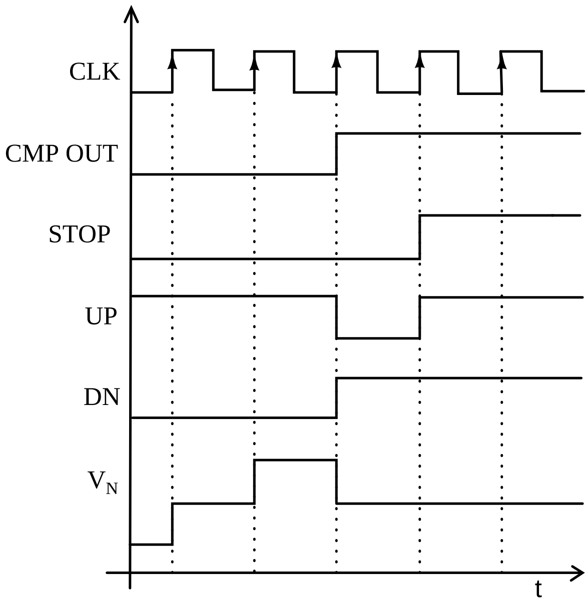


Figure B.5: Timing diagram showing various signal transitions with DAC settling to V_N.

CMP OUT to logic 1 which indicates change in sign of v_e. The control logic decrements the counter output by one bit by switching UP, DOWN signal. This is done to avoid overcorrecting for V_N from N-bit DAC. The STOP signal from sequence detector stops the tracking process by holding the DAC voltage to desired V_N. Similarly, for initial value of logic 1 for CMP OUT identifying V_N can be determined by counting downwards and incrementing by 1 after the comparator flips.

The OTA is configured in normal mode after calibration by turning S_1 OFF. In this mode, G_{SUB} is configured as buffer and the differential signals $V_{IN}+$, $V_{IN}-$ from previous buffer stage are applied to the differential inputs of G_{SUB}.

B.2.1 Transistor level implementation

Figure B.6 shows transistor level schematic of G_{SUB} with switch S_1 for controlling OTA modes of operation. As discussed earlier, in calibration mode Δv_i is generated internally. This is done by creating deterministic offset in subtractor by connecting M_{1a} in parallel via S_1 to M_1 of differential pair $M_{1,2}$ by closing S_1. In this case, the inputs $V_{IN}+$, $V_{IN}-$ are held at V_{DD}. In normal mode of OTA, the differential pair $M_{3,4}$ are made inactive by disabling the bias current I_B and disconnecting M_{1a} from M_1. Figure B.7 shows the MOS level schematic diagram of OTA

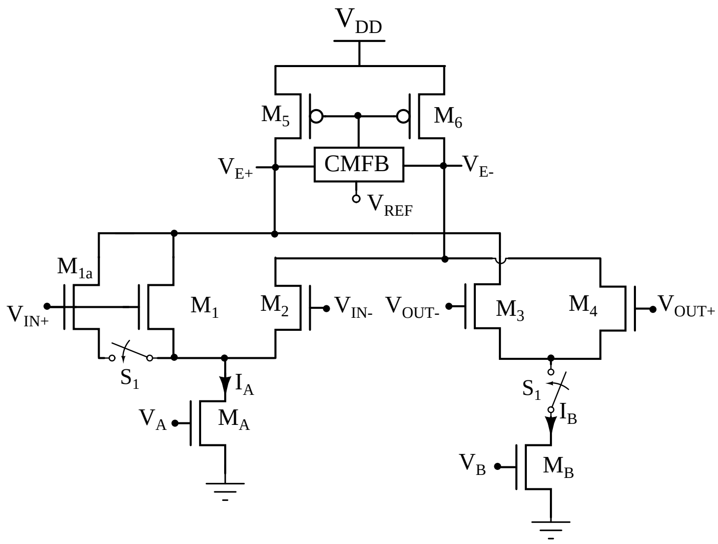


Figure B.6: Transistor level schematic diagram of G_{SUB} with configured as subtractor in calibration mode (S_1 ON) and buffer in normal mode (S_1 OFF).

with tunable negative conductance from differential pair $M_{5,6}$ in positive feedback.

The V_N detection block is implemented with strong arm dynamic comparator [108]. Offset correction of the comparator is not included in this simulation. Offset compensated comparators with residual offset under 1 mV, such as one reported in [107] could be used in actual implementation. An ideal model of 4-bit DAC controlled by 4-bit up/down counter is used in simulation for demonstrating the proposed technique.

B.3 Simulation Results

The proposed high-gain OTA is implemented in TSMC 65-nm technology with the supply voltage V_{DD} of 1.2 V. The OTA is biased with current I_O of 100 μA with input and output common mode voltage, $V_{CM} = 0.95\ V$. The subtractor block is biased with currents I_A and I_B of

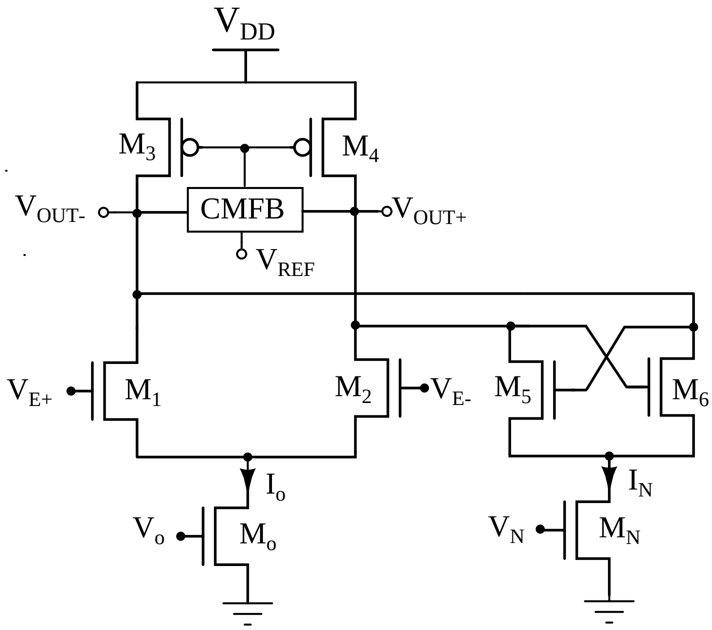


Figure B.7: MOS schematic of differential transconductor with negative conductance.

100 μA. The V_N detection block works with a clock frequency of 1 MHz and settles the DAC voltage V_N to 0.40 V after calibration. Simulation result (as seen in Figure B.8) shows that OTA without impedance cancellation has DC gain of 18dB and OTA with the proposed calibration method has a DC gain of 54 dB. The OTA has power dissipation of 0.45 mW with unity gain

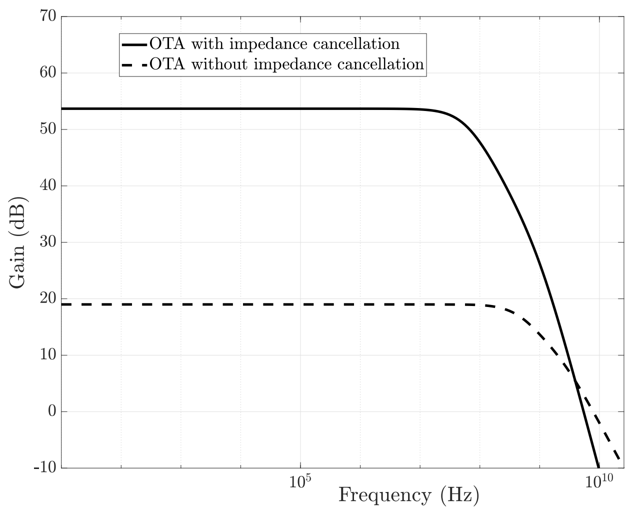


Figure B.8: Gain response of OTA with and without impedance cancellation.

bandwidth (UGB) of 5.4 GHz. The proposed OTA is compared with other existing architectures and observations are summarized in Table B.1. Process corner and temperature variation simulations are performed to check the robustness of OTA gain. Results show that the gain varies

Table B.1: Comparison of proposed OTA with other existing architectures

Parameters	Proposed OTA	OTA without impedance cancellation	[105]	[103]
Technology (nm)	65	65	130	350
Supply Voltage (V)	1.2	1.2	1.2	3
Gain (dB)	54	18	48	83
Power (mW)	0.45	0.16	9	2.4
UGB(GHz)	5.4	6.15	20	0.13 (C_L=2 pF)
Calibration type	Inline	Not applicable	Replica	Not mentioned
Immune to over correction	Yes	Not applicable	No	No

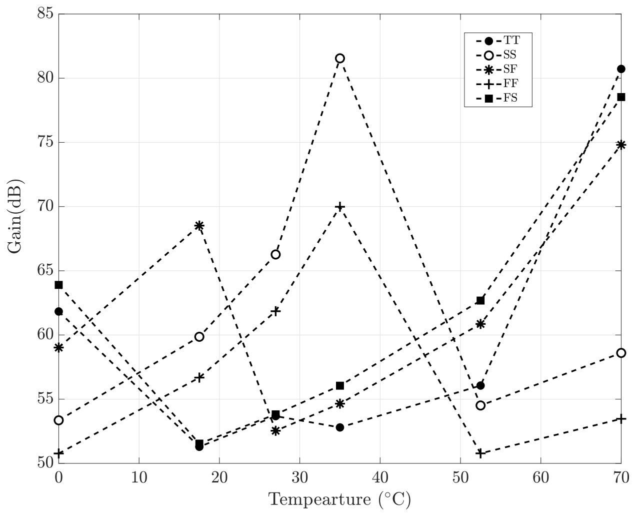


Figure B.9: Gain across all the process corners when temperature is varied from 0°C to 70°C

from a maximum value of 81 dB to a minimum value of 51 dB without reversal in gain polarity across all the process corners when temperature is varied from 0°C to 70°C.

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Patents and Publications from This Work

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Acknowledgments

I am deeply grateful to my supervisor, Prof. Naveen Kadayinti, for his guidance, encouragement, and patience at every stage of my work. His mentorship and expertise have been invaluable assets throughout this journey. I am thankful and honored to have had the opportunity to work with him. I sincerely thank my Research Progress Committee members: Prof. Naveen M.B., Prof. Ruma Ghosh, Prof. Saroj Mondal, Prof. Nagaveni S., and Prof. Samatha B. from the Department of EECE, IIT Dharwad for their valuable feedback and advice which have been integral to my research and academic progress. I would sincerely thank Prof. Bharath for his valuable input during the initial stage of my research. I would also like to thank Mr. Manjunath for his help in soldering during testing.

I want to express gratitude to my parents, and sister, especially my mother for her encouragement to pursue my research. I would like to thank my better half, my wife Priya for her moral support during my research. I would also like to thank my in-laws for their constant support during my study. Finally, thanks to my little princess Pranati for her cute smile which is a big source of motivation.

Mayur Shivamurthy